
O-RAN Work Group 5 (Open F1/W1/E1/X2/Xn interface Workgroup)

O1 Interface specification for O-CU-UP and O-CU-CP

Copyright © 2024 by O-RAN ALLIANCE e.V.

The copying or incorporation into any other work of part or all of the material available in this specification in any form without the prior written permission of O-RAN ALLIANCE e.V. is prohibited, save that you may print or download extracts of the material of this specification for your personal use, or copy the material of this specification for the purpose of sending to individual third parties for their information provided that you acknowledge O-RAN ALLIANCE as the source of the material and that you inform the third party that these conditions apply to them and that they must comply with them.

O-RAN ALLIANCE e.V.
Buschkauler Weg 27, 53347 Alfter, Germany
Register of Associations, Bonn VR 11238
VAT ID DE321720189.

Contents

1	Contents	2
2	List of Figures	6
3	List of Tables.....	7
4	Foreword	7
5	Modal verbs terminology	7
6	1 Scope.....	8
7	2 References	8
8	2.1 Normative references	8
9	2.2 Informative references.....	8
10	3 Definitions of terms symbols and abbreviations.....	9
11	3.1 Terms	9
12	3.2 Symbols.....	9
13	3.3 Abbreviations and acronyms	9
14	3.4 Conventions.....	9
15	3.5 Revision and compatibility handling.....	10
16	4 High Level Description	10
17	4.1 Top level functional description and terminology.....	10
18	4.1.1 Architecture for O-RAN WG5 functional split	10
19	4.1.2 O1 interface for O-CU architecture model	10
20	4.1.3 Transport Network.....	10
21	4.2 Interfaces.....	10
22	5 PNF Startup and Registration Management	11
23	5.1 PNF Plug-n-Play	11
24	5.2 PNF Registration.....	11
25	6 Heartbeat Management Services	11
26	6.1 Heartbeat Notification	11
27	6.2 Heartbeat Control.....	11
28	7 PNF Software Management Services.....	12
29	7.1 Software Package Naming and Content	12
30	7.2 Software Inventory.....	12
31	7.3 Software Download.....	12
32	7.4 Software Activation Pre-Check	12
33	7.5 Software Activate.....	12
34	8 Performance Management.....	12
35	8.1 Performance Data File Reporting.....	12
36	8.2 Performance Data Streaming.....	12
37	8.3 Measurement Job Control.....	12
38	9 Fault Management	13
39	9.1 Fault Notification.....	13
40	9.2 Fault Supervision Control.....	13
41	10 File Management	13
42	10.1 File Structure	13
43	10.2 Notification notifyFileReady	14
44	10.3 File Management Operation: listAvailableFiles.....	15
45	10.4 File Management Operation: transferFile.....	15
46	10.5 File Management Operation: downloadFile	15

1	11	Provisioning Management.....	16
2	11.1	Configuration for O-CU	16
3	11.2	Notification aspect	17
4	11.2.1	Subscribing for O-CU notification from SMO	17
5	11.3	Information model for O-CU-CP	17
6	11.3.1	Imported and associated information entities	17
7	11.3.2	Class diagram	17
8	11.3.3	Class definition.....	20
9	11.3.4	Attribute definitions.....	22
10	12	Trace Management Services.....	24
11	12.1	Call Trace	24
12	12.2	Minimization of Drive Testing (MDT).....	24
13	12.3	Radio Link Failure (RLF).....	24
14	12.4	RRC Connection Establishment Failure (RCEF)	25
15	12.5	Trace Control.....	25
16	12.6	Streaming Trace	25
17		Annex A (informative) : Counter definition.....	26
18	A.1	NR PDCP performance counters (Type A):.....	26
19	A.1.1	Distribution of UE per UL received data volume	26
20	A.1.2	Distribution of DL UE throughput in O-CU.....	27
21	A.1.3	Total discarded UL PDCP SDU volume	28
22	A.1.4	UL PDCP SDU volume discarded due to bearer release	28
23	A.1.5	Total discarded DL PDCP SDU volume discarded.....	29
24	A.1.6	DL PDCP SDU volume discarded due to bearer release	29
25	A.1.7	DL PDCP SDU volume discarded due to out of memory	30
26	A.1.8	PDCP transmission stop	30
27	A.1.9	UL F1-U packet loss rate	31
28	A.1.10	UL PDCP SDU Loss Rate	31
29	A.1.11	DL PDCP SDU Loss Rate	32
30	A.1.12	DL F1-U packet loss rate.....	32
31	A.2	NR PDCP performance counters (Type B):	33
32	A.2.1	UL PDCP SDU Data Volume	33
33	A.2.2	UL PDCP SDU Data Volume on X2 Interface	33
34	A.2.3	UL PDCP SDU Data Volume on Xn Interface.....	34
35	A.2.4	DL PDCP SDU Data Volume	35
36	A.2.5	DL PDCP SDU Data Volume on X2 Interface.....	35
37	A.2.6	DL PDCP SDU Data Volume on Xn Interface.....	36
38	A.2.7	UL PDCP SDU Data Volume per cell	37
39	A.2.8	UL PDCP SDU Data Volume on X2 Interface per cell.....	37
40	A.2.9	UL PDCP SDU Data Volume on Xn Interface per cell.....	38
41	A.2.10	DL PDCP SDU Data Volume per cell	38
42	A.2.11	DL PDCP SDU Data Volume on X2 Interface per cell.....	39
43	A.2.12	DL PDCP SDU Data Volume on Xn Interface per cell.....	39
44	A.3	NR S1-U Interface Performance Counters	40
45	A.3.1	UL PDCP SDU volume transmitted via S1-U UL GTP-U tunnel	40
46	A.3.2	DL PDCP SDU volume received via S1-U DL GTP-U tunnel.....	40
47	A.3.3	UL PDCP SDUs transmitted via S1-U UL GTP-U tunnel	41
48	A.3.4	DL PDCP SDUs transmitted via S1-U DL GTP-U tunnel	41
49	A.4	NR NG-U Interface Performance Counters.....	42
50	A.4.1	UL PDCP SDU volume transmitted via NG-U UL GTP-U tunnel.....	42
51	A.4.2	DL PDCP SDU volume received via NG-U DL GTP-U tunnel	42
52	A.5	NR X2-U Interface performance counters (O-CU).....	43
53	A.5.1	UL PDCP SDU volume received via X2-U UL GTP-U tunnel (X2-U UL data forwarding).....	43
54	A.5.2	DL PDCP SDU volume received via X2-U DL GTP-U tunnel (X2-U DL data forwarding).....	43
55	A.6	NR Xn-U Interface performance counters (O-CU).....	44
56	A.6.1	UL PDCP SDU volume received via Xn-U UL GTP-U tunnel (Xn-U UL data forwarding).....	44
57	A.6.2	DL PDCP SDU volume received via Xn-U DL GTP-U tunnel (Xn-U DL data forwarding).....	44
58	A.7	NR X2-C Interface performance counters (O-CU).....	45
59	A.7.1	Transmitted X2-C messages.....	45
60	A.7.2	Received X2-C messages.....	45

1	A.8 NR Xn-C Interface performance counters (O-CU).....	46
2	A.8.1 Transmitted Xn-C messages.....	46
3	A.8.2 Received Xn-C messages.....	46
4	A.9 NR F1-C Interface performance counters (O-CU-CP).....	47
5	A.9.1 Transmitted F1-C messages	47
6	A.9.2 Received F1-C messages.....	47
7	A.10 Number of UE Contexts for EN-DC.....	47
8	A.10.1 Max SN terminated split bearer UE Contexts.....	47
9	A.10.2 Max SN terminated MCG bearer UE Contexts.....	48
10	A.10.3 Average SN terminated split bearer UE Contexts.....	49
11	A.10.4 Average SN terminated MCG bearer UE Contexts.....	49
12	A.10.5 Total DL CA UE Contexts PSCell (100ms)	50
13	A.10.6 Total DL CA UE Contexts PSCell (10s).....	51
14	A.10.7 Total DL CA UE Contexts SCell (100ms)	52
15	A.10.8 Total DL CA UE Contexts SCell (10s)	52
16	A.10.9 Max DL CA UE Contexts PSCell (100ms)	53
17	A.10.10 Max DL CA UE Contexts PSCell (10s)	54
18	A.10.11 Max DL CA UE Contexts SCell (100ms)	55
19	A.10.12 Max DL CA UE Contexts SCell (10s)	55
20	A.11 Number of UE Contexts for SA.....	56
21	A.11.1 Total RRC Connected UE Contexts.....	56
22	A.11.2 Total UE Contexts Per QoS.....	57
23	A.11.3 Total SCell Configuration UE Contexts.....	58
24	A.11.4 Total Measurement Gap UE Contexts.....	59
25	A.11.5 Max RRC Connected UE Contexts.....	59
26	A.11.6 Max UE Contexts Per QoS.....	60
27	A.11.7 Max SCell Configuration UE Contexts.....	61
28	A.11.8 Max Measurement Gap UE Contexts.....	62
29	A.11.9 Total emergency UE Contexts.....	62
30	A.11.10 Total high priority access UE Contexts.....	63
31	A.11.11 Max emergency UE Contexts	64
32	A.11.12 Max high priority access UE Contexts	65
33	A.11.13 Total SN terminated split bearer UE Contexts in NR-DC	66
34	A.11.14 Max SN terminated split bearer UE Contexts in NR-DC	67
35	A.11.15 Total UE Contexts during voice call	68
36	A.12 Monitoring of procedure for EN-DC	68
37	A.12.1 Number of SgNB Addition procedure attempted for each Cell	68
38	A.12.2 Number of SgNB Addition procedure successfully for each Cell.....	69
39	A.12.3 Number of SgNB Addition procedure attempted for each neighbour eNB	69
40	A.12.4 Number of SgNB Addition procedure successfully for each neighbour eNB.....	70
41	A.12.5 Number of SgNB modification procedure attempted for each neighbour eNB	70
42	A.12.6 Number of SgNB modification procedure successfully for each neighbour eNB.....	71
43	A.13 Monitoring of RRC Connection for EN-DC.....	71
44	A.13.1 Number of Measurement Gap Coordination for per FR2 gap via SRB3 attempted	71
45	A.13.2 Number of Measurement Gap Coordination for per FR2 gap via SRB3 successfully	72
46	A.13.3 Number of Inter gNB-DU PSCell Change via SRB3 attempted.....	72
47	A.13.4 Number of Inter gNB-DU PSCell Change via SRB3 successfully	73
48	A.13.5 Number of Intra gNB-DU PSCell Change via SRB3 attempted.....	73
49	A.13.6 Number of Intra gNB-DU PSCell Change via SRB3 successfully	74
50	A.14 Monitoring of RRC Connection for SA	74
51	A.14.1 Number of initial RRC Connection requests	74
52	A.14.2 Number of RRC connection completions for initial RRC Connection requests	74
53	A.14.3 Number of retransmission RRC Connections requests.....	75
54	A.14.4 Number of RRC Connections completions for retransmission RRC Connections requests	75
55	A.14.5 Number of incomplete RRC Connection by failed resource allocations	76
56	A.14.6 Number of incomplete RRC Connection by O-CU internal error.....	76
57	A.14.7 Number of initial RRC Connection requests each Establishment Cause.....	76
58	A.14.8 Number of RRC connection completions for initial RRC Connection requests each Establishment Cause	77
59	A.14.9 Number of retransmission RRC Connections requests each Establishment Cause	78
60	A.14.10 Number of RRC Connections completions for retransmission RRC Connections requests each	
61	Establishment Cause	79

1	A.14.11 Number of incomplete RRC Connection by failed resource allocations each Establishment Cause.....	80
2	A.14.12 Number of incomplete RRC Connection by O-CU internal error each Establishment Cause.....	80
3	A.15 Monitoring of Establishment calls for SA	81
4	A.15.1 Number of attempted establishment calls for mo-Data	81
5	A.15.2 Number of successful establishment calls for mo-Data.....	82
6	A.15.3 Number of incomplete establishment calls for mo-Data by protocol error.....	82
7	A.15.4 Number of incomplete establishment calls for mo-Data by O-CU internal error.....	83
8	A.15.5 Number of incomplete establishment calls for mo-Data by O-DU error detection	83
9	A.15.6 Number of attempted establishment calls for mt-Access.....	83
10	A.15.7 Number of successful establishment calls for mt-Access.....	84
11	A.15.8 Number of incomplete establishment calls for mt-Access by protocol error.....	84
12	A.15.9 Number of incomplete establishment calls for mt-Access by O-CU internal error.....	85
13	A.15.10 Number of incomplete establishment calls for mt-Access by O-DU error detection	85
14	A.15.11 Number of attempted establishment calls for mo-Signalling.....	85
15	A.15.12 Number of successful establishment calls for mo-Signalling	86
16	A.15.13 Number of incomplete establishment calls for mo-Signalling by protocol error	86
17	A.15.14 Number of incomplete establishment calls for mo-Signalling by O-CU internal error	87
18	A.15.15 Number of incomplete establishment calls for mo-Signalling by O-DU error detection.....	87
19	A.15.16 Number of attempted establishment calls for mo-SMS	88
20	A.15.17 Number of successful establishment calls for mo-SMS	88
21	A.15.18 Number of incomplete establishment calls for mo-SMS by protocol error	89
22	A.15.19 Number of incomplete establishment calls for mo-SMS by O-CU internal error	89
23	A.15.20 Number of incomplete establishment calls for mo-SMS by O-DU error detection.....	89
24	A.15.21 Number of attempted establishment calls for high priority access	90
25	A.15.22 Number of successful establishment calls for high priority access	90
26	A.15.23 Number of incomplete establishment calls for high priority access by protocol error	91
27	A.15.24 Number of incomplete establishment calls for high priority access by O-CU internal error	91
28	A.15.25 Number of incomplete establishment calls for high priority access by O-DU error detection.....	92
29	A.15.26 Number of attempted establishment calls for emergency	92
30	A.15.27 Number of successful establishment calls for emergency	92
31	A.15.28 Number of incomplete establishment calls for emergency by protocol error	93
32	A.15.29 Number of incomplete establishment calls for emergency by O-CU internal error	93
33	A.15.30 Number of incomplete establishment calls for emergency by O-DU error detection.....	94
34	A.15.31 Discarded paging records	94
35	A.16 Monitoring of PDU session connection for SA	95
36	A.16.1 Accepted PDU Session Resource Configuration	95
37	A.16.2 Completed PDU Session Resource Configuration	95
38	A.16.3 Interrupted PDU Session Resource configuration by Protocol Error	96
39	A.16.4 Interrupted PDU Session Resource configuration by O-DU Error Detection.....	96
40	A.16.5 Interrupted PDU Session Resource configuration by O-CU internal error.....	97
41	A.16.6 Accepted PDU Session Resource Configuration for emergency and high priority access	98
42	A.16.7 Completed PDU Session Resource Configuration for emergency and high priority access.....	98
43	A.16.8 Interrupted PDU Session Resource configuration for emergency and high priority access by Protocol Error	99
44	A.16.9 Interrupted PDU Session Resource configuration for emergency and high priority access by O-DU Error	
45	Detection.....	100
46	A.16.10 Interrupted PDU Session Resource configuration for emergency and high priority access by O-CU	
47	internal Error	100
48	A.17 Monitoring of mobility for SA	101
49	A.17.1 Accepted Handover.....	101
50	A.17.2 Completed Handover	102
51	A.17.3 Accepted Inter RAT Handover to LTE	102
52	A.17.4 Completed Inter RAT Handover to LTE.....	102
53	A.17.5 Accepted EPS Fallback.....	103
54	A.17.6 Completed EPS Fallback.....	103
55	A.17.7 Normal interrupted EPS Fallback	104
56	A.17.8 Accepted Handover during Voice call	104
57	A.17.9 Completed Handover during Voice call	104
58	A.17.10 Accepted Inter RAT Handover to LTE during Voice call	105
59	A.17.11 Completed Inter RAT Handover to LTE during Voice call.....	105
60	A.18 Monitoring of RRC re-establishment for SA.....	106
61	A.18.1 Number of initial RRC re-establishment requests when UE context can be retrieved	106

A.18.2	Number of RRC re-establishment request completions for initial RRC re-establishment requests when UE context can be retrieved	106
A.18.3	Number of retransmission RRC re-establishment requests when UE context can be retrieved.....	107
A.18.4	Number of RRC re-establishment request completions for retransmission RRC re-establishment request when UE context can be retrieved	107
A.18.5	Number of initial RRC re-establishment requests when UE context cannot be retrieved.....	107
A.18.6	Number of RRC re-establishment request completions for initial RRC re-establishment requests when UE context cannot be retrieved.....	108
A.18.7	Number of retransmission RRC re-establishment requests when UE context cannot be retrieved	108
A.18.8	Number of RRC re-establishment request completions for retransmission RRC re-establishment requests when UE context cannot be retrieved	109
A.18.9	Number of initial RRC re-establishment requests during Voice call when UE context can be retrieved	109
A.18.10	Number of RRC re-establishment request completions for initial RRC re-establishment requests during Voice call when UE context can be retrieved.....	110
A.18.11	Number of retransmission RRC re-establishment requests during Voice call when UE context can be retrieved	110
A.18.12	Number of RRC re-establishment request completions for retransmission RRC re-establishment request during Voice call when UE context can be retrieved	111
A.19	Monitoring of connection status for SA	111
A.19.1	Normally Released Calls	111
A.19.2	Abnormally Released Calls	111
A.19.3	Another Cell HO UE Contexts	112
A.19.4	Normally Released Calls during voice call.....	112
A.19.5	Abnormally Released Calls during voice call.....	113
A.19.6	Another Cell HO UE Contexts during voice call.....	113
A.20	Monitoring of procedure for NR-DC	114
A.20.1	Number of S-NG-RAN Node Addition procedure attempted for each Cell	114
A.20.2	Number of S-NG-RAN Node Addition procedure successfully for each Cell.....	114
A.20.3	Number of S-NG-RAN Node Addition procedure attempted for each neighbour gNB	114
A.20.4	Number of S-NG-RAN Node Addition procedure successfully for each neighbour gNB.....	115
A.20.5	Number of S-NG-RAN Node modification procedure attempted for each Cell	116
A.20.6	Number of S-NG-RAN Node modification procedure successfully for each Cell.....	116
A.20.7	Number of S-NG-RAN Node modification procedure attempted for each neighbour gNB	117
A.20.8	Number of S-NG-RAN Node modification procedure successfully for each neighbour gNB.....	117
A.20.9	Number of S-NG-RAN Node Release procedure attempted for each Cell	118
A.20.10	Number of S-NG-RAN Node Release procedure successfully for each Cell	118
A.20.11	Number of S-NG-RAN Node Release procedure attempted for each neighbour gNB	119
A.20.12	Number of S-NG-RAN Node Release procedure successfully for each neighbour gNB	119
A.20.13	Number of SCGFailureInformation received.....	120
A.21	Monitoring of CA for SA	120
A.21.1	Number of SCell Addition procedure attempted.....	120
A.21.2	Number of SCell Addition procedure successfully	120
A.21.3	Number of SCell Change procedure attempted	121
A.21.4	Number of SCell Change procedure successfully	121
A.21.5	Number of SCell Deletion procedure attempted.....	122
A.21.6	Number of SCell Deletion procedure successfully	122
	Revision History	123

List of Figures

Figure 6-1:	Heartbeat Notification	11
Figure 10-1:	File Ready Notification.....	15
Figure 11-1:	Containment for SecurityHandling and CUCountGroup	18
Figure 11-2:	Containment for PDCCPConfig	19
Figure 11-3:	Inheritance for CUCountGroup, SecurityHandling, PDCCPConfig	20

List of Tables

Table 1: Parameters list for Notification notifyFileReady	14
---	----

Foreword

This Technical Specification (TS) has been produced by WG5 of the O-RAN Alliance.

The contents of the present document are subject to continuing work within O-RAN WG5 and may change following formal O-RAN approval. Should the O-RAN.org modify the contents of the present document, it will be re-released by O-RAN Alliance with an identifying change of release date and an increase in version number as follows:

Release x.y.z

where:

- x the first digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have x=01).
- y the second digit is incremented when editorial only changes have been incorporated in the document.
- z the third digit is included only in working versions of the document indicating incremental changes during the editing process.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are NOT allowed in O-RAN deliverables except when used in direct citation.

1 Scope

The present document specifies the O1 Interface specification for O-CU used over the interface linking the O-CU (O-RAN Central Unit) with Service Management and Orchestration (SMO).

2 References

2.1 Normative references

References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific. For a specific reference, subsequent revisions do not apply. For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document.

The following referenced documents are necessary for the application of the present document.

- [1] 3GPP TR 21.905: “Vocabulary for 3GPP Specifications”, Release 17, v17.0.0, 2020-07
- [2] O-RAN-WG10.OAM-Architecture-v09.00, O-RAN Alliance, Working Group 10
- [3] O-RAN.WG10.O1-Interface.0-v10.00, “O-RAN Operations and Maintenance Interface Specification”, O-RAN Alliance, Working Group 10
- [4] 3GPP TS 28.541, “Management and orchestration; 5G Network Resource Model (NRM); Stage 2 and stage 3”, Release 17, v17.2.1, 2021-04
- [5] 3GPP TS 32.342, “File Transfer (FT); Integration Reference Point (IRP); Information Service (IS)”, Release 16, v16.0.0, 2020-07
- [6] 3GPP TS 32.302, “Configuration Management (CM); Notification Integration Reference Point (IRP) ; Information Service (IS)”, Release 16, v16.0.0, 2020-07
- [7] 3GPP TS 28.622, “Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)”, Release 16, v16.7.1, 2021-04
- [8] 3GPP TS 28.552: “Management and orchestration; 5G performance measurements”, Release 17, v17.4.0, 2021-09
- [9] 3GPP TS 32.156: “Telecommunication management; Fixed Mobile Convergence (FMC) Model repertoire”, 3GPP, Release 17, v17.1.0, 2022-06
- [10] 3GPP TS 32.160: “Management and orchestration; Management service template”, 3GPP, Release 17, v17.4.0, 2022-06
- [11] 3GPP TS 38.401: “NG-RAN; Architecture description”, 3GPP, Release 17, v17.1.1, 2022-07
- [12] 3GPP TS 38.331: “NR; Radio Resource Control (RRC); Protocol specification”, 3GPP, Release 17
- [13] 3GPP TS 38.463: “NG-RAN; E1 Application Protocol (E1AP)”, 3GPP, Release 17

2.2 Informative references

References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific. For a specific reference, subsequent revisions do not apply. For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

3 Definitions of terms symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the O1 interface specification [3] apply.

3.2 Symbols

For the purposes of the present document, the symbols given in 3GPP TR 21.905 [1] and the following apply. A symbol defined in the present document takes precedence over the definition of the same symbol, if any, in 3GPP TR 21.905 [1].

3.3 Abbreviations and acronyms

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

5QI	5G QoS Identifier
MnS	Management Service
NETCONF	Network Configuration Protocol
NMS	Network Management System
O-CU	O-RAN Central Unit
O-CU-CP	O-RAN Central Unit Control Plane
O-CU-UP	O-RAN Central Unit User Plane
O-DU	O-RAN Distributed Unit
O-RU	O-RAN Radio Unit
OAM	Operations And Maintenance
PNF	Physical Network Function
QoS	Quality of Service
RPC	Remote Procedure Call
sFTP	Secure File Transfer Protocol or SSH File Transfer Protocol
SMO	Service Management and Orchestration
SSH	Secure Shell
VES	Virtual-function Event Streaming
VLAN	Virtual LAN
YANG	Yet Another Next Generation

3.4 Conventions

This specification includes cross references to a set of associated YANG models. Text may reference particular YANG leafs, notifications and remote procedure calls (RPCs). In order to assist in readability, all cross references to YANG defined elements will keep the identical case format as defined in the corresponding YANG model, with the font-weight set to **bold**. This convention applies only to text and not to YANG elements embedded into figures.

If there is any conflict between the YANG models and the accompanying text description in this specification, the definition of the YANG models shall take precedence.

3.5 Revision and compatibility handling

The revision statement in the YANG models shall be used to describe future revisions to the models that are backwards compatible. Backwards incompatible changes will be addressed by incrementing the number used as part of the model name and namespace, effectively creating a new YANG model. The format of the namespace used in all O-RAN YANG models is “urn:o-ran:”<model-name>“:”<model-number>, where the initial <model-number> used in a newly defined YANG model is “1.0”. Where this document makes reference to models, irrespective of their backward compatibility, a generic <model-number> of “x.y” is used to enable reference to all versions of the namespace for a particular <model-name>.

The revision statement in all YANG models includes a reference statement used to cross-reference to the first version of this document where the corresponding description was introduced. For example, the reference in all revision statements for the initial O-RAN models should include cross-reference to “O-RAN.WG5.O-CU-O1.0-v01.00”.

The revision statement of the YANG models also includes a description which is used to track the versioning of the YANG model. All revision statement descriptions will begin with “version ”<a>“.”“.”<c>, where <a>, and <c> are used to reflect the version of the YANG model, where

- <a> corresponds to the first digit of the O-RAN WG5 management plane specification version where the corresponding description was first introduced, corresponding to <x> in clause 1
- is incremented when errors in the YANG model have been corrected
- <c> is incremented only in working versions of the YANG model indicating incremental changes during the editing process

4 High Level Description

4.1 Top level functional description and terminology

4.1.1 Architecture for O-RAN WG5 functional split

This O1 interface facilitates the initialization, configuration and management of the O-CU-CP and O-CU-UP. Please refer to O-RAN OAM Architecture [2] for more details on the architecture.

4.1.2 O1 interface for O-CU architecture model

O1 interface for O-CU-CP and O-CU-UP is used for supporting the management features including “start up” installation, PNF software management, provisioning management, performance management, fault management and file management towards the O-CU.

4.1.3 Transport Network

Based on the transport topology, various modes of network connectivity are possible between O-CU-CP, O-CU-UP and SMO, as defined in the O-RAN OAM Architecture document [2].

4.2 Interfaces

The interface between the management system and the O-CU is as defined in the O1 specification [3].

5 PNF Startup and Registration Management

O-CU (including O-CU-CP and O-CU-UP) may be deployed as PNF as described in O1 Interface Specification [3], clause 6.7 PNF Startup and Registration Management Services.

5.1 PNF Plug-n-Play

PNF Plug-n-Play for O-CU is aligned with O1 Interface Specification [3], clause 6.7.1 PNF Plug-n-Connect.

5.2 PNF Registration

PNF Registration for O-CU is aligned with O1 Interface Specification [3], clause 6.7.2 PNF Registration.

6 Heartbeat Management Services

6.1 Heartbeat Notification

Heartbeat Notification for both O-CU-CP and O-CU-UP are aligned with O1 Interface Specification [3], clause 6.6.1 Heartbeat Notification. Figure 6-1 shows the example of notifyHeartbeat notification by using ‘SDO O1 format’.

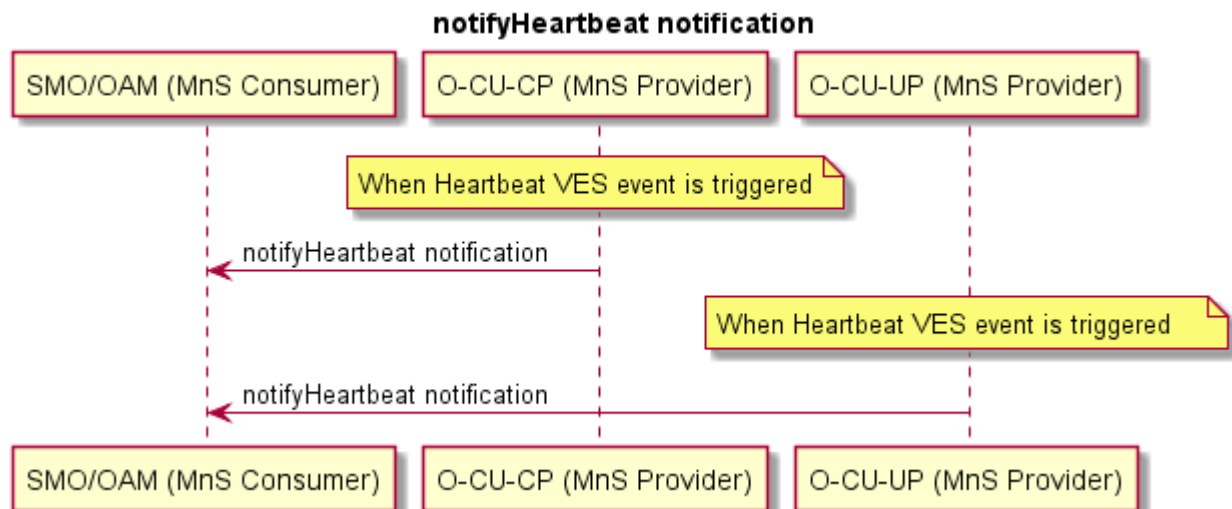


Figure 6-1: Heartbeat Notification

6.2 Heartbeat Control

Heartbeat Control for both O-CU-CP and O-CU-UP are aligned with O1 Interface Specification [3], clause 6.6.2 Heartbeat Control.

7 PNF Software Management Services

7.1 Software Package Naming and Content

Software Package Naming and Content for O-CU is aligned with O1 Interface Specification [3], clause 6.8.1 Software Package Naming and Content.

7.2 Software Inventory

Software Inventory for O-CU is aligned with O1 Interface Specification [3], clause 6.8.2 Software Inventory.

7.3 Software Download

Software Download for O-CU is aligned with O1 Interface Specification [3], clause 6.8.3 Software Download.

7.4 Software Activation Pre-Check

Software Activation Pre-Check for O-CU is aligned with O1 Interface Specification [3], clause 6.8.4 Software Activation Pre-Check.

7.5 Software Activate

Software Activate for O-CU is aligned with O1 Interface Specification [3], clause 6.8.5 Software Activate.

8 Performance Management

8.1 Performance Data File Reporting

Performance Data File Reporting for O-CU is aligned with O1 Interface Specification [3], clause 6.3.1 Performance Data File Reporting.

The Measurement Type and definitions for performance counters are described in both clause 6.3 of O1 interface specification [3] and Annex A of this specification. All measurements are optional and can be selected based on service deployment.

8.2 Performance Data Streaming

Performance Data Streaming for O-CU is aligned with O1 Interface Specification [3], clause 6.3.2 Performance Data Streaming.

8.3 Measurement Job Control

Measurement Job Control for O-CU is aligned with O1 Interface Specification [3], clause 6.3.3 Measurement Job Control.

9 Fault Management

Fault supervision management services allow a Fault Supervision MnS Provider to report errors and events to a Fault Supervision MnS Consumer and allow a Fault Supervision MnS Consumer to perform fault supervision operations on the Fault Supervision MnS Provider, such as get alarm list.

O-CU-UP and O-CU-CP Fault Supervision Management services are aligned with O1 Interface Specification [3], clause 6.2 Fault Supervision Management Services.

9.1 Fault Notification

O-CU-UP and O-CU-CP Fault notification events are aligned with O1 Interface Specification [3], clause 6.2.1 Fault Notification.

9.2 Fault Supervision Control

O-CU-UP and O-CU-CP Fault supervision control is aligned with O1 Interface Specification [3], clause 6.2.2 Fault Supervision Control.

10 File Management

This clause illustrate File Management between the O-CU-CP and SMO and between the O-CU-UP and SMO. The File Management Service is applied to manage file transfer for different types of data file such as Bulk CM file, Call and UE trace file, Performance Measurement result file, etc. The following file management use cases are supported for the O-CU-CP and O-CU-UP:

- List available files
- Transfer file
- Download file
- File ready notification

The detailed description of the use cases including file management operations and procedures as well as sequence diagram can be found in the O-RAN O1 interface specification [3], clause 6.5.

This clause shall describe the handling of files that are stored in the O-CU-CP which are generated by the O-CU-CP itself and O-CU-UP which are generated by the O-CU-UP itself.

The data files that are stored in the O-CU shall follow clause 10.1 of this specification.

10.1 File Structure

The file system structure for the O-CU-CP and O-CU-UP is defined as for file management procedures defined in the rest of this clause. If the O-CU-CP and O-CU-UP's physical file structure differs from the logical file structure defined below, the O-CU-CP and O-CU-UP shall perform the mapping related files into defined structure.

The O-CU-CP and O-CU-UP shall support the standardized folders for files generated by the O-CU-CP and O-CU-UP itself. The following standardized folders are defined:

O-CU-CP

- › O-RAN/O-CU-CP/PM/ for performance data files
- › O-RAN/O-CU-CP/CM/ for configuration files (except inventory)
- › O-RAN/O-CU-CP/NL/ for notification log file
- › O-RAN/O-CU-CP/CT/ for call trace files
- › O-RAN/O-CU-CP/OT/ for other files

O-CU-UP

- › O-RAN/O-CU-UP/PM/ for performance data files
- › O-RAN/O-CU-UP/CM/ for configuration files (except inventory)
- › O-RAN/O-CU-UP/NL/ for notification log file
- › O-RAN/O-CU-UP/CT/ for call trace files
- › O-RAN/O-CU-UP/OT/ for other files

The O-CU-CP and O-CU-UP may additionally support vendor defined folders which are out of scope of this specification.

10.2 Notification notifyFileReady

The file ready notification is applied to when the management data files have been prepared ready in O-CU-CP and O-CU-UP. The O-CU-CP and O-CU-UP will send notification using notifyFileReady VES event to subscribed SMO in order to notify the availability of the file(s) as defined in O1 Interface Specification [3].

The following input parameters are provided by the file ready notification (see also 3GPP TS 32.342 [5]):

objectClass	Notification header – see 3GPP TS 32.302[6]. It shall carry the FTIRP class name.
objectInstance	Notification header - see 3GPP TS 32.302[6]. It shall carry the DN of the FTIRP.
notificationId	Notification header - see 3GPP TS 32.302[6] .
eventTime	Notification header - see 3GPP TS 32.302[6] .
systemDN	Notification header - see 3GPP TS 32.302[6] .
notificationType	Notification header – see 3GPP TS 32.302[6] .
fileInfoList	It specifies the information of the available file.
additionalText	It carries vendor-specific semantics not defined in the present document.

Table 1: Parameters list for Notification notifyFileReady

When a file is available, the O-CU-CP and O-CU-UP sends a FileReady notification to the SMO using HTTP/TLS as defined in O1 Interface Specification [3]. The FileReady notification for O-CU-CP and O-CU-UP is performed independently.

```
@startuml
Title File Ready Notification
participant "SMO/OAM\n(File Management Service Consumer)" as L
participant "O-CU-CP\n(File Management Service Provider)" as M
participant "O-CU-UP\n(File Management Service Provider)" as N
note over ME : New file is available
```

```

1 L <- M: HTTP/TLS<notifyFileReady>
2 note over N: New file is available
3 L <- N: HTTP/TLS<notifyFileReady>
4 @enduml

```

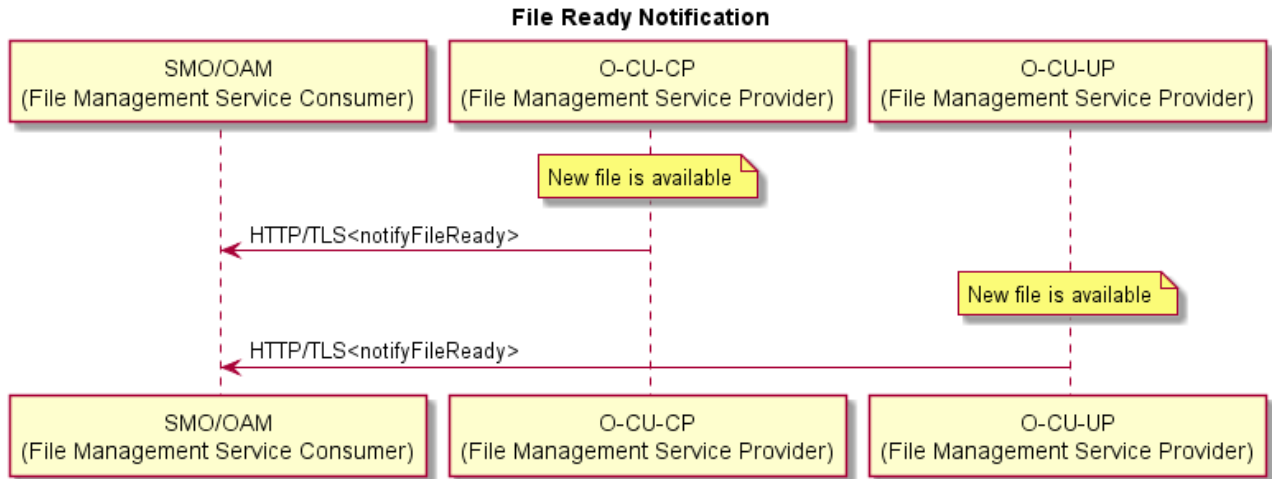


Figure 10-1: File Ready Notification

10.3 File Management Operation: listAvailableFiles

This listAvailableFiles operation for the SMO is applied to obtain the available file list with files location and type from the O-CU-CP and O-CU-UP by reading the AvailableFileList IOC as specified in O1 Interface Specification [3], clause 6.5.2. The listAvailableFiles operation for O-CU-CP and O-CU-UP can be performed independently.

10.4 File Management Operation: transferFile

The Transfer File Service provides the capability for the SMO to transfer files from or to the O-CU-CP and O-CU-UP. SMO/OAM may perform this action as a result of a FileReady notification from the O-CU-CP and O-CU-UP informing SMO that a file is now available to transfer. In any case, the file transfer is performed using a secure file transfer protocol (SFTP or FTPeS) from or to the location provided by O-CU-CP and O-CU-UP. The SMO triggers file transfer operation to O-CU-CP and O-CU-UP. Detailed procedure can be found in clause 6.5.3 of O1 Interface Specification [3]. The transferFile operation for O-CU-CP and O-CU-UP is performed independently.

10.5 File Management Operation: downloadFile

The Download File Service provides the capability for the SMO to request the O-CU-CP and O-CU-UP to download the file(s) when the SMO has a file that needs to be downloaded. In any case, the file download is performed using a secure file transfer protocol (SFTP or FTPeS) from the location provided by the SMO. The O-CU-CP and O-CU-UP replies to the SMO with the result of the download procedure. Detailed procedure can be found in clause 6.5.4 of O1 Interface Specification [3]. The downloadFile operation for O-CU-CP and O-CU-UP is performed independently.

11 Provisioning Management

This clause describes the Provisioning Management for O-CU specific aspects.

11.1 Configuration for O-CU

SMO configures the O-CU-CP and O-CU-UP.

Data Model configurations:

1. SA5 data model

The following 3GPP SA5 YANG modules are imported in the O-CU YANG modules published along with this specification. See TS 28.541 [4]

- _3gpp-common-top.yang
- _3gpp-common-yang-types.yang
- _3gpp-common-managed-element.yang
- _3gpp-nr-nrm-gnbcucpfunction.yang
- _3gpp-nr-nrm-nrcellcu.yang
- _3gpp-nr-nrm-nrfreqrelation.yang
- _3gpp-nr-nrm-gnbcuupfunction.yang
- _3gpp-nr-nrm-ep.yang

2. O-RAN augmentation of SA5 data models

O-CU-CP

- o-ran-cu-security-handling.yang

O-CU-UP

The O-CU YANG modules listed above, are published along with this specification document, and augment SA5 data models.

SupportQualifier(SQ) property for each attribute in the O-CU YANG modules, specifies whether it is mandatory for vendor to support the attribute in the implementation and in the data model. Meaning of this property, follows the definition in 3GPP TS 32.156 [9], clause 6. Values can be M, O, CM, CO.

If SQ=O, for an attribute, and if vendor does not support it, then this lack of support shall be indicated, as specified in 3GPP TS 32.160 [10] clause 6.2.26.

If SQ is not specified, then a default value of 'O' is to be assumed.

11.2 Notification aspect

11.2.1 Subscribing for O-CU notification from SMO

The O-CU would represent the parameters in its YANG model.

The SMO is responsible of subscribing to the parameters of interest from the O-CU data model using subscription mechanism described in O1 Interface specification [3] clause 6.1.7 Subscription Control.

11.3 Information model for O-CU-CP

11.3.1 Imported and associated information entities

11.3.1.1 Imported information entities and local labels

Label reference	Local label
3GPP TS 28.622 [7], IOC, Top	Top

11.3.1.2 Associated information entities and local labels

Label reference	Local label
3GPP TS 28.622 [7], IOC, ManagedElement	ManagedElement

11.3.2 Class diagram

11.3.2.1 Relationships

```

@startuml
skin rose
Class ManagedElement <<InformationObjectClass>> << (A, CadetBlue) >>
Class GNBCUCPFunction <<InformationObjectClass>> << (L, #FF7700) >>

class SecurityHandling <<InformationObjectClass>> << (L, #FF7700) >>
SecurityHandling : <color:Blue> cipheringAlgoPrio </color>
SecurityHandling : <color:Blue> integrityProtectAlgoPrio </color>

class CUCountGroup <<InformationObjectClass>> << (L, #FF7700) >>
CUCountGroup : <color:Blue> cu-count-group-index </color>
CUCountGroup : <color:Blue> drb-id </color>
CUCountGroup : <color:Blue> qci </color>
CUCountGroup : <color:Blue> srb-id </color>
CUCountGroup : <color:Blue> fiveQi </color>
CUCountGroup : <color:Blue> sd </color>
CUCountGroup : <color:Blue> sst </color>
CUCountGroup : <color:Blue> plmn-id </color>
CUCountGroup : <color:Blue> arp </color>

ManagedElement *-- "0..*" GNBCUCPFunction
GNBCUCPFunction *-- "0..1" SecurityHandling
GNBCUCPFunction *-- "0..30" CUCountGroup
@enduml

```

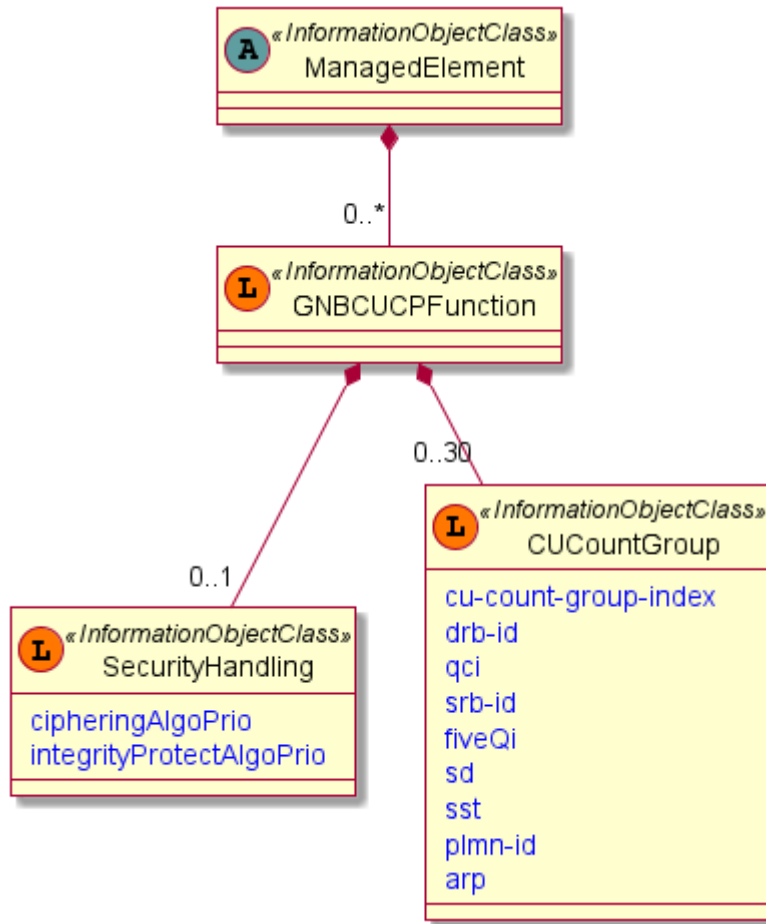


Figure 11-1: Containment for SecurityHandling and CUCCountGroup

```

@startuml
skin rose

class GNBCUCPFunction <<InformationObjectClass>>

class "PDCPConfig" <<InformationObjectClass>>
"PDCPConfig" : qci
"PDCPConfig" : fiveQI
"PDCPConfig" : discard-timer
"PDCPConfig" : pdcp-sn-size-ul
"PDCPConfig" : pdcp-sn-size-dl
"PDCPConfig" : t-reordering
"PDCPConfig" : t-reordering-ul
"PDCPConfig" : rlc-mode
"PDCPConfig" : header-compression-enabled
"PDCPConfig" : ul-data-split-threshold

GNBCUCPFunction *-- "0..32" "PDCPConfig" : <<names>>

@enduml

```

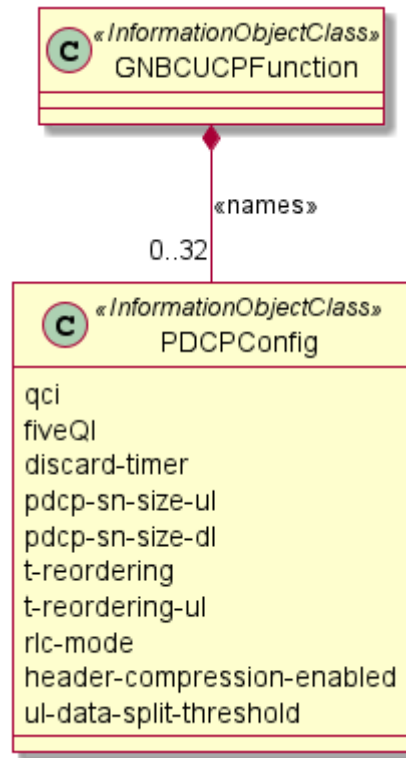


Figure 11-2: Containment for PDCPConfig

11.3.2.2 Inheritance

```

@startuml
skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members
abstract class Top <<InformationObjectClass>>
class CUCountGroup <<InformationObjectClass>>
Top <|-d- CUCountGroup
class SecurityHandling <<InformationObjectClass>>
Top <|-d- SecurityHandling
class "PDCPConfig" <<InformationObjectClass>>
Top <|-d- "PDCPConfig"
note right of Top
    defined in
        3GPP TS 28.622
        clause 4.3.29
end note
@enduml
    
```

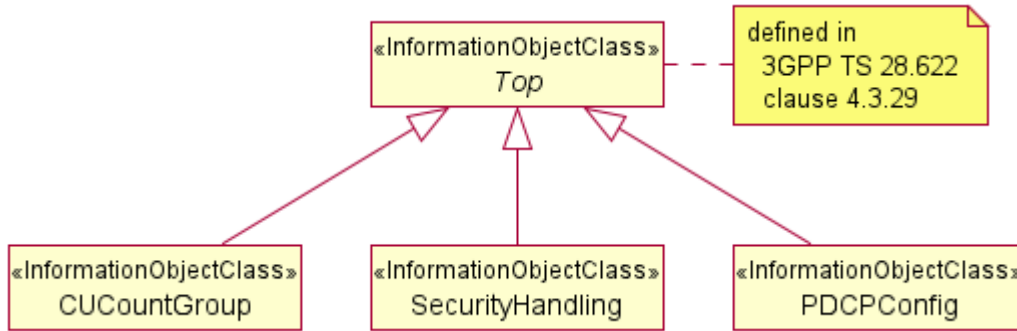


Figure 11-3: Inheritance for CUCountGroup, SecurityHandling, PDCPConfig

11.3.3 Class definition

11.3.3.1 SecurityHandling <<InformationObjectClass>>

11.3.3.1.1 Definition

This IOC represents the CU-CP security related handling function.

11.3.3.1.2 Attributes

Attribute name	Support Qualifier	isReadable	isWritable	isInvariant	isNotifiable
cipheringAlgoPrio	O	T	T	F	T
integrityProtectAlgoPrio	O	T	T	F	T

11.3.3.1.3 Attribute constraints

None

11.3.3.1.4 Notifications

There are no notifications defined.

11.3.3.1.5 State diagram

None

11.3.3.2 CUCountGroup <<InformationObjectClass>>

11.3.3.2.1 Definition

This IOC represents the cucountgroup function.

1 11.3.3.2.2 Attributes

Attribute name	Support Qualifier	isReadable	isWritable	isInvariant	isNotifiable
cu-count-group-index	O	T	T	F	T
drb-id	O	T	T	F	T
qci	O	T	T	F	T
srb-id	O	T	T	F	T
fiveQI	O	T	T	F	T
sd	O	T	T	F	T
sst	O	T	T	F	T
plmn-id	O	T	T	F	T
arp	O	T	T	F	T

2 11.3.3.2.3 Attribute constraints

3 None

4 11.3.3.2.4 Notifications

5 There are no notifications defined.

6 11.3.3.2.5 State diagram

7 None

8 11.3.3.3 PDCPConfig <<InformationObjectClass>>

9 11.3.3.3.1 Definition

10 This IOC represents PDCP configurations for O-CU-CP.

11 11.3.3.3.2 Attributes

Attribute name	Support Qualifier	isReadable	isWritable	isInvariant	isNotifiable
qci	O	T	T	F	T
fiveQI	O	T	T	F	T
discard-timer	O	T	T	F	T
pdcp-sn-size-ul	O	T	T	F	T
pdcp-sn-size-dl	O	T	T	F	T
t-reordering	O	T	T	F	T
t-reordering-ul	O	T	T	F	T
rlc-mode	O	T	T	F	T
header-compression-enabled	O	T	T	F	T
ul-data-split-threshold	O	T	T	F	T

12 11.3.3.3.3 Attribute constraints

13 None

14 11.3.3.3.4 Notifications

15 There are no notifications defined.

11.3.3.3.5 State diagram

None

11.3.4 Attribute definitions

11.3.4.1 Attribute properties

Attribute Name	Documentation and Allowed Values	Properties
<code>cipheringAlgoPrio</code>	The attribute provides allowed ciphering algorithms with priority order. Algorithm in index zero has highest priority. Each algorithm written only once in array. AllowedValues: NEA0, NEA1, NEA2, NEA3 Default value: NEA1,NEA2,NEA0	Type: ENUM CipheringAlgorithmID multiplicity: 1..4 isOrdered: True isUnique: True defaultValue: NEA1,NEA2,NEA0 isNullable: False
<code>integrityProtectAlgoPrio</code>	The attribute provides allowed integrity algorithms with priority order. Algorithm in index zero has highest priority. Each algorithm written only once in array. AllowedValues: NIA1, NIA2, NIA3 DefaultValue: NIA2,NIA1	Type: ENUM IntegrityAlgorithmID multiplicity: 1..3 isOrdered: True isUnique: True defaultValue: NIA2,NIA1 isNullable: False
<code>cu-count-group-index</code>	The attribute provides Cu Counter Group index. AllowedValues: 0..30	Type: uint8 multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: - isNullable: False
<code>drb-id</code>	The attribute provides DRB ID which is used for concerned CuCountGroup. Value 0 indicates that this DRB ID is not counted in this CuCountGroup AllowedValues: 0..32	Type: uint8 multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: - isNullable: False
<code>CuCountGroup.qci</code>	The attribute provides QoS Class Identifier defined in TS 23.401. Logical range and coding specified in TS 23.203. Value 0 indicates that this QCI shall NOT be counted in the CuCountGroup. AllowedValues: 0..255	Type: uint8 multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: - isNullable: False
<code>srb-id</code>	The attribute provides SRB ID used for concerned CuCountGroup. Value 0 indicates that this SRB ID is not counted in this CuCountGroup. AllowedValues; 0..3	Type: uint8 multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: - isNullable: False
<code>CuCountGroup.fiveqi</code>	The attribute provides 5QI identifier. Value 0 indicates that this 5QI shall NOT be counted in the CuCountGroup. AllowedValues; 0..255	Type: uint8 multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: - isNullable: False

sd	This parameter indicates the slice differentiator. Value 16777215 (0xFFFFF) means no SD value associated with the SST. AllowedValues; 0..16777215	Type: uint8 multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: - isNullable: False
sst	This parameter indicates the slice service type. AllowedValues; 0..255	Type: uint8 multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: - isNullable: False
plmn-id	3-octet string, representing PLMN ID.	Type: string multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: - isNullable: False
arp	This parameter indicates the allocation and retention priority. Value 0 indicates that this ARP is not counted in this CuCountGroup. AllowedValues; 0..15	Type: uint8 multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: - isNullable: False
PDCPConfig.qci	List of QCIs for which this PDCP configuration is applied. allowedValues: 0..255	Type: Integer multiplicity: 0..* isOrdered: False isUnique: True defaultValue: - isNullable: False
PDCPConfig.fiveQI	List of 5QIs for which this PDCP configuration is applied. allowedValues: 0..255	Type: Integer multiplicity: 0..* isOrdered: False isUnique: True defaultValue: - isNullable: False
discard-timer	Discard timer in milliseconds for this PDCP Configuration. See TS 38.331 [12]. If no value set, implies infinity. allowedValues: 10, 20, 30, 40, 50, 60, 75, 100, 150, 200, 250, 300, 500, 750, 1500	Type: Integer multiplicity: 0..1 isOrdered: N/A isUnique: N/A defaultValue: - isNullable: False
pdcp-sn-size-ul	PDCP SN size in Uplink for this PDCP Configuration. Not used when rlc-mode set to RLC-TM. See TS 38.331 [12]. allowedValues: LEN-12-BITS, LEN-18-BITS	Type: ENUM pdcpSNSize multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: - isNullable: False
pdcp-sn-size-dl	PDCP SN size in Downlink for this PDCP Configuration. Not used when rlc-mode set to RLC-TM. See TS 38.331 [12]. allowedValues: LEN-12-BITS, LEN-18-BITS	Type: ENUM pdcpSNSize multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: - isNullable: False
t-reordering	Reordering timer in milliseconds for this PDCP Configuration. This timer is used by the receiving side of PDCP entity to detect loss of RLC PDUs at lower layer. See TS 38.331 [12]. If no value set, implies infinity. allowedValues: 0, 1, 2, 4, 5, 8, 10, 15, 20, 30, 40, 50, 60, 80, 100, 120, 140, 160, 180, 200, 220, 240, 260, 280, 300, 500, 750, 1000, 1250, 1500, 1750, 2000, 2250, 2500, 2750, 3000	Type: Integer multiplicity: 0..1 isOrdered: N/A isUnique: N/A defaultValue: - isNullable: False

t-reordering-ul	Uplink Reordering Timer in milliseconds for this PDCP Configuration. Used by CU-UP for Uplink. See TS 38.463 [13]. If no value set, implies infinity. allowedValues: 0, 1, 2, 4, 5, 8, 10, 15, 20, 30, 40, 50, 60, 80, 100, 120, 140, 160, 180, 200, 220, 240, 260, 280, 300, 500, 750, 1000, 1250, 1500, 1750, 2000, 2250, 2500, 2750, 3000	Type: Integer multiplicity: 0..1 isOrdered: N/A isUnique: N/A defaultValue: - isNullable: False
rlc-mode	RLC mode for this PDCP Configuration. See TS 38.463 [13]. allowedValues: RLC-TM, RLC-AM, RLC-UM-BIDIRECTIONAL, RLC-UM-BIDIRECTIONAL-UL, RLC-UM-UNIDIRECTIONAL-DL	Type: ENUM rlcModeType multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: - isNullable: False
header-compression-enabled	Specifies whether ROHC should be enabled for this PDCP Configuration. See TS 38.331 [12].	Type: Boolean multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: - isNullable: False
ul-data-split-threshold	Uplink Data Split Threshold in bytes for this PDCP Configuration. See TS 38.331 [12]. If no value set, implies infinity. allowedValues: 0, 100, 200, 400, 800, 1600, 3200, 6400, 12800, 25600, 51200, 102400, 204800, 409600, 819200, 1228800, 1638400, 2457600, 3276800, 4096000, 4915200, 5734400, 6553600	Type: Integer multiplicity: 0..1 isOrdered: N/A isUnique: N/A defaultValue: - isNullable: False

12 Trace Management Services

Trace Management Service for both O-CU-CP and O-CU-UP are aligned with O1 Interface Specification [3], clause 6.4. The O-CU-CP and O-CU-UP can be Trance MnS Provider independently

12.1 Call Trace

Call Trace for both O-CU-CP and O-CU-UP are aligned with O1 Interface Specification [3], clause 6.4.1.

12.2 Minimization of Drive Testing (MDT)

Minimization of Drive Testing for both O-CU-CP and O-CU-UP are aligned with O1 Interface Specification [3], clause 6.4.2.

12.3 Radio Link Failure (RLF)

Radio Link Failure for both O-CU-CP and O-CU-UP are aligned with O1 Interface Specification [3], clause 6.4.3.

1

12.4 RRC Connection Establishment Failure (RCEF)

2

RRC Connection Establishment Failure for both O-CU-CP and O-CU-UP are aligned with O1 Interface Specification [3], clause 6.4.4.

3

4

5

12.5 Trace Control

6

Trace Control for both O-CU-CP and O-CU-UP are aligned with O1 Interface Specification [3], clause 6.4.5.

7

8

12.6 Streaming Trace

9

Streaming Trace for both O-CU-CP and O-CU-UP are aligned with O1 Interface Specification [3], clause 6.4.6.

Annex A (informative)

: Counter definition

A.1 NR PDCP performance counters (Type A):

A.1.1 Distribution of UE per UL received data volume

A.1.1.1 Performance Counter Table

Measurement Name	OR.PDCPA.DistUeUIRxData
Description	This counter provides the distribution of the UE per received uplink data volume. When more than one RLCs are configured, the data volume refers to the all volume regardless through which RLC the data is transferred. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of CuCountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	If O-CU connects 5GC, measurement subcounter is the number of the QoS flow of which x is mapped to bin of subcounter. <i>binX</i> . If O-CU connects EPC, measurement subcounter is the number of the bearer of which x is mapped to the bin of subcounter. <i>binX</i> . The number, x is acquired and calculated at the end of measurement period (100ms). x is incremented by the volume of the UL PDCP PDU volume whenever the UL PDCP PDU is received from lower layer when the UL PDCP PDU is mapped to group of subcounter. <i>QoS</i> , subcounter. <i>SNSSAI</i> , or subcounter. <i>CuCountgroup</i> .
Measurement Result	Integer (U16)
Measurement Type	OR.PDCPA.DistUeUIRxData. <i>binX</i> , or optionally OR.PDCPA.DistUeUIRxData. <i>binX.QoS</i> , where <i>QoS</i> identifies the target quality of service class, and OR.PDCPA.DistUeUIRxData. <i>binX.SNSSAI</i> , where <i>SNSSAI</i> identifies the S-NSSAI, and OR.PDCPA.DistUeUIRxData. <i>binX.CuCountgroup</i> , where <i>CuCountgroup</i> identifies an instance of CuCountGroup IOC, and where <i>binX</i> is the bin of the throughput, x: bin 1: 0 Byte < x < 1 kByte bin 2: 1 kByte ≤ x < 2 kByte bin 3: 2 kByte ≤ x < 5 kByte bin 4: 5 kByte ≤ x < 10 kByte bin 5: 10 kByte ≤ x < 20 kByte bin 6: 20 kByte ≤ x < 50 kByte bin 7: 50 kByte ≤ x < 100 kByte bin 8: 100 kByte ≤ x < 200 kByte bin 9: 200 kByte ≤ x < 500 kByte bin 10: 500 kByte ≤ x < 1 MByte bin 11: 1 MByte ≤ x < 1.5 MByte bin 12: 1.5 MByte ≤ x < 2 MByte bin 13: 2 MByte ≤ x < 2.5 MByte bin 14: 2.5 MByte ≤ x < 3 MByte bin 15: 3 MByte ≤ x < 3.5 MByte bin 16: 3.5 MByte ≤ x < 4 MByte bin 17: 4 MByte ≤ x < 4.5 MByte bin 18: 4.5 MByte ≤ x < 5 MByte bin 19: 5 MByte ≤ x < 10 MByte bin 20: 10 MByte ≤ x

Measurement Object Class	GNBCUUPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.1.2 Distribution of DL UE throughput in O-CU

A.1.2.1 Performance Counter Table

Measurement Name	OR.PDCPA.DistDIUEThroughput
Description	This counter provides the distribution of the UE throughput in downlink. When more than one RLCs are configured, the data volume refers to the all volume regardless through which RLC the data is transferred. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of <i>Cucountgroup</i> IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	If O-CU connects 5GC, measurement subcounter is the number of the QoS flow of which throughput, x/y is mapped to bin of subcounter.<i>binX</i>. If O-CU connects EPC, measurement subcounter is the number of the bearer of which throughput, x/y is mapped to bin of subcounter.<i>binX</i>. The number, x/y is acquired and calculated at the end of measurement period. x is incremented by the volume of the DL PDCP PDU volume which is acknowledged by the DDDS whenever DDDS is received when PDCP PDU is mapped to group of subcounter.<i>QoS</i>, subcounter.<i>SNSSAI</i>, or subcounter.<i>Cucountgroup</i>. y is the smaller value in the followings: $y = \Sigma(y1 - y2)$: where $y1$ is the point in time after $y2$ when data up until the second last piece of data in the transmitted data burst which emptied the PDCP SDU available for transmission for the group of subcounter.<i>Cucountgroup</i> was successfully transmitted, as acknowledged by the UE, and where $y2$ is the point in time when the first transmission begins after a PDCP SDU becomes available for transmission, where previously no PDCP SDUs were available for transmission for the particular subcounter.<i>QoS</i>, subcounter.<i>SNSSAI</i>, or subcounter.<i>Cucountgroup</i>. - The measurement report period is 60000 ms.
Measurement Result	Integer (U16)
Measurement Type	OR.PDCPA.DistDIUEThroughput.<i>binX</i>, or optionally OR.PDCPA.DistDIUEThroughput.<i>binX.QoS</i>, where <i>QoS</i> identifies the target quality of service class, and OR.PDCPA.DistDIUEThroughput.<i>binX.SNSSAI</i>, where <i>SNSSAI</i> identifies the S-NSSAI, and OR.PDCPA.DistDIUEThroughput.<i>binX.Cucountgroup</i>, where <i>Cucountgroup</i> identifies an instance of CuCountGroup IOC, and where <i>binX</i> is the bin of the throughput, x/y: bin 1: $0 \text{ Mbps} \leq x/y < 1 \text{ Mbps}$ bin 2: $1 \text{ Mbps} \leq x/y < 2 \text{ Mbps}$ bin 3: $2 \text{ Mbps} \leq x/y < 5 \text{ Mbps}$ bin 4: $5 \text{ Mbps} \leq x/y < 10 \text{ Mbps}$ bin 5: $10 \text{ Mbps} \leq x/y < 20 \text{ Mbps}$ bin 6: $20 \text{ Mbps} \leq x/y < 50 \text{ Mbps}$

	bin 7: $50 \text{ Mbps} \leq x/y < 100 \text{ Mbps}$ bin 8: $100 \text{ Mbps} \leq x/y < 200 \text{ Mbps}$ bin 9: $200 \text{ Mbps} \leq x/y < 500 \text{ Mbps}$ bin 10: $500 \text{ Mbps} \leq x/y < 1 \text{ Gbps}$ bin 11: $1 \text{ Gbps} \leq x/y < 1.5 \text{ Gbps}$ bin 12: $1.5 \text{ Gbps} \leq x/y < 2 \text{ Gbps}$ bin 13: $2 \text{ Gbps} \leq x/y < 2.5 \text{ Gbps}$ bin 14: $2.5 \text{ Gbps} \leq x/y < 3 \text{ Gbps}$ bin 15: $3 \text{ Gbps} \leq x/y < 3.5 \text{ Gbps}$ bin 16: $3.5 \text{ Gbps} \leq x/y < 4 \text{ Gbps}$ bin 17: $4 \text{ Gbps} \leq x/y < 4.5 \text{ Gbps}$ bin 18: $4.5 \text{ Gbps} \leq x/y < 5 \text{ Gbps}$ bin 19: $5 \text{ Gbps} \leq x/y < 10 \text{ Gbps}$ bin 20: $10 \text{ Gbps} \leq x/y$
Measurement Object Class	GNBCUUPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2 A.1.3 Total discarded UL PDCP SDU volume

3 A.1.3.1 Performance Counter Table

Measurement Name	OR.PDCPA.TotalDiscardedUIPdcPsdVol
Description	This counter provides the UL PDCP SDU volume discarded at PDCP due to any cause. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of Cucountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	The measurement counter is incremented by the volume of the UL PDCP SDU whenever the UL PDCP SDU volume is discarded when the UL PDCP SDU is mapped to group of subcounter. <i>Cucountgroup</i> .
Measurement Result	kilobyte (U32)
Measurement Type	OR.PDCPA.TotalDiscardedUIPdcPsdVol, or optionally OR.PDCPA.TotalDiscardedUIPdcPsdVol.QoS, where QoS identifies the target quality of service class, and OR.PDCPA.TotalDiscardedUIPdcPsdVol.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.PDCPA.TotalDiscardedUIPdcPsdVol.Cucountgroup, where Cucountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	GNBCUUPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

4

5 A.1.4 UL PDCP SDU volume discarded due to bearer release

6 A.1.4.1 Performance Counter Table

Measurement Name	OR.PDCPA.UIPdcPsdVolDiscardBearerRel
Description	This counter provides the UL PDCP SDU volume discarded at PDCP due to bearer release. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of Cucountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)

Condition	The measurement counter is incremented by the volume of the UL PDCP SDU whenever the UL PDCP SDU volume is discarded due to bearer release when the UL PDCP SDU is mapped to group of subcounter.
Measurement Result	kilobyte (U32)
Measurement Type	OR.PDCPA.UlPdcPsdVolDiscardBearerRel, or optionally OR.PDCPA.UlPdcPsdVolDiscardBearerRel.QoS, where QoS identifies the target quality of service class, and OR.PDCPA.UlPdcPsdVolDiscardBearerRel.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.PDCPA.UlPdcPsdVolDiscardBearerRel.Cucountgroup, where Cucountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	GNBCUUPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.1.5 Total discarded DL PDCP SDU volume discarded

3

A.1.5.1 Performance Counter Table

Measurement Name	OR.PDCPA.TotalDiscardedDlPdcPsdVol
Description	This counter provides the DL PDCP SDU volume discarded at PDCP due to any causes. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of Cucountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	The measurement counter is incremented by the volume of the DL PDCP SDU whenever the DL PDCP SDU volume is discarded when the DL PDCP SDU is mapped to group of subcounter.
Measurement Result	kilobyte (U32)
Measurement Type	OR.PDCPA.TotalDiscardedDlPdcPsdVol, or optionally OR.PDCPA.TotalDiscardedDlPdcPsdVol.QoS, where QoS identifies the target quality of service class, and OR.PDCPA.TotalDiscardedDlPdcPsdVol.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.PDCPA.TotalDiscardedDlPdcPsdVol.Cucountgroup, where Cucountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	GNBCUUPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

4

5

A.1.6 DL PDCP SDU volume discarded due to bearer release

6

A.1.6.1 Performance Counter Table

Measurement Name	OR.PDCPA.DlPdcPsdVolDiscardLessBearerRel
Description	This counter provides the DL PDCP SDU volume discarded at PDCP due to bearer release. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of Cucountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)

Condition	The measurement counter is incremented by the volume of the DL PDCP SDU whenever the DL PDCP SDU volume is discarded due to bearer release when the DL PDCP SDU is mapped to group of subcounter.
Measurement Result	kilobyte (U32)
Measurement Type	OR.PDCPA.DlPdcPsdVolDiscardLessBearerRel, or optionally OR.PDCPA.DlPdcPsdVolDiscardLessBearerRel.QoS, where QoS identifies the target quality of service class, and OR.PDCPA.DlPdcPsdVolDiscardLessBearerRel.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.PDCPA.DlPdcPsdVolDiscardLessBearerRel.Cucountgroup, where Cucountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	GNBCUUPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2 A.1.7 DL PDCP SDU volume discarded due to out of memory

3 A.1.7.1 Performance Counter Table

Measurement Name	OR.PDCPA.DlPdcPsdVolDiscardOutOfMemory
Description	This counter provides the DL PDCP SDU volume discarded at PDCP due to out of memory. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of Cucountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	The measurement counter is incremented by the volume of the DL PDCP SDU whenever the DL PDCP SDU volume is discarded due to out of memory when the DL PDCP SDU is mapped to group of subcounter.
Measurement Result	kilobyte (U32)
Measurement Type	OR.PDCPA.DlPdcPsdVolDiscardOthercauses, or optionally OR.PDCPA.DlPdcPsdVolDiscardOthercauses.QoS, where QoS identifies the target quality of service class, and OR.PDCPA.DlPdcPsdVolDiscardOthercauses.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.PDCPA.DlPdcPsdVolDiscardOthercauses.Cucountgroup, where Cucountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	GNBCUUPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

4

5 A.1.8 PDCP transmission stop

6 A.1.8.1 Performance Counter Table

Measurement Name	OR.PDCPA.PdcPTxStop
Description	This counter the number of the detection of PDCP transmission stop. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of Cucountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)

Condition	The measurement counter is incremented by 1 whenever PDCP transmission stop is occurred when the PDCP SDU is mapped to group of subcounter. It is incremented by 1 at most in each CU count group for one radio bearer until the transmission stop is resolved.
Measurement Result	Integer (U16)
Measurement Type	OR.PdcpTxStop, or optionally OR.PdcpTxStop.QoS, where QoS identifies the target quality of service class, and OR.PdcpTxStop.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.PdcpTxStop.Cucountgroup, where Cucountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	GNBCUUPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1 A.1.9 UL F1-U packet loss rate

3 A.1.9.1 Performance Counter Table

Measurement Name	DRB.F1UpacketLossRateUl
Description	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.2 <u>O-RAN addition:</u> The split into subcounters per S-NSSAI is recommended when the Slicing feature is supported. The split into subcounters per QoS is recommended where QoS identifies the target quality of service class.
Collection Method	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.2
Condition	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.2 <u>O-RAN addition:</u> The numbers are accumulated in the reference measurement period T (1000ms).
Measurement Result	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.2
Measurement Type	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.2
Measurement Object Class	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.2
Switching Technology	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.2
Generation	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.2
Purpose	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.2

4 A.1.10 UL PDCP SDU Loss Rate

6 A.1.10.1 Performance Counter Table

Measurement Name	DRB.PacketLossRateUl
Description	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.1 <u>O-RAN addition:</u> The split into subcounters per S-NSSAI is recommended when the Slicing feature is supported. The split into subcounters per QoS is recommended where QoS identifies the target quality of service class
Collection Method	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.1

Condition	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.1 <u>O-RAN addition:</u> The numbers are accumulated in the reference measurement period T (1000ms).
Measurement Result	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.1
Measurement Type	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.1
Measurement Object Class	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.1
Switching Technology	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.1
Generation	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.1
Purpose	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.1

1

2 A.1.11 DL PDCP SDU Loss Rate

3 A.1.11.1 Performance Counter Table

Measurement Name	DRB.PacketLossRateDL
Description	Refer to 3GPP TS 28.552 [8] clause 5.1.3.2.1 <u>O-RAN addition:</u> The split into subcounters per S-NSSAI is recommended when the Slicing feature is supported. The split into subcounters per QoS is recommended where QoS identifies the target quality of service class
Collection Method	Refer to 3GPP TS 28.552 [8] clause 5.1.3.2.1
Condition	Refer to 3GPP TS 28.552 [8] clause 5.1.3.2.1 <u>O-RAN addition:</u> The numbers are accumulated in the reference measurement period T (1000ms).
Measurement Result	Refer to 3GPP TS 28.552 [8] clause 5.1.3.2.1
Measurement Type	Refer to 3GPP TS 28.552 [8] clause 5.1.3.2.1
Measurement Object Class	Refer to 3GPP TS 28.552 [8] clause 5.1.3.2.1
Switching Technology	Refer to 3GPP TS 28.552 [8] clause 5.1.3.2.1
Generation	Refer to 3GPP TS 28.552 [8] clause 5.1.3.2.1
Purpose	Refer to 3GPP TS 28.552 [8] clause 5.1.3.2.1

4

5 A.1.12 DL F1-U packet loss rate

6 A.1.12.1 Performance Counter Table

Measurement Name	DRB.F1UpacketLossRateDL
Description	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.3 <u>O-RAN addition:</u> The split into subcounters per S-NSSAI is recommended when the Slicing feature is supported. The split into subcounters per QoS is recommended where QoS identifies the target quality of service class.
Collection Method	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.3

Condition	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.3 <u>O-RAN addition:</u> The numbers are accumulated in the reference measurement period T (1000ms).
Measurement Result	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.3
Measurement Type	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.3
Measurement Object Class	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.3
Switching Technology	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.3
Generation	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.3
Purpose	Refer to 3GPP TS 28.552 [8] clause 5.1.3.1.3

1

2

A.2 NR PDCP performance counters (Type B):

3

A.2.1 UL PDCP SDU Data Volume

4

A.2.1.1 Performance Counter Table

Measurement Name	OR.PDCPB.UlPdcPsdDataVol
Description	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.2 <u>O-RAN addition:</u> The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of CuCountgroup IOC may be used to define subcounter configurations.
Collection Method	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.2
Condition	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.2 <u>O-RAN addition:</u> The subcounter per QoS level should be regarded as subcounter QoS, SNSSAI or CuCountgroup. Note: excludes UL PDCP SDU received as data forwarding.
Measurement Result	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.2
Measurement Type	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.2 <u>O-RAN addition:</u> OR.PDCPB.UlPdcPsdDataVol, or optionally OR.PDCPB.UlPdcPsdDataVol.QoS, where QoS identifies the target quality of service class, and OR.PDCPB.UlPdcPsdDataVol.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.PDCPB.UlPdcPsdDataVol.CuCountgroup, where CuCountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.2
Switching Technology	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.2
Generation	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.2
Purpose	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.2

5

6

A.2.2 UL PDCP SDU Data Volume on X2 Interface

7

A.2.2.1 Performance Counter Table

Measurement Name	OR.PDCPB.UlPdcPsdDataVolX2
------------------	----------------------------

Description	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.4 <u>O-RAN addition:</u> The measurement is for X2 interface. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of Cucountgroup IOC may be used to define subcounter configurations.
Collection Method	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.4
Condition	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.4 <u>O-RAN addition:</u> The measurement is for X2 interface. The subcounter per QoS level should be regarded as subcounter QoS, SNSSAI or Cucountgroup. Note: excludes UL PDCP SDU received as data forwarding.
Measurement Result	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.4
Measurement Type	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.4 <u>O-RAN addition:</u> OR.PDCPB.UlPdcPsdDataVolX2, or optionally OR.PDCPB.UlPdcPsdDataVolX2.QoS, where QoS identifies the target quality of service class, and OR.PDCPB.UlPdcPsdDataVolX2.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.PDCPB.UlPdcPsdDataVolX2.Cucountgroup, where Cucountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.4
Switching Technology	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.4
Generation	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.4
Purpose	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.4

1

2

A.2.3 UL PDCP SDU Data Volume on Xn Interface

3

A.2.3.1 Performance Counter Table

Measurement Name	OR.PDCPB.UlPdcPsdDataVolXn
Description	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.4 <u>O-RAN addition:</u> The measurement is for Xn interface. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of Cucountgroup IOC may be used to define subcounter configurations.
Collection Method	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.4
Condition	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.4 <u>O-RAN addition:</u> The measurement is for Xn interface. The subcounter per QoS level should be regarded as subcounter QoS, SNSSAI or Cucountgroup. Note: excludes UL PDCP SDU received as data forwarding.
Measurement Result	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.4
Measurement Type	OR.PDCPB.UlPdcPsdDataVolXn, or optionally OR.PDCPB.UlPdcPsdDataVolXn.QoS, where QoS identifies the target quality of service class, and OR.PDCPB.UlPdcPsdDataVolXn.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.PDCPB.UlPdcPsdDataVolXn.Cucountgroup, where Cucountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.4

Switching Technology	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.4
Generation	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.4
Purpose	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.4

A.2.4 DL PDCP SDU Data Volume

A.2.4.1 Performance Counter Table

Measurement Name	OR.PDCPB.DlPdcPsdDataVol
Description	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.1 <u>O-RAN addition:</u> The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of CuCountgroup IOC may be used to define subcounter configurations.
Collection Method	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.1
Condition	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.1 <u>O-RAN addition:</u> The subcounter per QoS level should be regarded as subcounter QoS, SNSSAI or CuCountgroup. Note: excludes DL PDCP SDU transmitted as data forwarding.
Measurement Result	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.1
Measurement Type	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.1 <u>O-RAN addition:</u> OR.PDCPB.DlPdcPsdDataVol, or optionally OR.PDCPB.DlPdcPsdDataVol.QoS, where QoS identifies the target quality of service class, and OR.PDCPB.DlPdcPsdDataVol.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.PDCPB.DlPdcPsdDataVol.CuCountgroup, where CuCountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.1
Switching Technology	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.1
Generation	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.1
Purpose	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.1

A.2.5 DL PDCP SDU Data Volume on X2 Interface

A.2.5.1 Performance Counter Table

Measurement Name	OR.PDCPB.DlPdcPsdDataVolX2
Description	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.3 <u>O-RAN addition:</u> The measurement is for X2 interface. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in NR option 3) or per S-NSSAI. An instance of CuCountgroup IOC may be used to define subcounter configurations.
Collection Method	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.3
Condition	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.3 <u>O-RAN addition:</u> The measurement is for X2 interface. The subcounter per QoS level should be regarded as subcounter QoS, SNSSAI or CuCountgroup.

	Note: excludes DL PDCP SDU transmitted as data forwarding.
Measurement Result	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.3
Measurement Type	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.3 <u>O-RAN addition:</u> OR.PDCPB.DlPdcpSduDataVolX2, or optionally OR.PDCPB.DlPdcpSduDataVolX2.QoS, where QoS identifies the target quality of service class, and OR.PDCPB.DlPdcpSduDataVolX2.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.PDCPB.DlPdcpSduDataVolX2.Cucountgroup, where Cucountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.3
Switching Technology	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.3
Generation	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.3
Purpose	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.3

1

2

A.2.6 DL PDCP SDU Data Volume on Xn Interface

3

A.2.6.1 Performance Counter Table

Measurement Name	OR.PDCPB.DlPdcpSduDataVolXnCell
Description	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.3 <u>O-RAN addition:</u> The measurement is for Xn interface. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in NR option 3) or per S-NSSAI. An instance of Cucountgroup IOC may be used to define subcounter configurations.
Collection Method	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.3
Condition	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.3 <u>O-RAN addition:</u> The measurement is for Xn interface. The subcounter per QoS level should be regarded as subcounter QoS, SNSSAI or Cucountgroup. Note: excludes DL PDCP SDU transmitted as data forwarding.
Measurement Result	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.3
Measurement Type	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.3 <u>O-RAN addition:</u> OR.PDCPB.DlPdcpSduDataVolXn, or optionally OR.PDCPB.DlPdcpSduDataVolXn.QoS, where QoS identifies the target quality of service class, and OR.PDCPB.DlPdcpSduDataVolXn.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.PDCPB.DlPdcpSduDataVolXn.Cucountgroup, where Cucountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.3
Switching Technology	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.3
Generation	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.3
Purpose	Refer to 3GPP TS 28.552 [8] clause 5.1.3.6.2.3

4

1 A.2.7 UL PDCP SDU Data Volume per cell

2 A.2.7.1 Performance Counter Table

Measurement Name	OR.PDCPB.UlPdcPsdDataVolCell
Description	This counter provides the UL PDCP SDU volume received via X2, Xn or F1. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in NR option 3) or per S-NSSAI. An instance of CuCountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the volume of UL PDCP SDU whenever the UL PDCP PDU is received via X2-U, Xn-U or F1-U UL GTP-u tunnel when the UL PDCP SDU is mapped to group of subcounter. Note: excludes UL PDCP SDU received as data forwarding.
Measurement Result	Mbit (U32)
Measurement Type	OR.PDCPB.UlPdcPsdDataVolCell, or optionally OR.PDCPB.UlPdcPsdDataVolCell.QoS, where QoS identifies the target quality of service class, and OR.PDCPB.UlPdcPsdDataVolCell.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.PDCPB.UlPdcPsdDataVolCell.CuCountgroup, where CuCountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

3

4 A.2.8 UL PDCP SDU Data Volume on X2 Interface per cell

5 A.2.8.1 Performance Counter Table

Measurement Name	OR.PDCPB.UlPdcPsdDataVolX2Cell
Description	This counter provides the UL PDCP SDU volume received via X2. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in NR option 3) or per S-NSSAI. An instance of CuCountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the volume of UL PDCP SDU whenever the UL PDCP PDU is received via X2-U UL GTP-u tunnel when the UL PDCP SDU is mapped to group of subcounter. Note: excludes UL PDCP SDU received as data forwarding.
Measurement Result	Mbit (U32)
Measurement Type	OR.PDCPB.UlPdcPsdDataVolX2Cell, or optionally OR.PDCPB.UlPdcPsdDataVolX2Cell.QoS, where QoS identifies the target quality of service class, and OR.PDCPB.UlPdcPsdDataVolX2Cell.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.PDCPB.UlPdcPsdDataVolX2Cell.CuCountgroup, where CuCountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

6

A.2.9 UL PDCP SDU Data Volume on Xn Interface per cell

A.2.9.1 Performance Counter Table

Measurement Name	OR.PDCPB.UlPdcPsdDataVolXnCell
Description	This counter provides the UL PDCP SDU volume received via Xn. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in NR option 3) or per S-NSSAI. An instance of CuCountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the volume of UL PDCP SDU whenever the UL PDCP PDU is received via Xn-U UL GTP-u tunnel when the UL PDCP SDU is mapped to group of subcounter. Note: excludes UL PDCP SDU received as data forwarding.
Measurement Result	Mbit (U32)
Measurement Type	OR.PDCPB.UlPdcPsdDataVolXnCell, or optionally OR.PDCPB.UlPdcPsdDataVolXnCell.QoS, where QoS identifies the target quality of service class, and OR.PDCPB.UlPdcPsdDataVolXnCell.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.PDCPB.UlPdcPsdDataVolXnCell.CuCountgroup, where CuCountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.2.10 DL PDCP SDU Data Volume per cell

A.2.10.1 Performance Counter Table

Measurement Name	OR.PDCPB.DlPdcPsdDataVolCell
Description	This counter provides the DL PDCP SDU volume transmitted via X2, Xn or F1. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in NR option 3) or per S-NSSAI. An instance of CuCountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the volume of DL PDCP SDU whenever the DL PDCP PDU is transmitted via X2-U, Xn-U or F1-U DL GTP-u tunnel when the DL PDCP SDU is mapped to group of subcounter. Note: excludes DL PDCP SDU transmitted as data forwarding.
Measurement Result	Mbit (U32)
Measurement Type	OR.PDCPB.DlPdcPsdDataVolCell, or optionally OR.PDCPB.DlPdcPsdDataVolCell.QoS, where QoS identifies the target quality of service class, and OR.PDCPB.DlPdcPsdDataVolCell.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.PDCPB.DlPdcPsdDataVolCell.CuCountgroup, where CuCountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1 A.2.11 DL PDCP SDU Data Volume on X2 Interface per cell

2 A.2.11.1 Performance Counter Table

Measurement Name	OR.PDCPB.DlPdcPsdDataVolX2Cell
Description	This counter provides the DL PDCP SDU volume transmitted via X2. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in NR option 3) or per S-NSSAI. An instance of CuCountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the volume of DL PDCP SDU whenever the DL PDCP PDU is transmitted via X2-U DL GTP-u tunnel when the DL PDCP SDU is mapped to group of subcounter. Note: excludes DL PDCP SDU transmitted as data forwarding.
Measurement Result	Mbit (U32)
Measurement Type	OR.PDCPB.DlPdcPsdDataVolX2Cell, or optionally OR.PDCPB.DlPdcPsdDataVolX2Cell.QoS, where QoS identifies the target quality of service class, and OR.PDCPB.DlPdcPsdDataVolX2Cell.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.PDCPB.DlPdcPsdDataVolX2Cell.CuCountgroup, where CuCountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

3

4 A.2.12 DL PDCP SDU Data Volume on Xn Interface per cell

5 A.2.12.1 Performance Counter Table

Measurement Name	OR.PDCPB.DlPdcPsdDataVolXnCell
Description	This counter provides the DL PDCP SDU volume transmitted via Xn. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in NR option 3) or per S-NSSAI. An instance of CuCountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the volume of DL PDCP SDU whenever the DL PDCP PDU is transmitted via Xn-U DL GTP-u tunnel when the DL PDCP SDU is mapped to group of subcounter. Note: excludes DL PDCP SDU transmitted as data forwarding.
Measurement Result	Mbit (U32)
Measurement Type	OR.PDCPB.DlPdcPsdDataVolXnCell, or optionally OR.PDCPB.DlPdcPsdDataVolXnCell.QoS, where QoS identifies the target quality of service class, and OR.PDCPB.DlPdcPsdDataVolXnCell.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.PDCPB.DlPdcPsdDataVolXnCell.CuCountgroup, where CuCountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

6

A.3 NR S1-U Interface Performance Counters

A.3.1 UL PDCP SDU volume transmitted via S1-U UL GTP-U tunnel

A.3.1.1 Performance Counter Table

Measurement Name	OR.S1.UIPdcPsdUVolTxS1UUl
Description	This counter provides the UL PDCP SDU volume transmitted via S1-U UL GTP-U tunnel The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of Cucountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the volume of UL PDCP SDU whenever the UL PDCP SDU is transmitted via S1-U UL GTP-u tunnel when the UL PDCP SDU is group of subcounter.
Measurement Result	kilobyte (U32)
Measurement Type	OR.S1.UIPdcPsdUVolTxS1UUl, or optionally OR.S1.UIPdcPsdUVolTxS1UUl.QoS, where QoS identifies the target quality of service class, and OR.S1.UIPdcPsdUVolTxS1UUl.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.S1.UIPdcPsdUVolTxS1UUl.Cucountgroup, where Cucountgroup identifies ann instance of CuCountGroup IOC.
Measurement Object Class	GNBCUUPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.3.2 DL PDCP SDU volume received via S1-U DL GTP-U tunnel

A.3.2.1 Performance Counter Table

Measurement Name	OR.S1.DlPdcPsdUVolRxS1UDl
Description	This counter provides the DL PDCP SDU volume received via S1-U DL GTP-U tunnel The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of Cucountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the volume of DL PDCP SDU whenever the DL PDCP SDU is received via S1-U DL GTP-u tunnel when the DL PDCP SDU is mapped to group of subcounter.
Measurement Result	kilobyte (U32)
Measurement Type	OR.S1.DlPdcPsdUVolRxS1UDl, or optionally OR.S1.DlPdcPsdUVolRxS1UDl.QoS, where QoS identifies the target quality of service class, and OR.S1.DlPdcPsdUVolRxS1UDl.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.S1.DlPdcPsdUVolRxS1UDl.Cucountgroup, where Cucountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	GNBCUUPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1 A.3.3 UL PDCP SDUs transmitted via S1-U UL GTP-U tunnel

2 A.3.3.1 Performance Counter Table

Measurement Name	OR.S1.UIPdcPduTxS1UUl
Description	This counter provides the number of the UL PDCP SDUs transmitted via S1-U UL GTP-U tunnel. The counter is split into subcounters per GTP Path. The counter is split into subcounter of subcounter per CuCountGroup.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever the UL PDCP SDU is transmitted via S1 U UL GTP-u tunnel. GTP Path subcounter is maintained, the number of measurements is accumulated per the number or supported GTP Path. CuCountGroup subcounter of subcounter is maintained, the number of measurements is accumulated per the number of supported CuCountGroup.
Measurement Result	Integer number (U32)
Measurement Type	OR.S1.UIPdcPduTxS1UUl.GTPPath.Cugroup where <i>GTPPath</i> is GTP Path. <i>Cugroup</i> identifies an instance of CuCountGroup IOC.
Measurement Object Class	GNBCUUPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

4 A.3.4 DL PDCP SDUs transmitted via S1-U DL GTP-U tunnel

5 A.3.4.1 Performance Counter Table

Measurement Name	OR.S1.DlPdcPduTxS1UDl
Description	This counter provides the number of the DL PDCP SDUs transmitted via S1-U DL GTP-U tunnel. The counter is split into subcounters per GTP Path. The counter is split into subcounter of subcounter per CuCountGroup.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever the DL PDCP SDU is transmitted via S1 U DL GTP-u tunnel. GTP Path subcounter is maintained, the number of measurements is accumulated per the number or supported GTP Path. CuCountGroup subcounter of subcounter is maintained, the number of measurements is accumulated per the number of supported CuCountGroup.
Measurement Result	Integer number (U32)
Measurement Type	OR.S1.DlPdcPduTxS1UDl.GTPPath.Cugroup where <i>GTPPath</i> is GTP Path. <i>Cugroup</i> identifies an instance of CuCountGroup IOC.
Measurement Object Class	GNBCUUPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.4 NR NG-U Interface Performance Counters

A.4.1 UL PDCP SDU volume transmitted via NG-U UL GTP-U tunnel

A.4.1.1 Performance Counter Table

Measurement Name	OR.NG.UlPdcPsdUVolTxNGUUI
Description	This counter provides the UL PDCP SDU volume transmitted via NG-U UL GTP-U tunnel. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of CuCountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the volume of UL PDCP SDU whenever the UL PDCP SDU is transmitted via NG-U UL GTP-u tunnel when the UL PDCP SDU is mapped to group of subcounter.
Measurement Result	kilobyte (U32)
Measurement Type	OR.NG.UlPdcPsdUVolTxNGUUI, or optionally OR.NG.UlPdcPsdUVolTxNGUUI.QoS, where QoS identifies the target quality of service class, and OR.NG.UlPdcPsdUVolTxNGUUI.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.NG.UlPdcPsdUVolTxNGUUI.CuCountgroup, where CuCountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	GNBCUUPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.4.2 DL PDCP SDU volume received via NG-U DL GTP-U tunnel

A.4.2.1 Performance Counter Table

Measurement Name	OR.NG.DlPdcPsdUVolRxNGUDl
Description	This counter provides the DL PDCP SDU volume received via NG-U DL GTP-U tunnel. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of CuCountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the volume of DL PDCP SDU whenever the DL PDCP SDU is received via NG-U DL GTP-u tunnel when the DL PDCP SDU is mapped to group of subcounter.
Measurement Result	kilobyte (U32)
Measurement Type	OR.NG.DlPdcPsdUVolRxNGUDl, or optionally OR.NG.DlPdcPsdUVolRxNGUDl.QoS, where QoS identifies the target quality of service class, and OR.NG.DlPdcPsdUVolRxNGUDl.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.NG.DlPdcPsdUVolRxNGUDl.CuCountgroup, where CuCountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	GNBCUUPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.5 NR X2-U Interface performance counters (O-CU)

A.5.1 UL PDCP SDU volume received via X2-U UL GTP-U tunnel (X2-U UL data forwarding)

A.5.1.1 Performance Counter Table

Measurement Name	OR.X2.UlPdcPsdUVolRxX2UulDataForward
Description	This counter provides the UL PDCP SDU volume received via X2-U UL GTP-U tunnel for data forwarding. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of CuCountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the volume of UL PDCP SDU whenever the UL PDCP PDU is received via X2-U UL GTP-u tunnel for data forwarding when the UL PDCP SDU is mapped to group of subcounter.
Measurement Result	kilobyte (U32)
Measurement Type	OR.X2.UlPdcPsdUVolRxX2UulDataForward, or optionally OR.X2.UlPdcPsdUVolRxX2UulDataForward.QoS, where QoS identifies the target quality of service class, and OR.X2.UlPdcPsdUVolRxX2UulDataForward.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.X2.UlPdcPsdUVolRxX2UulDataForward.CuCountgroup, where CuCountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	GNBCUUPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.5.2 DL PDCP SDU volume received via X2-U DL GTP-U tunnel (X2-U DL data forwarding)

A.5.2.1 Performance Counter Table

Measurement Name	OR.X2.DlPdcPsdUVolTxX2UdlDataForward
Description	This counter provides the DL PDCP SDU volume received via X2-U DL GTP-U tunnel for data forwarding. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of CuCountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the volume of DL PDCP SDU whenever the DL PDCP PDU is transmitted via X2-U DL GTP-u tunnel for data forwarding when the DL PDCP SDU is mapped to group of subcounter.
Measurement Result	kilobyte (U32)
Measurement Type	OR.X2.DlPdcPsdUVolRxX2UdlDataForward, or optionally OR.X2.DlPdcPsdUVolRxX2UdlDataForward.QoS, where QoS identifies the target quality of service class, and OR.X2.DlPdcPsdUVolRxX2UdlDataForward.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.X2.DlPdcPsdUVolRxX2UdlDataForward.CuCountgroup, where CuCountgroup identifies an instance of CuCountGroup IOC.

Measurement Object Class	GNBCUUPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.6 NR Xn-U Interface performance counters (O-CU)

A.6.1 UL PDCP SDU volume received via Xn-U UL GTP-U tunnel (Xn-U UL data forwarding)

A.6.1.1 Performance Counter Table

Measurement Name	OR.Xn.UIPdcPsdUVolRxXnUUIDataForward
Description	This counter provides the UL PDCP SDU volume received via Xn-U UL GTP-U tunnel for data forwarding. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of CuCountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the volume of UL PDCP SDU whenever the UL PDCP PDU is received via Xn-U UL GTP-u tunnel for data forwarding when the UL PDCP SDU is mapped to group of subcounter.
Measurement Result	kilobyte (U32)
Measurement Type	OR.Xn.UIPdcPsdUVolRxXnUUIDataForward, or optionally OR.Xn.UIPdcPsdUVolRxXnUUIDataForward.QoS, where QoS identifies the target quality of service class, and OR.Xn.UIPdcPsdUVolRxXnUUIDataForward.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.Xn.UIPdcPsdUVolRxXnUUIDataForward.CuCountgroup, where CuCountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	GNBCUUPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.6.2 DL PDCP SDU volume received via Xn-U DL GTP-U tunnel (Xn-U DL data forwarding)

A.6.2.1 Performance Counter Table

Measurement Name	OR.Xn.DIPdcPsdUVolTxXnUDIDataForward
Description	This counter provides the DL PDCP SDU volume received via Xn-U DL GTP-U tunnel for data forwarding. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC) or per S-NSSAI. An instance of CuCountgroup IOC may be used to define subcounter configurations.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the volume of DL PDCP SDU whenever the DL PDCP PDU is transmitted via Xn-U DL GTP-u tunnel for data forwarding when the DL PDCP SDU is mapped to group of subcounter.
Measurement Result	kilobyte (U32)

Measurement Type	OR.Xn.OR.Xn.DlPdcPduVolTxXnUDlDataForward, or optionally OR.Xn.OR.Xn.DlPdcPduVolTxXnUDlDataForward.QoS, where QoS identifies the target quality of service class, and OR.Xn.OR.Xn.DlPdcPduVolTxXnUDlDataForward.SNSSAI, where SNSSAI identifies the S-NSSAI, and OR.Xn.OR.Xn.DlPdcPduVolTxXnUDlDataForward.Cucountgroup, where Cucountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	GNBCUUPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.7 NR X2-C Interface performance counters (O-CU)

3

A.7.1 Transmitted X2-C messages

4

A.7.1.1 Performance Counter Table

Measurement Name	OR.X2.TxX2CMesg
Description	This counter provides the number of the transmitted X2-C messages per signal type that is non UE-associated or UE-associated signaling TS 38.401 [11].
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever X2 C-plane message is transmitted when the signal type of the X2 C-plane message is group of <i>Signaltype</i> .
Measurement Result	Integer (32)
Measurement Type	OR.X2.TxX2CMesg.Signaltype where Signaltype is signal type: 0: non UE-associated 1: UE associated
Measurement Object Class	GNBCUCPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

5

6

A.7.2 Received X2-C messages

7

A.7.2.1 Performance Counter Table

Measurement Name	OR.X2.RxX2CMesg
Description	This counter provides the number of the received X2-C messages per signal type that is non UE-associated or UE-associated signaling TS 38.401 [11].
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever X2 C-plane message is received when the signal type of the X2 C-plane message is group of <i>Signaltype</i> .
Measurement Result	Integer (32)
Measurement Type	OR.X2.RxX2CMesg.Signaltype where Signaltype is signal type: 0: non UE-associated 1: UE associated
Measurement Object Class	GNBCUCPFunction
Switching Technology	Packet Switched

Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.8 NR Xn-C Interface performance counters (O-CU)

A.8.1 Transmitted Xn-C messages

A.8.1.1 Performance Counter Table

Measurement Name	OR.Xn.TxXnCMesg
Description	This counter provides the number of the transmitted Xn-C messages per signal type that is non UE-associated or UE-associated signaling TS 38.401 [11].
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever Xn C-plane message is transmitted when the signal type of Xn C-plane message is group of <i>Signaltype</i> .
Measurement Result	Integer (32)
Measurement Type	OR.Xn.TxXnCMesg. <i>Signaltype</i> where <i>Signaltype</i> is signal type: 0: non UE-associated 1: UE associated
Measurement Object Class	GNBCUCPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.8.2 Received Xn-C messages

A.8.2.1 Performance Counter Table

Measurement Name	OR.Xn.RxXnCMesg
Description	This counter provides the number of the received Xn-C messages per signal type that is non UE-associated or UE-associated signaling TS 38.401 [11].
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever Xn C-plane message is received when the signal type of the Xn C-plane message is group of <i>Signaltype</i> .
Measurement Result	Integer (32)
Measurement Type	OR.Xn.RxXnCMesg. <i>Signaltype</i> where <i>Signaltype</i> is signal type: 0: non UE-associated 1: UE associated
Measurement Object Class	GNBCUCPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.9 NR F1-C Interface performance counters (O-CU-CP)

A.9.1 Transmitted F1-C messages

A.9.1.1 Performance Counter Table

Measurement Name	OR.F1.TxF1CMesg
Description	This counter provides the number of the transmitted F1-C messages per signal type that is non UE-associated or UE-associated signaling TS 38.401 [11].
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever F1 C-plane message is transmitted when the signal type of the F1 C-plane message is group of <i>Signaltype</i> .
Measurement Result	Integer (32)
Measurement Type	OR.F1.TxF1CMesg. <i>Signaltype</i> where <i>Signaltype</i> is signal type: 0: non UE-associated 1: UE associated
Measurement Object Class	GNBCUCPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.9.2 Received F1-C messages

A.9.2.1 Performance Counter Table

Measurement Name	OR.F1.RxF1CMesg
Description	This counter provides the number of the received F1-C messages per signal type that is non UE-associated or UE-associated signaling TS 38.401 [11].
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever F1 C-plane message is received when the signal type of the F1 C-plane message is group of <i>Signaltype</i> .
Measurement Result	Integer (32)
Measurement Type	OR.F1.RxF1CMesg. <i>Signaltype</i> where <i>Signaltype</i> is signal type: 0: non UE-associated 1: UE associated
Measurement Object Class	GNBCUCPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.10 Number of UE Contexts for EN-DC

A.10.1 Max SN terminated split bearer UE Contexts

A.10.1.1 Performance Counter Table

Measurement Name	OR.UEENDC.MaxSnSplitBearer
Description	This counter provides the maximum number of UE Contexts that are configured SN terminated split bearer.

Collection Method	SI
Condition	Measurement subcounter is the maximum number of x. It is assigned with period of 100ms by the volume x, if x is bigger than previously counted volume. x is the number of the UE Context that are configured as SN terminated split bearer. x is acquired as an instantaneous value at every 100ms. - the triggers of addition - received X2-AP: SgNB Reconfiguration Complete (to SN terminated split bearer) - received X2-AP: SgNB Modification Confirm (Intra/Inter DU PSCell change using SRB1 (for target cell)) - received RRC: RRC Reconfiguration Complete (Intra/Inter DU PSCell change using SRB3 (for target cell)) - the triggers of subtraction - received X2-AP: SgNB Reconfiguration Complete (to SN terminated MCG bearer) - received X2-AP: SgNB Modification Confirm (Intra/Inter DU PSCell change using SRB1 (for source cell)) - received F1-AP: UE Context Release Complete (Intra/Inter DU PSCell change using SRB3 (for source cell)) - received X2-AP: UE Context Release
Measurement Result	Integer number (U32)
Measurement Type	OR.UEENDC.MaxSnSplitBearer
Measurement Object Class	NRCelICU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.10.2 Max SN terminated MCG bearer UE Contexts

3

A.10.2.1 Performance Counter Table

Measurement Name	OR.UEENDC.MaxSnMcgBearer
Description	This counter provides the maximum number of UE Contexts that are configured SN terminated MCG bearer.
Collection Method	SI
Condition	Measurement subcounter is the maximum number of x. It is assigned with period of 100ms by the volume x, if x is bigger than previously counted volume. x is the number of the UE Contexts that are configured as SN terminated MCG bearer. x is acquired as an instantaneous value at every 100ms. - the triggers of addition - received X2-AP: SgNB Reconfiguration Complete (to SN terminated MCG bearer) - received X2-AP: SgNB Reconfiguration Confirm (to SN terminated MCG bearer) - the triggers of subtraction - received X2-AP: SgNB Reconfiguration Complete (to SN terminated split bearer)

	- received X2-AP: UE Context Release
Measurement Result	Integer number (U32)
Measurement Type	OR.UEENDC.MaxSnMcgBearer
Measurement Object Class	NRCelICU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.10.3 Average SN terminated split bearer UE Contexts

3

A.10.3.1 Performance Counter Table

Measurement Name	OR.UEENDC.AveSnSplitBearer
Description	This counter provides the average number of UE Contexts that are configured SN terminated split bearer.
Collection Method	SI
Condition	Measurement subcounter is x/y. x is incremented by the number of UE Contexts that are configured SN terminated split bearer. This counter obtains the number of the UE Contexts every 100ms. y is calculated as measurement period divided by 100ms. - the triggers of addition - received X2-AP: SgNB Reconfiguration Complete (to SN terminated split bearer) - received X2-AP: SgNB Modification Confirm (Intra/Inter DU PSCell change using SRB1 (for target cell)) - received RRC: RRC Reconfiguration Complete (Intra/Inter DU PSCell change using SRB3 (for target cell)) - the triggers of subtraction - received X2-AP: SgNB Reconfiguration Complete (to SN terminated MCG bearer) - received X2-AP: SgNB Modification Confirm (Intra/Inter DU PSCell change using SRB1 (for source cell)) - received F1-AP: UE Context Release Complete (Intra/Inter DU PSCell change using SRB3 (for source cell)) - received X2-AP: UE Context Release
Measurement Result	Integer number (U32)
Measurement Type	OR.UEENDC.AveSnSplitBearer
Measurement Object Class	NRCelICU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

4

5

A.10.4 Average SN terminated MCG bearer UE Contexts

6

A.10.4.1 Performance Counter Table

Measurement Name	OR.UEENDC.AveSnMcgBearer
Description	This counter provides the average number of UE Contexts that are configured SN terminated MCG bearer that are already setup.

Collection Method	SI
Condition	Measurement subcounter is x/y. x is incremented by the number of UE Contexts that are configured SN terminated MCG bearer. This counter obtains the number of the UE Contexts every 100ms. y is calculated as measurement period divided by 100ms. - the triggers of addition - received X2-AP: SgNB Reconfiguration Complete (to SN terminated MCG bearer) - received X2-AP: SgNB Reconfiguration Confirm (to SN terminated MCG bearer) - the triggers of subtraction - received X2-AP: SgNB Reconfiguration Complete (to SN terminated split bearer) - received X2-AP: UE Context Release
Measurement Result	Integer number (U32)
Measurement Type	OR.UEENDC.AveSnMcgBearer
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2 A.10.5 Total DL CA UE Contexts PSCell (100ms)

3 A.10.5.1 Performance Counter Table

Measurement Name	OR.UEENDC.TotalDlCaUePscell100ms
Description	This counter provides the total number of UE Contexts that are set of DL CA as PSCell in units of NR Cell.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is the total number of x by the measurement period. x is the number of the UE Contexts using this cell as PSCell per the number of CC configured for DL CA in unit of NR Cell group as subcounter.Ccnum by every 100ms. - the triggers of addition - received X2-AP: SgNB Reconfiguration Complete (SCell addition) - received RRC: RRC Reconfiguration Complete (SCell addition/change using SRB3) - received X2-AP: SgNB Modification Confirm (SCell addition/change) - the triggers of subtraction - received X2-AP: SgNB Reconfiguration Complete (SCell release) - received RRC: RRC Reconfiguration Complete (SCell release/change using SRB3) - received X2-AP: SgNB Modification Confirm (SCell release/change) - received X2-AP: UE Context Release
Measurement Result	integer(U32)
Measurement Type	OR.UEENDC.TotalDlCaUePscell100ms.Ccnum where Ccnum is the number of CC:

	0: #0CC (number of SCell excluding PSCell in the cell group) 1: #1CC (number of SCell excluding PSCell in the cell group) ... 7: #7CC (number of SCell excluding PSCell in the cell group)
Measurement Object Class	NRCelICU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.10.6 Total DL CA UE Contexts PSCell (10s)

3

A.10.6.1 Performance Counter Table

Measurement Name	OR.UENDC.TotalDlCaUePscell10s
Description	This counter provides the total number of UE Contexts that are set of DL CA as PSCell in units of NR Cell.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is the total number of x by the measurement period. x is the number of the UE Contexts using this cell as PSCell per the number of CC configured for DL CA in unit of NR Cell group as subcounter.Ccnum by every 10s. - the triggers of addition - received X2-AP: SgNB Reconfiguration Complete (SCell addition) - received RRC: RRC Reconfiguration Complete (SCell addition/change using SRB3) - received X2-AP: SgNB Modification Confirm (SCell addition/change) - the triggers of subtraction - received X2-AP: SgNB Reconfiguration Complete (SCell release) - received F1-AP: UE Context Release Complete (SCell release/change using SRB3) - received X2-AP: SgNB Modification Confirm (SCell release/change) - received X2-AP: UE Context Release
Measurement Result	integer(U32)
Measurement Type	OR.UENDC.TotalDlCaUePscell10s.Ccnum where Ccnum is the number of CC: 0: #0CC (number of SCell excluding PSCell in the cell group) 1: #1CC (number of SCell excluding PSCell in the cell group) ... 7: #7CC (number of SCell excluding PSCell in the cell group)
Measurement Object Class	NRCelICU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

4

1 A.10.7 Total DL CA UE Contexts SCell (100ms)

2 A.10.7.1 Performance Counter Table

Measurement Name	OR.UEENDC.TotalDlCaUeScell100ms
Description	This counter provides the total number of UE Contexts that are set of DL CA as SCell in units of NR Cell.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is the total number of x by the measurement period. x is the number of the UE Contexts using this cell as SCell per the number of CC configured for DL CA in unit of NR Cell group as subcounter. <i>Ccnum</i> by every 100ms. - the triggers of addition - received X2-AP: SgNB Reconfiguration Complete (SCell addition) - received RRC: RRC Reconfiguration Complete (SCell addition/change using SRB3) - received X2-AP: SgNB Modification Confirm (SCell addition/change) - the triggers of subtraction - received X2-AP: SgNB Reconfiguration Complete (SCell release) - received F1-AP: UE Context Release Complete (SCell release/change using SRB3) - received X2-AP: SgNB Modification Confirm (SCell release/change) - received X2-AP: UE Context Release
Measurement Result	integer(U32)
Measurement Type	OR.UEENDC.TotalDlCaUeScell100ms. <i>Ccnum</i> where <i>Ccnum</i> is the number of CC: 0: #1CC (number of SCell excluding PSCell in the cell group) 1: #2CC (number of SCell excluding PSCell in the cell group) ... 6: #7CC (number of SCell excluding PSCell in the cell group)
Measurement Object Class	NRCeLU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

3 A.10.8 Total DL CA UE Contexts SCell (10s)

5 A.10.8.1 Performance Counter Table

Measurement Name	OR.UEENDC.TotalDlCaUeScell10s
Description	This counter provides the total number of UE Contexts that are set of DL CA as SCell in units of NR Cell.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is the total number of x by the measurement period. x is the number of the UE Contexts using this cell as SCell per the number of CC configured for DL CA in unit of NR Cell group as subcounter. <i>Ccnum</i> by every 10s.

	- the triggers of addition - received X2-AP: SgNB Reconfiguration Complete (SCell addition) - received RRC: RRC Reconfiguration Complete (SCell addition/change using SRB3) - received X2-AP: SgNB Modification Confirm (SCell addition/change) - the triggers of subtraction - received X2-AP: SgNB Reconfiguration Complete (SCell release) - received F1-AP: UE Context Release Complete (SCell release/change using SRB3) - received X2-AP: SgNB Modification Confirm (SCell release/change) - received X2-AP: UE Context Release
Measurement Result	integer(U32)
Measurement Type	OR.UEENDC.TotalDlCaUeScell10s.Ccnum where <i>Ccnum</i> is the number of CC: 0: #1CC (number of SCell excluding PSCell in the cell group) 1: #2CC (number of SCell excluding PSCell in the cell group) ... 6: #7CC (number of SCell excluding PSCell in the cell group)
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.10.9 Max DL CA UE Contexts PSCell (100ms)

3

A.10.9.1 Performance Counter Table

Measurement Name	OR.UEENDC.MaxDlCaUePscell100ms
Description	This counter provides the maximum number of UE Contexts that are set of DL CA as PSCell in units of NR Cell.
Collection Method	SI
Condition	Measurement subcounter is the maximum number of x. It is assigned with period of 100ms by the volume x, if x is bigger than previously counted volume. x is the number of the UE Contexts using this cell as PSCell per the number of CC configured for DL CA in unit of NR Cell group as subcounter.<i>Ccnum</i>. The number is acquired as an instance value at every 100ms. - the triggers of addition - received X2-AP: SgNB Reconfiguration Complete (SCell addition) - received RRC: RRC Reconfiguration Complete (SCell addition/change using SRB3) - received X2-AP: SgNB Modification Confirm (SCell addition/change) - the triggers of subtraction - received X2-AP: SgNB Reconfiguration Complete (SCell release) - received F1-AP: UE Context Release Complete (SCell release/change using SRB3)

	- received X2-AP: SgNB Modification Confirm (SCell release/change) - received X2-AP: UE Context Release
Measurement Result	Users (integer)
Measurement Type	OR.UEENDC.MaxDlCaUePscell100ms.Ccnum where <i>Ccnum</i> is the number of CC: 0: #0CC (number of SCell excluding PSCell in the cell group) 1: #1CC (number of SCell excluding PSCell in the cell group) ... 7: #7CC (number of SCell excluding PSCell in the cell group)
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.10.10 Max DL CA UE Contexts PSCell (10s)

3

A.10.10.1 Performance Counter Table

Measurement Name	OR.UEENDC.MaxDlCaUePscell10s
Description	This counter provides the maximum number of UE Contexts that are set of DL CA as PSCell in units of NR Cell.
Collection Method	SI
Condition	Measurement subcounter is the maximum number of x. It is assigned with period of 10s by the volume x, if x is bigger than previously counted volume. x is the number of the UE Contexts using this cell as PSCell per the number of CC configured for DL CA in unit of NR Cell group as subcounter.<i>Ccnum</i>. The number is acquired as an instance value at every 10s. - the triggers of addition - received X2-AP: SgNB Reconfiguration Complete (SCell addition) - received RRC: RRC Reconfiguration Complete (SCell addition/change using SRB3) - received X2-AP: SgNB Modification Confirm (SCell addition/change) - the triggers of subtraction - received X2-AP: SgNB Reconfiguration Complete (SCell release) - received F1-AP: UE Context Release Complete (SCell release/change using SRB3) - received X2-AP: SgNB Modification Confirm (SCell release/change) - received X2-AP: UE Context Release
Measurement Result	Users (integer)
Measurement Type	OR.UEENDC.MaxDlCaUePscell10s.Ccnum where <i>Ccnum</i> is the number of CC: 0: #0CC (number of SCell excluding PSCell in the cell group) 1: #1CC (number of SCell excluding PSCell in the cell group) ... 7: #7CC (number of SCell excluding PSCell in the cell group)
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched

Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.10.11 Max DL CA UE Contexts SCell (100ms)

A.10.11.1 Performance Counter Table

Measurement Name	OR.UEENDC.MaxDlCaUeScell100ms
Description	This counter provides the maximum number of UE Contexts that are set of DL CA as SCell in units of NR Cell.
Collection Method	SI
Condition	Measurement subcounter is the maximum number of x. It is assigned with period of 100ms by the volume x, if x is bigger than previously counted volume. x is the number of the UE Contexts using this cell as SCell per the number of CC configured for DL CA in unit of NR Cell group as subcounter. <i>Ccnum</i>. The number is acquired as an instance value at every 100ms. - the triggers of addition - received X2-AP: SgNB Reconfiguration Complete (SCell addition) - received RRC: RRC Reconfiguration Complete (SCell addition/change using SRB3) - received X2-AP: SgNB Modification Confirm (SCell addition/change) - the triggers of subtraction - received X2-AP: SgNB Reconfiguration Complete (SCell release) - received F1-AP: UE Context Release Complete (SCell release/change using SRB3) - received X2-AP: SgNB Modification Confirm (SCell release/change) - received X2-AP: UE Context Release
Measurement Result	Users (integer)
Measurement Type	OR.UEENDC.MaxDlCaUeScell100ms. <i>Ccnum</i> where <i>Ccnum</i> is the number of CC: 0: #1CC (number of SCell excluding PSCell in the cell group) 1: #2CC (number of SCell excluding PSCell in the cell group) ... 7: #6CC (number of SCell excluding PSCell in the cell group)
Measurement Object Class	NRCeLU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.10.12 Max DL CA UE Contexts SCell (10s)

A.10.12.1 Performance Counter Table

Measurement Name	OR.UEENDC.MaxDlCaUeScell10s
Description	This counter provides the maximum number of UE Contexts that are set of DL CA as SCell in units of NR Cell.
Collection Method	SI

Condition	Measurement subcounter is the maximum number of x. It is assigned with period of 10s by the volume x, if x is bigger than previously counted volume. x is the number of the UE Contexts using this cell as SCell per the number of CC configured for DL CA in unit of NR Cell group as subcounter. <i>Ccnum</i>. The number is acquired as an instance value at every 10s. - the triggers of addition - received X2-AP: SgNB Reconfiguration Complete (SCell addition) - received RRC: RRC Reconfiguration Complete (SCell addition/change using SRB3) - received X2-AP: SgNB Modification Confirm (SCell addition/change) - the triggers of subtraction - received X2-AP: SgNB Reconfiguration Complete (SCell release) - received F1-AP: UE Context Release Complete (SCell release/change using SRB3) - received X2-AP: SgNB Modification Confirm (SCell release/change) - received X2-AP: UE Context Release
Measurement Result	Users (integer)
Measurement Type	OR.UEENDC.MaxDlCaUeScell10s. <i>Ccnum</i> where <i>Ccnum</i> is the number of CC: 0: #1CC (number of SCell excluding PSCell in the cell group) 1: #2CC (number of SCell excluding PSCell in the cell group) ... 7: #6CC (number of SCell excluding PSCell in the cell group)
Measurement Object Class	NRCeLU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.11 Number of UE Contexts for SA

3

A.11.1 Total RRC Connected UE Contexts

4

A.11.1.1 Performance Counter Table

Measurement Name	OR.UESA.TotalRrcConnectedUes
Description	This counter provides the total number of the RRC connected UE Contexts in PCell.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is the total number of x by the measurement period. x is the number of the RRC connected UE Contexts by every 10s. - the triggers of addition - received RRC: RRC Setup Complete - received RRC: RRC Reconfiguration Complete (Inter RAT HO to NR, Inter gNB-CU HO, Intra gNB-CU Inter gNB-DU HO, Intra-DU Inter Cell HO, RRC Connection Re-establishment)

	(Intra gNB-CU Inter gNB-DU), RRC Connection Re-establishment (Intra gNB-DU Inter Cell)) - received RRC: RRC Resume Complete (RRC inactive to RRC Connected) - the triggers of subtraction - received Xn-AP: UE Context Release (Inter gNB-CU HO, RRC Connection Re-establishment (Inter gNB-CU)) - received NG-AP: UE Context Release Command (Inter-RAT HO to LTE, Intra gNB-CU Inter gNB-DU HO, EPS fallback) - received RRC: RRC Reconfiguration Complete (Intra gNB-DU Inter-Cell HO, RRC Connection Re-establishment (Intra gNB-DU Inter Cell))
Measurement Result	Integer number (U32)
Measurement Type	OR.UESA.TotalRrcConnectedUes
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2 A.11.2 Total UE Contexts Per QoS

3 A.11.2.1 Performance Counter Table

Measurement Name	OR.UESA.TotalUesCucountgroup
Description	This counter provides the total number of the UE Contexts per CU count group.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is the total number of x by the measurement period. x is the number of the UE Contexts per CU count group for PCell and SCell as subcountner. <i>Celltype</i> by every 10s. Note: multiple 5QIs can be set per UE Context, each 5QI is counted per QoS flow. - the triggers of addition for PCell - received NG-AP: Initial Context Setup Response - received RRC: RRC Reconfiguration Complete (Inter RAT HO to NR, Inter gNB-CU HO, Intra gNB-CU Inter gNB-DU HO, Intra-DU Inter Cell HO, RRC Connection Re-establishment (Intra gNB-CU Inter gNB-DU), RRC Connection Re-establishment (Intra gNB-DU Inter Cell)) - received RRC: RRC Resume Complete (RRC inactive to RRC Connected) - send NG-AP: PDU Session Resource Setup Response (PDU Session Establishment) - send NG-AP: PDU Session Resource Modify Response (PDU Session Modification) - the triggers of subtraction for PCell - send RRC: RRC Release (UE Context Release, RRC Connected to RRC inactive, SN Release without keeping UE) - received Xn-AP: UE Context Release (Inter gNB-CU HO, RRC Connection Re-establishment (Inter gNB-CU)) - received NG-AP: UE Context Release Command (Inter-RAT HO to LTE, Intra gNB-CU Inter gNB-DU HO, EPS fallback)

	- received RRC: RRC Reconfiguration Complete (Intra gNB-DU Inter-Cell HO, RRC Connection Re-establishment (Intra gNB-DU Inter Cell)) - received F1-AP: UE Context Release Command (RRC Connection Re-establishment (Intra gNB-CU Inter gNB-DU)) - send NG-AP: PDU Session Resource Release Response (PDU Session Release) - send NG-AP: PDU Session Resource Modify Response (PDU Session Modification) - the triggers of addition for SCell - received RRC: RRC Reconfiguration Complete (SCell addition/change) - received RRC: RRC Resume Complete (RRC inactive to RRC Connected with SCell addition) - the triggers of subtraction for SCell - received RRC: RRC Reconfiguration Complete (SCell release/change) - received Xn-AP: UE Context Release (Inter gNB-CU HO w/ Xn) - received NG-AP: UE Context Release Command
Measurement Result	Integer number (U32)
Measurement Type	OR.UESA.TotalUesCucountgroup.Cucountgroup.Celltype where <i>Cucountgroup</i> identifies an instance of CuCountGroup IOC. <i>Celltype</i> is the cell type: 0: PCell 1: SCell
Measurement Object Class	NRCeIlCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.11.3 Total SCell Configuration UE Contexts

3

A.11.3.1 Performance Counter Table

Measurement Name	OR.UESA.TotalScellConfigUes.SCell
Description	This counter provides the total number of each SCell configuration UE Contexts.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is the total number of x by the measurement period. x is the number of each SCell configuration UE Contexts by every 10s. - the triggers of addition - received RRC: RRC Reconfiguration Complete (SCell addition/change) - received RRC: RRC Resume Complete (RRC inactive to RRC Connected with SCell addition) - the triggers of subtraction - received RRC: RRC Reconfiguration Complete (SCell release/change) - received Xn-AP: UE Context Release (Inter gNB-CU HO w/ Xn) - received NG-AP: UE Context Release Command
Measurement Result	Integer number (U32)
Measurement Type	OR.UESA.TotalScellConfigUes.SCell

Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.11.4 Total Measurement Gap UE Contexts

A.11.4.1 Performance Counter Table

Measurement Name	OR.UESA.TotalMeasGapUes
Description	This counter provides the total number of UE Contexts which is configured with measurement gap.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is the total number of x by the measurement period. x is the number of UE Contexts which is configured measurement gap by every 10s. - the triggers of addition - received RRC: RRC Reconfiguration Complete (setup measurement gap) - the triggers of subtraction - received RRC: RRC Reconfiguration Complete (release measurement gap)
Measurement Result	Integer number (U32)
Measurement Type	OR.UESA.TotalMeasGapUes
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.11.5 Max RRC Connected UE Contexts

A.11.5.1 Performance Counter Table

Measurement Name	OR.UESA.MaxRrcConnectedUes
Description	This counter provides the maximum number of the RRC connected UE Contexts in PCell.
Collection Method	SI
Condition	Measurement subcounter is the maximum number of x. It is assigned with period of 10s by the volume x, if x is bigger than previously counted volume. x is the number of the RRC connected UE Contexts in PCell. The number is acquired as an instance value at every 10s. - the triggers of addition - received RRC: RRC Setup Complete - received RRC: RRC Reconfiguration Complete (Inter RAT HO to NR, Inter gNB-CU HO, Intra gNB-CU Inter gNB-DU HO, Intra-DU Inter Cell HO, RRC Connection Re-establishment (Intra gNB-CU Inter gNB-DU), RRC Connection Re-establishment (Intra gNB-DU Inter Cell)) - received RRC: RRC Resume Complete (RRC inactive to RRC Connected)

	- the triggers of subtraction - received Xn-AP: UE Context Release (Inter gNB-CU HO, RRC Connection Re-establishment (Inter gNB-CU)) - received NG-AP: UE Context Release Command (Inter-RAT HO to LTE, Intra gNB-CU Inter gNB-DU HO, EPS fallback) - received RRC: RRC Reconfiguration Complete (Intra gNB-DU Inter-Cell HO, RRC Connection Re-establishment (Intra gNB-DU Inter Cell))
Measurement Result	Integer number (U32)
Measurement Type	OR.UESA.MaxRrcConnectedUes
Measurement Object Class	NRCelICU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.11.6 Max UE Contexts Per QoS

3

A.11.6.1 Performance Counter Table

Measurement Name	OR.UESA.MaxUesPerCuCountgroup
Description	This counter provides the maximum number of the UE Contexts per CU count group.
Collection Method	SI
Condition	Measurement subcounter is the maximum number of x. It is assigned with period of 10s by the volume x, if x is bigger than previously counted volume. x is the number of the UE Contexts per CU count group for PCell and SCell as subcounter. <i>Celltype</i>. Note: multiple 5QIs can be set per UE Context, each 5QI is counted per QoS flow. - the triggers of addition for PCell - received NG-AP: Initial Context Setup Response - received RRC: RRC Reconfiguration Complete (Inter RAT HO to NR, Inter gNB-CU HO, Intra gNB-CU Inter gNB-DU HO, Intra-DU Inter Cell HO, RRC Connection Re-establishment (Intra gNB-CU Inter gNB-DU), RRC Connection Re-establishment (Intra gNB-DU Inter Cell)) - received RRC: RRC Resume Complete (RRC inactive to RRC Connected) - send NG-AP: PDU Session Resource Setup Response (PDU Session Establishment) - send NG-AP: PDU Session Resource Modify Response (PDU Session Modification) - the triggers of subtraction for PCell - send RRC: RRC Release (UE Context Release, RRC Connected to RRC inactive, SN Release without keeping UE) - received Xn-AP: UE Context Release (Inter gNB-CU HO, RRC Connection Re-establishment (Inter gNB-CU)) - received NG-AP: UE Context Release Command (Inter-RAT HO to LTE, Intra gNB-CU Inter gNB-DU HO, EPS fallback) - received RRC: RRC Reconfiguration Complete (Intra gNB-DU Inter-Cell HO, RRC Connection Re-establishment (Intra gNB-DU Inter Cell)) - received F1-AP: UE Context Release Command (RRC Connection Re-establishment (Intra gNB-CU Inter gNB-DU)) - send NG-AP: PDU Session Resource Release Response (PDU Session Release)

	- send NG-AP: PDU Session Resource Modify Response (PDU Session Modification) - the triggers of addition for SCell - received RRC: RRC Reconfiguration Complete (SCell addition/change) - received RRC: RRC Resume Complete (RRC inactive to RRC Connected with SCell addition) - the triggers of subtraction for SCell - received RRC: RRC Reconfiguration Complete (SCell release/change) - received Xn-AP: UE Context Release (Inter gNB-CU HO w/ Xn) - received NG-AP: UE Context Release Command
Measurement Result	Integer number (U32)
Measurement Type	OR.UESA.MaxUesPerCuCountgroup.CuCountgroup.Celltype where CuCountgroup identifies an instance of CuCountGroup IOC. Celltype is the cell type: 0: PCell 1: SCell
Measurement Object Class	NRCCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2 A.11.7 Max SCell Configuration UE Contexts

3 A.11.7.1 Performance Counter Table

Measurement Name	OR.UESA.MaxScellConfigUes.SCell
Description	This counter provides the maximum number of each SCell configuration UE Contexts.
Collection Method	SI
Condition	Measurement subcounter is the maximum number of x. It is assigned with period of 10s by the volume x, if x is bigger than previously counted volume. x is the number of each SCell configuration UE Contexts. The number is acquired as an instance value at every 10s. - the triggers of addition - received RRC: RRC Reconfiguration Complete (SCell addition/change) - received RRC: RRC Resume Complete (RRC inactive to RRC Connected with SCell addition) - the triggers of subtraction - received RRC: RRC Reconfiguration Complete (SCell release/change) - received Xn-AP: UE Context Release (Inter gNB-CU HO w/ Xn) - received NG-AP: UE Context Release Command
Measurement Result	Integer number (U32)
Measurement Type	OR.UESA.MaxScellConfigUes.SCell
Measurement Object Class	NRCCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1 A.11.8 Max Measurement Gap UE Contexts

2 A.11.8.1 Performance Counter Table

Measurement Name	OR.UESA.MaxMeasGapUes
Description	This counter provides the maximum number of UE Contexts which is configured with measurement gap.
Collection Method	SI
Condition	Measurement subcounter is the maximum number of x. It is assigned with period of 10s by the volume x, if x is bigger than previously counted volume. x is the number of the UE Contexts which is configured with measurement gap. The number is acquired as an instance value at every 10s. - the triggers of addition - received RRC: RRC Reconfiguration Complete (setup measurement gap) - the triggers of subtraction - received RRC: RRC Reconfiguration Complete (release measurement gap)
Measurement Result	Integer number (U32)
Measurement Type	OR.UESA.MaxMeasGapUes
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

3 A.11.9 Total emergency UE Contexts

5 A.11.9.1 Performance Counter Table

Measurement Name	OR.UESA.TotalEmergencyUes
Description	This counter provides the total number of the emergency UE Contexts in a Cell.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is the total number of x by the measurement period. x is the number of the emergency UE Contexts per PCell or SCell as subcounter. <i>Celltype</i> by every 10s. emergency UE Contexts: UE Contexts corresponding to ARP number assigned for emergency call or Establishment cause (or Resume cause); emergency is assigned. - the triggers of addition for PCell - send NG-AP: Initial Context Setup Response - received RRC: RRC Reconfiguration Complete (Inter RAT HO to NR, Inter gNB-CU HO, Intra gNB-CU Inter gNB-DU HO, Intra-DU Inter Cell HO, RRC Connection Re-establishment (Intra gNB-CU Inter gNB-DU), RRC Connection Re-establishment (Intra gNB-DU Inter Cell)) - received RRC: RRC Resume Complete (RRC inactive to RRC Connected) - the triggers of subtraction for PCell

	- send RRC: RRC Release (UE Context Release, RRC Connected to RRC inactive, SN Release without keeping UE) - received Xn-AP: UE Context Release (Inter gNB-CU HO, RRC Connection Re-establishment (Inter gNB-CU)) - received NG-AP: UE Context Release Command (Inter-RAT HO to LTE, Intra gNB-CU Inter gNB-DU HO, EPS fallback) - received RRC: RRC Reconfiguration Complete (Intra gNB-DU Inter-Cell HO, RRC Connection Re-establishment (Intra gNB-DU Inter Cell)) - received F1-AP: UE Context Release Command (RRC Connection Re-establishment (Intra gNB-CU Inter gNB-DU)) - the triggers of addition for SCell - received RRC: RRC Reconfiguration Complete (SCell addition/change) - received RRC: RRC Resume Complete (RRC inactive to RRC Connected with SCell addition) - the triggers of subtraction for SCell - received RRC: RRC Reconfiguration Complete (SCell release/change) - received Xn-AP: UE Context Release (Inter gNB-CU HO w/ Xn) - received NG-AP: UE Context Release Command
Measurement Result	Integer number (U32)
Measurement Type	OR.UESA.TotalEmergencyUes. <i>Celltype</i> where <i>Celltype</i> is the cell type: 0: PCell 1: SCell
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.11.10 Total high priority access UE Contexts

A.11.10.1 Performance Counter Table

Measurement Name	OR.UESA.TotalHighPriAccessUes
Description	This counter provides the total number of the high priority access UE Contexts in a Cell.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is the total number of x by the measurement period. x is the number of the high priority access UE Contexts per PCell or SCell as subcounter.<i>Celltype</i> by every 10s. high priority access UE Contexts: UE Contexts corresponding to ARP number assigned for high priority access call or Establishment cause (or Resume cause); highPriorityAccess is assigned. - the triggers of addition for PCell - send NG-AP: Initial Context Setup Response - received RRC: RRC Reconfiguration Complete (Inter RAT HO to NR, Inter gNB-CU HO, Intra gNB-CU Inter gNB-DU HO, Intra-DU Inter Cell HO, RRC Connection Re-establishment (Intra gNB-CU Inter gNB-DU), RRC Connection Re-establishment (Intra gNB-DU Inter Cell)) - received RRC: RRC Resume Complete (RRC inactive to RRC Connected)

	- the triggers of subtraction for PCell - send RRC: RRC Release (UE Context Release, RRC Connected to RRC inactive, SN Release without keeping UE) - received Xn-AP: UE Context Release (Inter gNB-CU HO, RRC Connection Re-establishment (Inter gNB-CU)) - received NG-AP: UE Context Release Command (Inter-RAT HO to LTE, Intra gNB-CU Inter gNB-DU HO, EPS fallback) - received RRC: RRC Reconfiguration Complete (Intra gNB-DU Inter-Cell HO, RRC Connection Re-establishment (Intra gNB-DU Inter Cell)) - received F1-AP: UE Context Release Command (RRC Connection Re-establishment (Intra gNB-CU Inter gNB-DU)) - the triggers of addition for SCell - received RRC: RRC Reconfiguration Complete (SCell addition/change) - received RRC: RRC Resume Complete (RRC inactive to RRC Connected with SCell addition) - the triggers of subtraction for SCell - received RRC: RRC Reconfiguration Complete (SCell release/change) - received Xn-AP: UE Context Release (Inter gNB-CU HO w/ Xn) - received NG-AP: UE Context Release Command
Measurement Result	Integer number (U32)
Measurement Type	OR.UESA.TotalHighPriAccessUes <i>Celltype</i> where <i>Celltype</i> is the cell type: 0: PCell 1: SCell
Measurement Object Class	NRCCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2 A.11.11 Max emergency UE Contexts

3 A.11.11.1 Performance Counter Table

Measurement Name	OR.UESA.MaxEmergencyUes
Description	This counter provides the maximum number of the Emergency UE Contexts in a Cell.
Collection Method	SI
Condition	Measurement subcounter is the maximum number of x. It is assigned with period of 10s by the volume x, if x is bigger than previously counted volume. X is the number of the emergency UE Contexts per PCell or SCell as subcounter.<i>Celltype</i>. The number is acquired as an instance value at every 10s. Emergency UE Contexts: UEs corresponding to ARP number assigned for emergency call or Establishment cause (or Resume cause): emergency is assigned. - the triggers of addition for PCell - send NG-AP: Initial Context Setup Response - received RRC: RRC Reconfiguration Complete (Inter RAT HO to NR, Inter gNB-CU HO, Intra gNB-CU Inter gNB-DU HO, Intra-DU Inter Cell HO, RRC Connection Re-establishment

	(Intra gNB-CU Inter gNB-DU), RRC Connection Re-establishment (Intra gNB-DU Inter Cell)) - received RRC: RRC Resume Complete (RRC inactive to RRC Connected) - the triggers of subtraction for PCell - send RRC: RRC Release (UE Context Release, RRC Connected to RRC inactive, SN Release without keeping UE) - received Xn-AP: UE Context Release (Inter gNB-CU HO, RRC Connection Re-establishment (Inter gNB-CU)) - received NG-AP: UE Context Release Command (Inter-RAT HO to LTE, Intra gNB-CU Inter gNB-DU HO, EPS fallback) - received RRC: RRC Reconfiguration Complete (Intra gNB-DU Inter-Cell HO, RRC Connection Re-establishment (Intra gNB-DU Inter Cell)) - received F1-AP: UE Context Release Command (RRC Connection Re-establishment (Intra gNB-CU Inter gNB-DU)) - the triggers of addition for SCell - received RRC: RRC Reconfiguration Complete (SCell addition/change) - received RRC: RRC Resume Complete (RRC inactive to RRC Connected with SCell addition) - the triggers of subtraction for SCell - received RRC: RRC Reconfiguration Complete (SCell release/change) - received Xn-AP: UE Context Release (Inter gNB-CU HO w/ Xn) - received NG-AP: UE Context Release Command
Measurement Result	Integer number (U32)
Measurement Type	OR.UESA.MaxEmergencyUes <i>Celltype</i> where <i>Celltype</i> is the cell type: 0: PCell 1: SCell
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2 A.11.12 Max high priority access UE Contexts

3 A.11.12.1 Performance Counter Table

Measurement Name	OR.UESA.MaxHighPriAccessUes
Description	This counter provides the maximum number of the high priority access UE Contexts in a Cell.
Collection Method	SI
Condition	Measurement subcounter is the maximum number of x. It is assigned with period of 10s by the volume x, if x is bigger than previously counted volume. x is the number of the high priority access UE Contexts per PCell or SCell as subcounter.<i>Celltype</i>. The number is acquired as an instance value at every 10s. high priority access UE Contexts: UE Contexts corresponding to ARP number assigned for high priority access call or Establishment cause (or Resume cause); highPriorityAccess is assigned. - the triggers of addition for PCell - send NG-AP: Initial Context Setup Response

	- received RRC: RRC Reconfiguration Complete (Inter RAT HO to NR, Inter gNB-CU HO, Intra gNB-CU Inter gNB-DU HO, Intra-DU Inter Cell HO, RRC Connection Re-establishment (Intra gNB-CU Inter gNB-DU), RRC Connection Re-establishment (Intra gNB-DU Inter Cell)) - received RRC: RRC Resume Complete (RRC inactive to RRC Connected) - the triggers of subtraction for PCell - send RRC: RRC Release (UE Context Release, RRC Connected to RRC inactive, SN Release without keeping UE) - received Xn-AP: UE Context Release (Inter gNB-CU HO, RRC Connection Re-establishment (Inter gNB-CU)) - received NG-AP: UE Context Release Command (Inter-RAT HO to LTE, Intra gNB-CU Inter gNB-DU HO, EPS fallback) - received RRC: RRC Reconfiguration Complete (Intra gNB-DU Inter-Cell HO, RRC Connection Re-establishment (Intra gNB-DU Inter Cell)) - received F1-AP: UE Context Release Command (RRC Connection Re-establishment (Intra gNB-CU Inter gNB-DU)) - the triggers of addition for SCell - received RRC: RRC Reconfiguration Complete (SCell addition/change) - received RRC: RRC Resume Complete (RRC inactive to RRC Connected with SCell addition) - the triggers of subtraction for SCell - received RRC: RRC Reconfiguration Complete (SCell release/change) - received Xn-AP: UE Context Release (Inter gNB-CU HO w/ Xn) - received NG-AP: UE Context Release Command
Measurement Result	Integer number (U32)
Measurement Type	OR.UESA.MaxHighPriAccessUes <i>Celltype</i> where <i>Celltype</i> is the cell type: 0: PCell 1: SCell
Measurement Object Class	NRCelICU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.11.13 Total SN terminated split bearer UE Contexts in NR-DC

3

A.11.13.1 Performance Counter Table

Measurement Name	OR.UENRDC.TotalSnSplitbearUes
Description	This counter provides the total number of UE Contexts configured as SN terminated split bearer in the NR-DC.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is the total number of x by the measurement period. x is the number of the UE Contexts configured as SN terminated Split bearer in the NR-DC. The number is acquired as an instance value at every 10s. - the triggers of addition

	- received Xn-AP: SN Reconfiguration Complete (to SN terminated split bearer) - received Xn-AP: S-Node Modification Confirm (Intra/Inter DU PSCell change using SRB1 (for target cell)) - received RRC: RRC Reconfiguration Complete (Intra DU PSCell change using SRB3 (for target cell)) - the triggers of subtraction - received Xn-AP: S-Node Modification Confirm (Intra/Inter DU PSCell change using SRB1 (for source cell)) - received RRC: RRC Reconfiguration Complete (Intra DU PSCell change using SRB3 (for source cell)) - received Xn-AP: UE Context Release
Measurement Result	Integer number (U32)
Measurement Type	OR.UENRDC.TotalSnSplitbearUes
Measurement Object Class	NRCelICU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.11.14 Max SN terminated split bearer UE Contexts in NR-DC

3

A.11.14.1 Performance Counter Table

Measurement Name	OR.UENRDC.MaxSnSplitbearUes
Description	This counter provides the maximum number of UE Contexts configured as SN terminated split bearer in the NR-DC.
Collection Method	SI
Condition	Measurement subcounter is the maximum number of x. It is assigned with period of 100ms by the volume x, if x is bigger than previously counted volume. x is the number of the UE Contexts configured as SN terminated split bearer in the NR-DC. x is acquired as an instantaneous value at every 100ms. - the triggers of addition - received Xn-AP: SN Reconfiguration Complete (to SN terminated split bearer) - received Xn-AP: S-Node Modification Confirm (Intra/Inter DU PSCell change using SRB1 (for target cell)) - received RRC: RRC Reconfiguration Complete (Intra DU PSCell change using SRB3 (for target cell)) - the triggers of subtraction - received Xn-AP: S-Node Modification Confirm (Intra/Inter DU PSCell change using SRB1 (for source cell)) - received RRC: RRC Reconfiguration Complete (Intra DU PSCell change using SRB3 (for source cell)) - received Xn-AP: UE Context Release
Measurement Result	Integer number (U32)
Measurement Type	OR.UENRDC.MaxSnSplitbearUes
Measurement Object Class	NRCelICU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

4

A.11.15 Total UE Contexts during voice call

A.11.15.1 Performance Counter Table

Measurement Name	OR.UESA.TotalUesVoice
Description	This counter provides the total number of the UE Contexts during voice call.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is the total number of x by the measurement period. x is the number of the UE Contexts during voice call by every 10s. - the triggers of addition - send NG-AP: Initial Context Setup Response - received RRC: RRC Reconfiguration Complete (Inter RAT HO to NR, Inter gNB-CU HO, Intra gNB-CU Inter gNB-DU HO, Intra-DU Inter Cell HO, RRC Connection Re-establishment (Intra gNB-CU Inter gNB-DU), RRC Connection Re-establishment (Intra gNB-DU Inter Cell)) - received RRC: RRC Resume Complete (RRC inactive to RRC Connected) - the triggers of subtraction - send RRC: RRC Release (UE Context Release, RRC Connected to RRC inactive, SN Release without keeping UE) - received Xn-AP: UE Context Release (Inter gNB-CU HO, RRC Connection Re-establishment (Inter gNB-CU)) - received NG-AP: UE Context Release Command (Inter-RAT HO to LTE, Intra gNB-CU Inter gNB-DU HO, EPS fallback) - received RRC: RRC Reconfiguration Complete (Intra gNB-DU Inter-Cell HO, RRC Connection Re-establishment (Intra gNB-DU Inter Cell)) - received F1-AP: UE Context Release Command (RRC Connection Re-establishment (Intra gNB-CU Inter gNB-DU))
Measurement Result	Integer number (U32)
Measurement Type	OR.UESA.TotalUesVoice
Measurement Object Class	NRCeIIcU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.12 Monitoring of procedure for EN-DC

A.12.1 Number of SgNB Addition procedure attempted for each Cell

A.12.1.1 Performance Counter Table

Measurement Name	OR.ENDCPROCEDURE.SgnbAddAttemptCell
Description	This counter provides the number of the transmitted X2-AP:SGNB ADDITION REQUEST ACKNOWLEDGE messages.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever X2-AP:SGNB ADDITION REQUEST ACKNOWLEDGE message is transmitted as PSCell.
Measurement Result	Integer number (U32)
Measurement Type	OR.ENDCPROCEDURE.SgnbAddAttemptCell

Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.12.2 Number of SgNB Addition procedure successfully for each Cell

A.12.2.1 Performance Counter Table

Measurement Name	OR.ENDCPROCEDURE.SgnbAddSuccessCell
Description	This counter provides the number of the completed SgNB addition Procedure.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever SgNB addition procedure is completed as PSCell (i.e. X2-AP:SGNB RECONFIGURATION COMPLETE message (SgNB addition complete) is received and RACH processing with UE is completed).
Measurement Result	Integer number (U32)
Measurement Type	OR.ENDCPROCEDURE.SgnbAddSuccessCell
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.12.3 Number of SgNB Addition procedure attempted for each neighbour eNB

A.12.3.1 Performance Counter Table

Measurement Name	OR.ENDCPROCEDURE.SgnbAddAttemptNeiEnb
Description	This counter provides the number of the transmitted X2-AP:SGNB ADDITION REQUEST ACKNOWLEDGE messages. This counter is split into subcounters per neighbour eNB.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever X2-AP:SGNB ADDITION REQUEST ACKNOWLEDGE message is transmitted when the eNB configured by the message is group of subcounter. <i>neighboureNB</i> .
Measurement Result	Integer number (U32)
Measurement Type	OR.ENDCPROCEDURE.SgnbAddAttemptNeiEnb. <i>neighboureNB</i> where <i>neighboureNB</i> is neighbour eNB index: 0: #0 1: #1 ... 63: #63
Measurement Object Class	GNBCUCPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.12.4 Number of SgNB Addition procedure successfully for each neighbour eNB

A.12.4.1 Performance Counter Table

Measurement Name	OR.ENDCPROCEDURE.SgnbAddSuccessNeiEnb
Description	This counter provides the number of the completed SgNB addition Procedure. This counter is split into subcounters per neighbour eNB.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever SgNB addition procedure is completed (i.e. X2-AP:SGNB RECONFIGURATION COMPLETE message (SgNB addition complete) is received and RACH processing with UE is completed) when the eNB configured by the message is group of subcounter. <i>neighboureNB</i> .
Measurement Result	Integer number (U32)
Measurement Type	OR.ENDCPROCEDURE.SgnbAddSuccessNeiEnb. <i>neighboureNB</i> where <i>neighboureNB</i> is neighbour eNB index: 0: #0 1: #1 ... 63: #63
Measurement Object Class	GNBCUCPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.12.5 Number of SgNB modification procedure attempted for each neighbour eNB

A.12.5.1 Performance Counter Table

Measurement Name	OR.ENDCPROCEDURE.SgnbModAttemptNeiEnb
Description	This counter provides the number of the received X2-AP:SGNB MODIFICATION REQUEST messages or transmitted X2-AP:SGNB MODIFICATION REQUIRED messages. This counter is split into subcounters per neighbour eNB.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever X2-AP:SGNB MODIFICATION REQUEST message is received or X2-AP:SGNB MODIFICATION REQUIRED message is transmitted , when the eNB configured by the message is group of subcounter. <i>neighboureNB</i> . Double counting is not performed if the procedure is changed to the MN initiated SN Modification procedure after transmitted X2-AP:SGNB MODIFICATION REQUIRED message (e.g. Measurement gap Coordination(SN initiated)). This subcounter will not be incremented if received X2-AP:SGNB MODIFICATION REQUEST message which True is set for SCG Configuration Query IE.
Measurement Result	Integer number (U32)
Measurement Type	OR.ENDCPROCEDURE.SgnbModAttemptNeiEnb. <i>neighboureNB</i> where <i>neighboureNB</i> is neighbour eNB index:

	0: #0 1: #1 ... 63: #63
Measurement Object Class	GNBCUCPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

A.12.6 Number of SgNB modification procedure successfully for each neighbour eNB

3

A.12.6.1 Performance Counter Table

4

Measurement Name	OR.ENDCPROCEDURE.SgnbModSuccessNeiEnb
Description	This counter provides the number of the completed SgNB modification Procedure. This counter is split into subcounters per neighbour eNB.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever SgNB Modification procedure is completed (i.e. X2-AP:SGNB RECONFIGURATION COMPLETE message(SgNB modification complete) is received or X2-AP:SGNB MODIFICATION CONFIRM is received and RACH processing with UE is completed) when the eNB configured by the message is group of subcounter. <i>neighboureNB</i> .
Measurement Result	Integer number (U32)
Measurement Type	OR.ENDCPROCEDURE.SgnbModSuccessNeiEnb. <i>neighboureNB</i> where <i>neighboureNB</i> is neighbour eNB index: 0: #0 1: #1 ... 63: #63
Measurement Object Class	GNBCUCPFunction
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

5

A.13 Monitoring of RRC Connection for EN-DC

6

A.13.1 Number of Measurement Gap Coordination for per FR2 gap via SRB3 attempted

8

A.13.1.1 Performance Counter Table

9

Measurement Name	OR.RRCCONENDC.ReconfSrb3AttemptGap
Description	This counter provides the number of the transmitted RRC:RRC Reconfiguration messages via SRB3 when the following procedure.

	- Measurement Gap Coordination for per FR2 gap (without MN involvement) Procedure (SN initiated).
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever RRC:RRCReconfiguration message is transmitted via SRB3 when the following procedure. - Measurement Gap Coordination for per FR2 gap (without MN involvement) Procedure (SN initiated).
Measurement Result	Integer number (U32)
Measurement Type	OR.RRCCONENDC.ReconfSrb3AttemptGap
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

A.13.2 Number of Measurement Gap Coordination for per FR2 gap via SRB3 successfully

3

A.13.2.1 Performance Counter Table

4

Measurement Name	OR.RRCCONENDC.ReconfSrb3SuccessGap
Description	This counter provides the number of the received RRC:RRC Reconfiguration Complete messages via SRB3 when the following procedure. - Measurement Gap Coordination for per FR2 gap (without MN involvement) Procedure (SN initiated).
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever RRC:RRCReconfiguration Complete message is received via SRB3 when the following procedure. - Measurement Gap Coordination for per FR2 gap (without MN involvement) Procedure (SN initiated).
Measurement Result	Integer number (U32)
Measurement Type	OR.RRCCONENDC.ReconfSrb3SuccessGap
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

5

A.13.3 Number of Inter gNB-DU PSCell Change via SRB3 attempted

6

A.13.3.1 Performance Counter Table

7

Measurement Name	OR.RRCCONENDC.ReconfSrb3AttemptInterDuPscellch
Description	This counter provides the number of the transmitted RRC:RRC Reconfiguration messages via SRB3 when the following procedure. - Inter gNB-DU PSCell Change using SRB3 for RRC Reconfiguration
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever RRC:RRCReconfiguration message is transmitted via SRB3 when the following procedure.

	- Inter gNB-DU PSCell Change using SRB3 for RRC Reconfiguration.
Measurement Result	Integer number (32)
Measurement Type	OR.RRCCONENDC.ReconfSrb3AttemptInterDuPscellch
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.13.4 Number of Inter gNB-DU PSCell Change via SRB3 successfully

3

A.13.4.1 Performance Counter Table

Measurement Name	OR.RRCCONENDC.ReconfSrb3SuccessInterDuPscellch
Description	This counter provides the number of the received RRC:RRC Reconfiguration Complete messages via SRB3 when the following procedure. - Inter gNB-DU PSCell Change using SRB3 for RRC Reconfiguration.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever RRC:RRCReconfiguration Complete message is received via SRB3 when the following procedure. - Inter gNB-DU PSCell Change using SRB3 for RRC Reconfiguration.
Measurement Result	Integer number (U32)
Measurement Type	OR.RRCCONENDC.ReconfSrb3SuccessInterDuPscellch
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

4

5

A.13.5 Number of Intra gNB-DU PSCell Change via SRB3 attempted

6

A.13.5.1 Performance Counter Table

Measurement Name	OR.RRCCONENDC.ReconfSrb3AttemptIntraDuPscellch
Description	This counter provides the number of the transmitted RRC:RRC Reconfiguration messages via SRB3 when the following procedure. - Intra gNB-DU PSCell Change using SRB3 for RRC Reconfiguration.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever RRC:RRCReconfiguration message is transmitted via SRB3 when the following procedure. - Intra gNB-DU PSCell Change using SRB3 for RRC Reconfiguration.
Measurement Result	Integer number (U32)
Measurement Type	OR.RRCCONENDC.ReconfSrb3AttemptIntraDuPscellch
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

7

A.13.6 Number of Intra gNB-DU PSCell Change via SRB3 successfully

A.13.6.1 Performance Counter Table

Measurement Name	OR.RRCCONENDC.ReconfSrb3SuccessIntraDuPscellch
Description	This counter provides the number of the received RRC:RRC Reconfiguration Complete messages via SRB3 when the following procedure. - Intra gNB-DU PSCell Change using SRB3 for RRC Reconfiguration.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever RRC:RRCReconfiguration Complete message is received via SRB3 when the following procedure. - Intra gNB-DU PSCell Change using SRB3 for RRC Reconfiguration.
Measurement Result	Integer number (U32)
Measurement Type	OR.RRCCONENDC.ReconfSrb3SuccessIntraDuPscellch
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.14 Monitoring of RRC Connection for SA

A.14.1 Number of initial RRC Connection requests

A.14.1.1 Performance Counter Table

Measurement Name	OR.RRCCONSA.IniConReq
Description	This counter provides the number of the received RRC: RRC Setup Request (or RRC: RRC Resume Request) messages via F1-C which is in the first time for each ue-Identity.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Setup Request (or RRC: RRC Resume Request) message is received via F1-C, if RRC: RRC Setup Request (or RRC: RRC Resume Request) message of the same ue-Identity is not received for a certain period of time from the reception timing.
Measurement Result	Integer number (U32)
Measurement Type	OR.RRCCONSA.IniConReq
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.14.2 Number of RRC connection completions for initial RRC Connection requests

A.14.2.1 Performance Counter Table

Measurement Name	OR.RRCCONSA.IniConComp
------------------	------------------------

Description	This counter provides the number of the received RRC: RRC Setup Complete (or RRC: RRC Resume Response) messages via F1-C which corresponds to the initial RRC: RRC Setup Request (or RRC: RRC Resume Request) except fall-back procedure.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Setup Complete (or RRC: RRC Resume Response) message which corresponds to the initial RRC: RRC Setup Request (or RRC: RRC Resume Request) except fall-back procedure is received via F1-C.
Measurement Result	Integer number (U32)
Measurement Type	OR.RRCCONSA.IniConComp
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2 A.14.3 Number of retransmission RRC Connections requests

3 A.14.3.1 Performance Counter Table

Measurement Name	OR.RRCCONSA.RetConReq
Description	This counter provides the number of the received RRC: RRC Setup Request (or RRC: RRC Resume Request) messages via F1-C which is retransmission for each ue-Identity.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Setup Request (or RRC: RRC Resume Request) message is received via F1-C, if RRC: RRC Setup Request (or RRC: RRC Resume Request) message of the same ue-Identity is received for a certain period of time from the reception timing.
Measurement Result	Integer number (U32)
Measurement Type	OR.RRCCONSA.RetConReq
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

4

5 A.14.4 Number of RRC Connections completions for retransmission RRC 6 Connections requests

7 A.14.4.1 Performance Counter Table

Measurement Name	OR.RRCCONSA.RetransConComp
Description	This counter provides the number of the received RRC: RRC Setup Complete (or RRC: RRC Resume Response) messages via F1-C which corresponds to the retransmission RRC: RRC Setup Request (or RRC: RRC Resume Request) except fall-back procedure.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Setup Complete (or RRC: RRC Resume Response) message which corresponds to the retransmission RRC: RRC Setup Request (or RRC: RRC Resume Request) except fall-back procedure is received via F1-C.
Measurement Result	Integer number (U32)

Measurement Type	OR.RRCCONSA.RetransConComp
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.14.5 Number of incomplete RRC Connection by failed resource allocations

A.14.5.1 Performance Counter Table

Measurement Name	OR.RRCCONSA.IncompConResourAllo
Description	This counter provides the number of the RRC Connection configuration interrupted by failed resource allocations.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC Connection configuration is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to RRC: RRC Setup Complete (or RRC: RRC Resume Complete) by failed resource allocations.
Measurement Result	Integer number (U32)
Measurement Type	OR.RRCCONSA.IncompConResourAllo
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.14.6 Number of incomplete RRC Connection by O-CU internal error

A.14.6.1 Performance Counter Table

Measurement Name	OR.RRCCONSA.IncompConOcuInterErr
Description	This counter provides the number of the RRC Connection configuration interrupted by O-CU internal error (e.g. L3 message sending NG).
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC Connection configuration is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to RRC: RRC Setup Complete (or RRC: RRC Resume Complete) by O-CU internal error.
Measurement Result	Integer number (U32)
Measurement Type	OR.RRCCONSA.IncompConOcuInterErr
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.14.7 Number of initial RRC Connection requests each Establishment Cause

A.14.7.1 Performance Counter Table

Measurement Name	OR.RRCCONSA.IniConReqEstaCause
------------------	--------------------------------

Description	This counter provides the number of the received RRC: RRC Setup Request (or RRC: RRC Resume Request) messages via F1-C which is in the first time for each ue-Identity. This counter is split into subcounters per EstablishmentCause.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever RRC: RRC Setup Request (or RRC: RRC Resume Request) message is received via F1-C, if RRC: RRC Setup Request (or RRC: RRC Resume Response) message of the same ue-Identity is not received for a certain period of time from the reception timing when the cell configured by the message is group of subcounter. <i>EstablishmentCause</i> .
Measurement Result	Integer number (U32)
Measurement Type	OR.RRCCONSA.IniConReqEstaCause. <i>EstablishmentCause</i> where <i>EstablishmentCause</i> is EstablishmentCause: 0: EstablishmentCause = emergency 1: EstablishmentCause = highPriorityAccess 2: EstablishmentCause = mt-Access 3: EstablishmentCause = mo-Signalling 4: EstablishmentCause = mo-Data 5: EstablishmentCause = mo-VoiceCall 6: EstablishmentCause = mo-VideoCall 7: EstablishmentCause = mo-SMS 8: EstablishmentCause = mps-PriorityAccess 9: EstablishmentCause = mcs-PriorityAccess
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

A.14.8 Number of RRC connection completions for initial RRC Connection requests each Establishment Cause

3

A.14.8.1 Performance Counter Table

4

Measurement Name	OR.RRCCONSA.IniConCompEstaCause
Description	This counter provides the number of the received RRC: RRC Setup Complete (or RRC: RRC Resume Response) messages via F1-C which corresponds to the initial RRC: RRC Setup Request (or RRC: RRC Resume Request) except fall-back procedure. This counter is split into subcounters per EstablishmentCause.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever RRC:RRC Setup Complete (or RRC: RRC Resume Response) message which corresponds to the initial RRC: RRC Setup Request (or RRC: RRC Resume Request) except fall-back procedure is received via F1-C when the cell configured by the message is group of subcounter. <i>EstablishmentCause</i> .
Measurement Result	Integer number (U32)

Measurement Type	OR.RRCCONSA.IniConCompEstaCause. <i>EstablishmentCause</i> where <i>EstablishmentCause</i> is EstablishmentCause: 0: EstablishmentCause = emergency 1: EstablishmentCause = highPriorityAccess 2: EstablishmentCause = mt-Access 3: EstablishmentCause = mo-Signalling 4: EstablishmentCause = mo-Data 5: EstablishmentCause = mo-VoiceCall 6: EstablishmentCause = mo-VideoCall 7: EstablishmentCause = mo-SMS 8: EstablishmentCause = mps-PriorityAccess 9: EstablishmentCause = mcs-PriorityAccess
Measurement Object Class	NRCeLU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

3

A.14.9 Number of retransmission RRC Connections requests each Establishment Cause

4

A.14.9.1 Performance Counter Table

Measurement Name	OR.RRCCONSA.RetransConReqEstaCause
Description	This counter provides the number of the received RRC: RRC Setup Request (or RRC: RRC Resume Request) messages via F1-C which is retransmission for each ue-Identity. This counter is split into subcounters per EstablishmentCause.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever RRC: RRC Setup Request (or RRC: RRC Resume Request) message is received via F1-C, if RRC: RRC Setup Request (or RRC: RRC Resume Request) message of the same ue-Identity is received for a certain period of time from the reception timing when the cell configured by the message is group of subcounter. <i>EstablishmentCause</i> .
Measurement Result	Integer number (U32)
Measurement Type	OR.RRCCONSA.RetransConReqEstaCause. <i>EstablishmentCause</i> where <i>EstablishmentCause</i> is EstablishmentCause: 0: EstablishmentCause = emergency 1: EstablishmentCause = highPriorityAccess 2: EstablishmentCause = mt-Access 3: EstablishmentCause = mo-Signalling 4: EstablishmentCause = mo-Data 5: EstablishmentCause = mo-VoiceCall

	6: EstablishmentCause = mo-VideoCall 7: EstablishmentCause = mo-SMS 8: EstablishmentCause = mps-PriorityAccess 9: EstablishmentCause = mcs-PriorityAccess
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

A.14.10 Number of RRC Connections completions for retransmission RRC Connections requests each Establishment Cause

3

A.14.10.1 Performance Counter Table

4

Measurement Name	OR.RRCCONSA.RetransConCompEstaCause
Description	This counter provides the number of the received RRC: RRC Setup Complete (or RRC: RRC Resume Response) messages via F1-C which corresponds to the retransmission RRC: RRC Setup Request (or RRC: RRC Resume Request) except fall-back procedure. This counter is split into subcounters per EstablishmentCause.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever RRC: RRC Setup Complete (or RRC: RRC Resume Response) message which corresponds to the retransmission RRC: RRC Setup Request (or RRC: RRC Resume Request) except fall-back procedure is received via F1-C when the cell configured by the message is group of subcounter. <i>EstablishmentCause</i> .
Measurement Result	Integer number (U32)
Measurement Type	OR.RRCCONSA.RetransConCompEstaCause. <i>EstablishmentCause</i> where <i>EstablishmentCause</i> is EstablishmentCause: 0: EstablishmentCause = emergency 1: EstablishmentCause = highPriorityAccess 2: EstablishmentCause = mt-Access 3: EstablishmentCause = mo-Signalling 4: EstablishmentCause = mo-Data 5: EstablishmentCause = mo-VoiceCall 6: EstablishmentCause = mo-VideoCall 7: EstablishmentCause = mo-SMS 8: EstablishmentCause = mps-PriorityAccess 9: EstablishmentCause = mcs-PriorityAccess
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS

Purpose	Network Operator's Traffic Engineering Community
---------	--

A.14.11 Number of incomplete RRC Connection by failed resource allocations each Establishment Cause

A.14.11.1 Performance Counter Table

Measurement Name	OR.RRCCONSA.IncompConResourAlloEstaCause
Description	This counter provides the number of the RRC Connection configuration interrupted by failed resource allocations. This counter is split into subcounters per EstablishmentCause.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever RRC Connection configuration is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to RRC: RRC Setup Complete (or RRC: RRC Resume Complete) by failed resource allocations when the cell configured by the message is group of subcounter. <i>EstablishmentCause</i> .
Measurement Result	Integer number (U32)
Measurement Type	OR.RRCCONSA.IncompConResourAlloEstaCause. <i>EstablishmentCause</i> where <i>EstablishmentCause</i> is EstablishmentCause: 0: EstablishmentCause = emergency 1: EstablishmentCause = highPriorityAccess 2: EstablishmentCause = mt-Access 3: EstablishmentCause = mo-Signalling 4: EstablishmentCause = mo-Data 5: EstablishmentCause = mo-VoiceCall 6: EstablishmentCause = mo-VideoCall 7: EstablishmentCause = mo-SMS 8: EstablishmentCause = mps-PriorityAccess 9: EstablishmentCause = mcs-PriorityAccess
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.14.12 Number of incomplete RRC Connection by O-CU internal error each Establishment Cause

A.14.12.1 Performance Counter Table

Measurement Name	OR.RRCCONSA.IncompConOcuInterErrEstaCause
Description	This counter provides the number of the RRC Connection configuration interrupted by O-CU internal error (e.g. L3 message sending NG). This counter is split into subcounters per EstablishmentCause.

Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever RRC Connection configuration is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to RRC: RRC Setup Complete (or RRC: RRC Resume Complete) by O-CU internal error when the cell configured by the message is group of subcounter. <i>EstablishmentCause</i> .
Measurement Result	Integer number (U32)
Measurement Type	OR.RRCCONSA.IncompConOcuInterErrEstaCause. <i>EstablishmentCause</i> where <i>EstablishmentCause</i> is EstablishmentCause: 0: EstablishmentCause = emergency 1: EstablishmentCause = highPriorityAccess 2: EstablishmentCause = mt-Access 3: EstablishmentCause = mo-Signalling 4: EstablishmentCause = mo-Data 5: EstablishmentCause = mo-VoiceCall 6: EstablishmentCause = mo-VideoCall 7: EstablishmentCause = mo-SMS 8: EstablishmentCause = mps-PriorityAccess 9: EstablishmentCause = mcs-PriorityAccess
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.15 Monitoring of Establishment calls for SA

3

A.15.1 Number of attempted establishment calls for mo-Data

4

A.15.1.1 Performance Counter Table

Measurement Name	OR.ESTACALL.AttemptModata
Description	This counter provides the number of the received RRC: RRC Setup Complete (or RRC: RRC Resume Complete) messages via F1-C after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (Resume Cause):mo-Data).
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Setup Complete (or RRC: RRC Resume Complete) message is received via F1-C after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): mo-Data).
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.AttemptModata
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS

Purpose	Network Operator's Traffic Engineering Community
---------	--

A.15.2 Number of successful establishment calls for mo-Data

A.15.2.1 Performance Counter Table

Measurement Name	OR.ESTACALL.SuccessModata
Description	This counter provides the number of the following message after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): mt-Data): - send NG-AP: Initial Context Setup Response - received NG-AP: UE Context Release Command (for deregistration) before sending NG-AP: Initial Context Setup Response
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever the following event occurs after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): mo-Data): - send NG-AP: Initial Context Setup Response - received NG-AP: UE Context Release Command (for deregistration) before sending NG-AP: Initial Context Setup Response Measurement counter is incremented by 1 whenever NG-AP:Initial Context Setup Response message is transmitted after received RRC: RRC Setup Request (Establishment Cause: mo-Data).
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.SuccessModata
Measurement Object Class	NRCeIlCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.3 Number of incomplete establishment calls for mo-Data by protocol error

A.15.3.1 Performance Counter Table

Measurement Name	OR.ESTACALL.IncompModataProtocolErr
Description	This counter provides the number of the Call establishment for mo-Data interrupted by protocol error.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever Call establishment for mo-Data is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) by protocol error.
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.IncompModataProtocolErr
Measurement Object Class	NRCeIlCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.4 Number of incomplete establishment calls for mo-Data by O-CU internal error

A.15.4.1 Performance Counter Table

Measurement Name	OR.ESTACALL.IncompModataOcuInterErr
Description	This counter provides the number of the call establishment for mo-Data interrupted by O-CU internal error.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever call establishment for mo-Data is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) by O-CU internal error.
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.IncompModataOcuInterErr
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.5 Number of incomplete establishment calls for mo-Data by O-DU error detection

A.15.5.1 Performance Counter Table

Measurement Name	OR.ESTACALL.IncompModataOduErrDetect
Description	This counter provides the number of the call establishment for mo-Data interrupted by O-DU Error Detection.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever call establishment for mo-Data is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) by O-DU Error Detection.
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.IncompModataOduErrDetect
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.6 Number of attempted establishment calls for mt-Access

A.15.6.1 Performance Counter Table

Measurement Name	OR.ESTACALL.AttemptMtaccess
Description	This counter provides the number of the received RRC: RRC Setup Complete (or RRC: RRC Resume Complete) messages via F1-C after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): mt-Access).
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Setup Complete (or RRC: RRC Resume Complete) message is received via F1-

	C after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): mt-Access).
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.AttemptMtaccess
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.15.7 Number of successful establishment calls for mt-Access

3

A.15.7.1 Performance Counter Table

Measurement Name	OR.ESTACALL.SuccessMtaccess
Description	This counter provides the number of the following message after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause):mt-Access): - send NG-AP: Initial Context Setup Response - received NG-AP: UE Context Release Command (for deregistration) before sending NG-AP: Initial Context Setup Response
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever the following event occurs after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause):mt-Access): - send NG-AP: Initial Context Setup Response - received NG-AP: UE Context Release Command (for deregistration) before sending NG-AP: Initial Context Setup Response
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.SuccessMtaccess
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

4

5

A.15.8 Number of incomplete establishment calls for mt-Access by protocol error

6

7

A.15.8.1 Performance Counter Table

Measurement Name	OR.ESTACALL.IncompMtaccessProtocolErr
Description	This counter provides the number of the call establishment for mt-Access interrupted by protocol error.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever call establishment for mt-Access is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) by protocol error.
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.IncompMtaccessProtocolErr
Measurement Object Class	NRCellCU

Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.9 Number of incomplete establishment calls for mt-Access by O-CU internal error

A.15.9.1 Performance Counter Table

Measurement Name	OR.ESTACALL.IncompMtaccessOcuInterErr
Description	This counter provides the number of the call establishment for mt-Access interrupted by O-CU internal error.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever call establishment for mt-Access is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) by O-CU internal error.
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.IncompMtaccessOcuInterErr
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.10 Number of incomplete establishment calls for mt-Access by O-DU error detection

A.15.10.1 Performance Counter Table

Measurement Name	OR.ESTACALL.IncompMtaccessOduErrDetect
Description	This counter provides the number of the call establishment for mt-Access is interrupted by O-DU Error Detection.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever call establishment for mt-Access is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) by DU Error Detection.
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.IncompMtaccessOduErrDetect
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.11 Number of attempted establishment calls for mo-Signalling

A.15.11.1 Performance Counter Table

Measurement Name	OR.ESTACALL.AcceptMosignal
Description	This counter provides the number of the received RRC: RRC Setup Complete (or RRC: RRC Resume Complete) messages via F1-C after

	received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause):mo-Signalling).
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Setup Complete (or RRC: RRC Resume Complete) message is received via F1-C after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause):mo-Signalling).
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.AcceptMosignal
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2 A.15.12 Number of successful establishment calls for mo-Signalling

3 A.15.12.1 Performance Counter Table

Measurement Name	OR.ESTACALL.SuccessMosignal
Description	This counter provides the number of the following message after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): mo-Signalling): - send NG-AP: Initial Context Setup Response - received NG-AP: UE Context Release Command (for deregistration) before sending NG-AP: Initial Context Setup Response
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever the following event occurs after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): mo-Signalling): - send NG-AP: Initial Context Setup Response - received NG-AP: UE Context Release Command (for deregistration) before sending NG-AP: Initial Context Setup Response
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.SuccessMosignal
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

4

5 A.15.13 Number of incomplete establishment calls for mo-Signalling by 6 protocol error

7 A.15.13.1 Performance Counter Table

Measurement Name	OR.ESTACALL.IncompMosignalProtocolErr
Description	This counter provides the number of the call establishment for mo-Signalling interrupted by protocol error.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever call establishment for mo-Signalling is interrupted between RRC: RRC Setup Request (or

	RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) by protocol error.
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.IncompMosignalProtocolErr
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.14 Number of incomplete establishment calls for mo-Signalling by O-CU internal error

A.15.14.1 Performance Counter Table

Measurement Name	OR.ESTACALL.IncompMosignalOcuInterErr
Description	This counter provides the number of the call establishment for mo-Signalling interrupted by O-CU internal error.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever mo-Signalling configuration is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) by O-CU internal error.
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.IncompMosignalOcuInterErr
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.15 Number of incomplete establishment calls for mo-Signalling by O-DU error detection

A.15.15.1 Performance Counter Table

Measurement Name	OR.ESTACALL.IncompMosignalDuErrDetect
Description	This counter provides the number of the call establishment for mo-Signalling interrupted by O-DU error detection.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever call establishment for mo-Signalling is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) by O-DU error detection.
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.IncompMosignalDuErrDetect
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.16 Number of attempted establishment calls for mo-SMS

A.15.16.1 Performance Counter Table

Measurement Name	OR.ESTACALL.AttemptMossm
Description	This counter provides the number of the received RRC: RRC Setup Complete (or RRC: RRC Resume Complete) messages via F1-C after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): mo-SMS).
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Setup Complete (or RRC: RRC Resume Complete) message is received via F1-C after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): mo-SMS).
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.AttemptMossm
Measurement Object Class	NRCCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.17 Number of successful establishment calls for mo-SMS

A.15.17.1 Performance Counter Table

Measurement Name	OR.ESTACALL.SuccessMossm
Description	This counter provides the number of the following message after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): mo-SMS): - send NG-AP: Initial Context Setup Response - received NG-AP: UE Context Release Command (for deregistration) before sending NG-AP: Initial Context Setup Response
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever the following event occurs after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): mo-SMS): - send NG-AP: Initial Context Setup Response - received NG-AP: UE Context Release Command (for deregistration) before sending NG-AP: Initial Context Setup Response
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.SuccessMossm
Measurement Object Class	NRCCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.18 Number of incomplete establishment calls for mo-SMS by protocol error

A.15.18.1 Performance Counter Table

Measurement Name	OR.ESTACALL.IncompMossmProtocolErr
Description	This counter provides the number of the call establishment for mo-SMS interrupted by protocol error.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever call establishment for mo-SMS is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) by protocol error.
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.IncompMossmProtocolErr
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.19 Number of incomplete establishment calls for mo-SMS by O-CU internal error

A.15.19.1 Performance Counter Table

Measurement Name	OR.ESTACALL.IncompMossmOcuInterErr
Description	This counter provides the number of the call establishment for mo-SMS interrupted by O-CU internal error.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever call establishment for mo-SMS is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) by O-CU internal error.
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.IncompMossmOcuInterErr
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.20 Number of incomplete establishment calls for mo-SMS by O-DU error detection

A.15.20.1 Performance Counter Table

Measurement Name	OR.ESTACALL.IncompMossmOduErrDetect
Description	This counter provides the number of the call establishment for mo-SMS interrupted by O-DU error detection.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever call establishment for mo-SMS is interrupted between RRC: RRC Setup Request (or RRC:

	RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) by O-DU error detection.
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.IncompMossmsOduErrDetect
Measurement Object Class	NRCCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.15.21 Number of attempted establishment calls for high priority access

3

A.15.21.1 Performance Counter Table

Measurement Name	OR.ESTACALL.AttemptHighPriAccess
Description	This counter provides the number of the received RRC: RRC Setup Complete messages (or RRC: RRC Resume Complete) via F1-C after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): highPriorityAccess).
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Setup Complete (or RRC: RRC Resume Complete) message is received via F1-C after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): highPriorityAccess).
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.AttemptHighPriAccess
Measurement Object Class	NRCCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

4

5

A.15.22 Number of successful establishment calls for high priority access

6

A.15.22.1 Performance Counter Table

Measurement Name	OR.ESTACALL.SuccessHighPriAccess
Description	This counter provides the number of the following message after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): highPriorityAccess): - NG-AP: Initial Context Setup Response is transmitted. - NG-AP: UE Context Release Command (for detach) is received before NG-AP: Initial Context Setup Response is transmitted.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever the following event occurs after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): highPriorityAccess): - send NG-AP: Initial Context Setup Response - NG-AP: UE Context Release Command (for detach) is received before NG-AP: Initial Context Setup Response is transmitted.
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.SuccessHighPriAccess
Measurement Object Class	NRCCellCU
Switching Technology	Packet Switched

Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.23 Number of incomplete establishment calls for high priority access by protocol error

A.15.23.1 Performance Counter Table

Measurement Name	OR.ESTACALL.IncompHighPriAccessProtocolErr
Description	This counter provides the number of the call establishment for high priority access interrupted by protocol error.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever call establishment for high priority access is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) by protocol error (Call acceptance condition of CU/DU is not met).
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.IncompHighPriAccessProtocolErr
Measurement Object Class	NRCe11CU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.24 Number of incomplete establishment calls for high priority access by O-CU internal error

A.15.24.1 Performance Counter Table

Measurement Name	OR.ESTACALL.IncompHighPriAccessOcuInterErr
Description	This counter provides the number of the call establishment for high priority access interrupted by O-CU internal error.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever call establishment for high priority access is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) by O-CU internal error (Call acceptance condition of CU/DU is not met).
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.IncompHighPriAccessOcuInterErr
Measurement Object Class	NRCe11CU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.25 Number of incomplete establishment calls for high priority access by O-DU error detection

A.15.25.1 Performance Counter Table

Measurement Name	OR.ESTACALL.IncompHighPriAccessOduErrDetect
Description	This counter provides the number of the call establishment for high priority access interrupted by O-DU error detection.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever call establishment for high priority access is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) by O-DU error detection (Call acceptance condition of CU/DU is not met).
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.IncompHighPriAccessOduErrDetect
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.26 Number of attempted establishment calls for emergency

A.15.26.1 Performance Counter Table

Measurement Name	OR.ESTACALL.AttemptEmergency
Description	This counter provides the number of the received RRC: RRC Setup Complete (or RRC: RRC Resume Complete) messages via F1-C after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): emergency).
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Setup Complete (or RRC: RRC Resume Complete) message is received via F1-C after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): emergency).
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.AttemptEmergency
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.27 Number of successful establishment calls for emergency

A.15.27.1 Performance Counter Table

Measurement Name	OR.ESTACALL.SuccessEmergency
Description	This counter provides the number of the following message after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): emergency): - send NG-AP: Initial Context Setup Response

	- NG-AP: UE Context Release Command (for detach) is received before NG-AP: Initial Context Setup Response is transmitted.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever the following event occurs after received RRC: RRC Setup Request (or RRC: RRC Resume Request) (Establishment Cause (or Resume Cause): emergency): - NG-AP: Initial Context Setup Response is transmitted. - NG-AP: UE Context Release Command (for detach) is received before NG-AP: Initial Context Setup Response is transmitted.
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.SuccessEmergency
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.28 Number of incomplete establishment calls for emergency by protocol error

A.15.28.1 Performance Counter Table

Measurement Name	OR.ESTACALL.IncompEmergencyProtocolErr
Description	This counter provides the number of the call establishment for emergency interrupted by protocol error.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever call establishment for emergency is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) by protocol error (Call acceptance condition of CU/DU is not met).
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.IncompEmergencyProtocolErr
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.29 Number of incomplete establishment calls for emergency by O-CU internal error

A.15.29.1 Performance Counter Table

Measurement Name	OR.ESTACALL.IncompEmergencyOcuInterErr
Description	This counter provides the number of the call establishment for emergency interrupted by O-CU internal error.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever call establishment for emergency is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) by O-CU internal error (Call acceptance condition of CU/DU is not met).
Measurement Result	Integer number (U32)

Measurement Type	OR.ESTACALL.IncompEmergencyOcuInterErr
Measurement Object Class	NRCCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.30 Number of incomplete establishment calls for emergency by O-DU error detection

A.15.30.1 Performance Counter Table

Measurement Name	OR.ESTACALL.IncompEmergencyOduErrDetect
Description	This counter provides the number of the call establishment for emergency interrupted by O-DU error detection.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever call establishment for emergency is interrupted between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) by O-DU error detection (Call acceptance condition of CU/DU is not met).
Measurement Result	Integer number (U32)
Measurement Type	OR.ESTACALL.IncompEmergencyOduErrDetect
Measurement Object Class	NRCCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.15.31 Discarded paging records

A.15.31.1 Performance Counter Table

Measurement Name	PAG.DiscardedNbrCnInitiated
Description	Refer to 3GPP TS 28.552 [8] clause 5.1.1.27.4
Collection Method	Refer to 3GPP TS 28.552 [8] clause 5.1.1.27.4
Condition	Refer to 3GPP TS 28.552 [8] clause 5.1.1.27.4
Measurement Result	Refer to 3GPP TS 28.552 [8] clause 5.1.1.27.4
Measurement Type	Refer to 3GPP TS 28.552 [8] clause 5.1.1.27.4
Measurement Object Class	Refer to 3GPP TS 28.552 [8] clause 5.1.1.27.4
Switching Technology	Refer to 3GPP TS 28.552 [8] clause 5.1.1.27.4
Generation	Refer to 3GPP TS 28.552 [8] clause 5.1.1.27.4
Purpose	Refer to 3GPP TS 28.552 [8] clause 5.1.1.27.4

A.16 Monitoring of PDU session connection for SA

A.16.1 Accepted PDU Session Resource Configuration

A.16.1.1 Performance Counter Table

Measurement Name	OR.PDUSESSION.AcceptConfig
Description	This counter provides the number of the accepted PDU session resource configuration per CU count group.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever gNB received the following message per CU count group. Note: multiple 5QIs can be set per PDU session, each 5QI is counted per QoS flow. - "NG-AP: Initial Context Setup Request" is received - "NG-AP: PDU Session Resource Setup Request" - "NG-AP: PDU Session Resource Modify Request"
Measurement Result	Integer number (U32)
Measurement Type	OR.PDUSESSION.AcceptConfig.Cucountgroup where Cucountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.16.2 Completed PDU Session Resource Configuration

A.16.2.1 Performance Counter Table

Measurement Name	OR.PDUSESSION.CompConfig
Description	This counter provides the number of the completed PDU session resource configuration per CU count group.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever gNB transmitted the following message per CU count group. Note: multiple 5QIs can be set per PDU session, each 5QI is counted per QoS flow. - "NG-AP: Initial Context Setup Response" - "NG-AP: PDU Session Resource Setup Response" - "NG-AP: PDU Session Resource Modify Response"
Measurement Result	Integer number (U32)
Measurement Type	OR.PDUSESSION.CompConfig.Cucountgroup where Cucountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.16.3 Interrupted PDU Session Resource configuration by Protocol Error

A.16.3.1 Performance Counter Table

Measurement Name	OR.PDUSESSION.InterruptConfigProcolErr
Description	This counter provides the number of the PDU session resource configuration interrupted by protocol error per CU count group.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever the PDU session resource configuration interrupted in the following interval by protocol error per CU count group. Note: multiple 5QIs can be set per PDU session, each 5QI is counted per QoS flow. - between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) - between Xn-AP: Handover Request to NG-AP: Path Switch Acknowledge (Inter gNB-CU HO w Xn) - between NG-AP: Handover Request to NG-AP: Path Switch Acknowledge (Inter gNB-CU HO w/o Xn) - between NG-AP: Handover Request to NG-AP: Handover Notify (Inter RAT HO to NR) - between NG-AP: PDU Session Resource Setup Request to NG-AP: PDU Session Resource Setup Response (PDU Session Establishment) - between NG-AP: PDU Session Resource Modify Request to NG-AP: PDU Session Resource Modify Response (PDU Session Modification) - between NG-AP: PDU Session Resource Release Request to NG-AP: PDU Session Resource Release Response (PDU Session Release)
Measurement Result	Integer number (U32)
Measurement Type	OR.PDUSESSION.InterruptConfigProcolErr.Cucountgroup where <i>Cucountgroup</i> identifies an instance of CuCountGroup IOC.
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.16.4 Interrupted PDU Session Resource configuration by O-DU Error Detection

A.16.4.1 Performance Counter Table

Measurement Name	OR.PDUSESSION.InterruptConfigOduErrDetect
Description	This counter provides the number of the PDU session resource configuration interrupted by O-DU error detection per CU count group.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever the PDU session resource configuration interrupted in the following interval by O-DU error detection per CU count group. Note: multiple 5QIs can be set per PDU session, each 5QI is counted per QoS flow.

	- between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) - between Xn-AP: Handover Request to NG-AP: Path Switch Acknowledge (Inter gNB-CU HO w Xn) - between NG-AP: Handover Request to NG-AP: Path Switch Acknowledge (Inter gNB-CU HO w/o Xn) - between NG-AP: Handover Request to NG-AP: Handover Notify (Inter RAT HO to NR) - between NG-AP: PDU Session Resource Setup Request to NG-AP: PDU Session Resource Setup Response (PDU Session Establishment) - between NG-AP: PDU Session Resource Modify Request to NG-AP: PDU Session Resource Modify Response (PDU Session Modification) - between NG-AP: PDU Session Resource Release Request to NG-AP: PDU Session Resource Release Response (PDU Session Release)
Measurement Result	Integer number (U32)
Measurement Type	OR.PDUSESSION.InterruptConfigOduErrDetect.Cucountgroup where Cucountgroup identifies an instance of CuCountGroup IOC.
Measurement Object Class	NRCelICU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

A.16.5 Interrupted PDU Session Resource configuration by O-CU internal error

3

A.16.5.1 Performance Counter Table

4

Measurement Name	OR.PDUSESSION.InterruptConfigOcuInterErr
Description	This counter provides the number of the PDU session resource configuration interrupted by O-CU internal error per CU count group.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever the PDU session resource configuration interrupted in the following interval by O-CU internal error per CU count group. Note: multiple 5QIs can be set per PDU session, each 5QI is counted per QoS flow. - between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) - between Xn-AP: Handover Request to NG-AP: Path Switch Acknowledge (Inter gNB-CU HO w Xn) - between NG-AP: Handover Request to NG-AP: Path Switch Acknowledge (Inter gNB-CU HO w/o Xn) - between NG-AP: Handover Request to NG-AP: Handover Notify (Inter RAT HO to NR) - between NG-AP: PDU Session Resource Setup Request to NG-AP: PDU Session Resource Setup Response (PDU Session Establishment) - between NG-AP: PDU Session Resource Modify Request to NG-AP: PDU Session Resource Modify Response (PDU Session Modification)

	- between NG-AP: PDU Session Resource Release Request to NG-AP: PDU Session Resource Release Response (PDU Session Release)
Measurement Result	Integer number (U32)
Measurement Type	OR.PDUSESSION.InterruptConfigOcuInterErr. <i>Cucountgroup</i> where <i>Cucountgroup</i> identifies an instance of CuCountGroup IOC.
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2 A.16.6 Accepted PDU Session Resource Configuration for emergency and 3 high priority access

4 A.16.6.1 Performance Counter Table

Measurement Name	OR.PDUSESSION.AcceptConfigEmergencyHighPriAccess
Description	This counter provides the number of the accepted PDU session resource configuration for emergency and high priority access per CU count group.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever gNB received the following message after received RRC: RRC Setup Request (Establishment Cause: emergency or high priority access) per CU count group. Note: multiple 5QIs can be set per PDU session, each 5QI is counted per QoS flow. - received NG-AP: Initial Context Setup Request - received NG-AP: PDU Session Resource Setup Request - received NG-AP: PDU Session Resource Modify Request
Measurement Result	Integer number (U32)
Measurement Type	OR.PDUSESSION.AcceptConfigEmergencyHighPriAccess. <i>Cucountgroup</i> where <i>Cucountgroup</i> identifies an instance of CuCountGroup IOC.
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

5

6 A.16.7 Completed PDU Session Resource Configuration for emergency and 7 high priority access

8 A.16.7.1 Performance Counter Table

Measurement Name	OR.PDUSESSION.CompConfigEmergencyHighPriAccess
Description	This counter provides the number of the completed PDU session resource configuration for emergency and high priority access per CU count group.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever gNB transmitted the following message after received RRC: RRC Setup Request (Establishment Cause: emergency or high priority access) per CU count group.

	Note: multiple 5QIs can be set per PDU session, each 5QI is counted per QoS flow. - send NG-AP: Initial Context Setup Response - send NG-AP: PDU Session Resource Setup Response - send NG-AP: PDU Session Resource Modify Response
Measurement Result	Integer number (U32)
Measurement Type	OR.PDUSESSION.CompConfigEmergencyHighPriAccess.Cucountgroup where <i>Cucountgroup</i> identifies an instance of CuCountGroup IOC.
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

3

A.16.8 Interrupted PDU Session Resource configuration for emergency and high priority access by Protocol Error

4

A.16.8.1 Performance Counter Table

Measurement Name	OR.PDUSESSION.InterruptConfigProcolErrEmergencyHighPriAccess
Description	This counter provides the number of the PDU session resource configuration for emergency and high priority access interrupted by protocol error per CU count group.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever the PDU session resource configuration for emergency and high priority access interrupted in the following interval by protocol error per CU count group. Note: multiple 5QIs can be set per PDU session, each 5QI is counted per QoS flow. - between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) - between Xn-AP: Handover Request to NG-AP: Path Switch Acknowledge (Inter gNB-CU HO w Xn) - between NG-AP: Handover Request to NG-AP: Path Switch Acknowledge (Inter gNB-CU HO w/o Xn) - between NG-AP: Handover Request to NG-AP: Handover Notify (Inter RAT HO to NR) - between NG-AP: PDU Session Resource Setup Request to NG-AP: PDU Session Resource Setup Response (PDU Session Establishment) - between NG-AP: PDU Session Resource Modify Request to NG-AP: PDU Session Resource Modify Response (PDU Session Modification) - between NG-AP: PDU Session Resource Release Request to NG-AP: PDU Session Resource Release Response (PDU Session Release)
Measurement Result	Integer number (U32)
Measurement Type	OR.PDUSESSION.InterruptConfigProcolErrEmergencyHighPriAccess.Cucountgroup where <i>Cucountgroup</i> identifies an instance of CuCountGroup IOC.
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.16.9 Interrupted PDU Session Resource configuration for emergency and high priority access by O-DU Error Detection

A.16.9.1 Performance Counter Table

Measurement Name	OR.PDUSESSION.InterruptConfigOduErrDetectEmergencyHighPriAccess
Description	This counter provides the number of the PDU session resource configuration for emergency and high priority access interrupted by O-DU error detection per CU count group.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever the PDU session resource configuration for emergency and high priority access interrupted in the following interval by O-DU error detection per CU count group. Note: multiple 5QIs can be set per PDU session, each 5QI is counted per QoS flow. - between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) - between Xn-AP: Handover Request to NG-AP: Path Switch Acknowledge (Inter gNB-CU HO w Xn) - between NG-AP: Handover Request to NG-AP: Path Switch Acknowledge (Inter gNB-CU HO w/o Xn) - between NG-AP: Handover Request to NG-AP: Handover Notify (Inter RAT HO to NR) - between NG-AP: PDU Session Resource Setup Request to NG-AP: PDU Session Resource Setup Response (PDU Session Establishment) - between NG-AP: PDU Session Resource Modify Request to NG-AP: PDU Session Resource Modify Response (PDU Session Modification) - between NG-AP: PDU Session Resource Release Request to NG-AP: PDU Session Resource Release Response (PDU Session Release)
Measurement Result	Integer number (U32)
Measurement Type	OR.PDUSESSION.InterruptConfigOduErrDetectEmergencyHighPriAccess.Cucountgroup where <i>Cucountgroup</i> identifies an instance of CuCountGroup IOC.
Measurement Object Class	NRCellicu
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.16.10 Interrupted PDU Session Resource configuration for emergency and high priority access by O-CU internal Error

A.16.10.1 Performance Counter Table

Measurement Name	OR.PDUSESSION.InterruptConfigOcuInterErrEmergencyHighPriAccess
------------------	--

Description	This counter provides the number of the PDU session resource configuration for emergency and high priority access interrupted by O-CU internal error per CU count group.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever the PDU session resource configuration for emergency and high priority access interrupted in the following interval by O-CU internal error per CU count group. Note: multiple 5QIs can be set per PDU session, each 5QI is counted per QoS flow. - between RRC: RRC Setup Request (or RRC: RRC Resume Request) to NG-AP: Initial Context Setup Response (or RRC: RRC Resume Complete) - between Xn-AP: Handover Request to NG-AP: Path Switch Acknowledge (Inter gNB-CU HO w Xn) - between NG-AP: Handover Request to NG-AP: Path Switch Acknowledge (Inter gNB-CU HO w/o Xn) - between NG-AP: Handover Request to NG-AP: Handover Notify (Inter RAT HO to NR) - between NG-AP: PDU Session Resource Setup Request to NG-AP: PDU Session Resource Setup Response (PDU Session Establishment) - between NG-AP: PDU Session Resource Modify Request to NG-AP: PDU Session Resource Modify Response (PDU Session Modification) - between NG-AP: PDU Session Resource Release Request to NG-AP: PDU Session Resource Release Response (PDU Session Release)
Measurement Result	Integer number (U32)
Measurement Type	OR.PDUSESSION.InterruptConfigOcuInterErrEmergencyHighPriAccess.Cucountgroup where <i>Cucountgroup</i> identifies an instance of <i>CuCountGroup</i> IOC.
Measurement Object Class	NRCelICU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2 A.17 Monitoring of mobility for SA

3 A.17.1 Accepted Handover

4 A.17.1.1 Performance Counter Table

Measurement Name	OR.MOBILITY.AcceptHo
Description	This counter provides the number of the accepted Intra RAT handover.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever source gNB decides to perform Intra RAT HO procedure by the received (transmitted) following messages: - Intra-Cell HO: transmitted "RRC: RRC Reconfiguration" - HO procedure other than those above: received "RRC: Measurement report"
Measurement Result	Integer number (U32)
Measurement Type	OR.MOBILITY.AcceptHo
Measurement Object Class	NRCelICU

Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.17.2 Completed Handover

A.17.2.1 Performance Counter Table

Measurement Name	OR.MOBILITY.CompHo
Description	This counter provides the number of the completed Intra RAT handover.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever source gNB detects successful Intra RAT handover procedure triggered by source gNB by the received following messages: - Intra-gNB HO: received "RRC: RRC Reconfiguration Complete" - HO procedure other than those above: received "XnAP:UE CONTEXT RELEASE" from the target gNB at Xn HO or "NGAP:UE CONTEXT RELEASE COMMAND" from AMF at NG HO
Measurement Result	Integer number (U32)
Measurement Type	OR.MOBILITY.CompHo
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.17.3 Accepted Inter RAT Handover to LTE

A.17.3.1 Performance Counter Table

Measurement Name	OR.MOBILITY.AcceptHotoLte
Description	This counter provides the number of the accepted Inter RAT Handover to LTE.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever source gNB decides to perform Inter RAT HO to LTE procedure by the received following messages: - received RRC: Measurement report (Inter RAT HO to LTE)
Measurement Result	Integer number (U32)
Measurement Type	OR.MOBILITY.AcceptHotoLte
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.17.4 Completed Inter RAT Handover to LTE

A.17.4.1 Performance Counter Table

Measurement Name	OR.MOBILITY.CompHotoLte
Description	This counter provides the number of the completed Inter RAT Handover to LTE.
Collection Method	CC (Cumulative Counter)

Condition	Measurement subcounter is incremented by 1 whenever source gNB detects successful handover procedure triggered by source gNB by the received following messages: - received NG-AP: UE Context Release Command (Inter RAT HO to LTE)
Measurement Result	Integer number (U32)
Measurement Type	OR.MOBILITY.CompHotoLte
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2 A.17.5 Accepted EPS Fallback

3 A.17.5.1 Performance Counter Table

Measurement Name	OR.MOBILITY.AcceptEpsFallback
Description	This counter provides the number of the accepted EPS Fallback.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever PCell decides to perform EPS Fallback procedure by the received following messages: - received NG-AP: PDU Session Resource Modify Request (EPS Fallback)
Measurement Result	Integer number (U32)
Measurement Type	OR.MOBILITY.AcceptEpsFallback
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

4

5 A.17.6 Completed EPS Fallback

6 A.17.6.1 Performance Counter Table

Measurement Name	OR.MOBILITY.CompEpsFallback
Description	This counter provides the number of the completed EPS Fallback.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever PCell detects successful EPS Fallback procedure triggered by the received following messages: - received NG-AP: UE Context Release Command (EPS Fallback)
Measurement Result	Integer number (U32)
Measurement Type	OR.MOBILITY.CompEpsFallback
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

7

A.17.7 Normal interrupted EPS Fallback

A.17.7.1 Performance Counter Table

Measurement Name	OR.MOBILITY.NormalInterruptedEpsFallback
Description	This counter provides the number of the completed EPS Fallback.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever PCell detects interrupted EPS Fallback procedure triggered by the received following messages: - received NG-AP: UE Context Release Command (Cause:Deregister) during EPS Fallback procedure
Measurement Result	Integer number (U32)
Measurement Type	OR.MOBILITY.NormalInterruptedEpsFallback
Measurement Object Class	NRCeLU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.17.8 Accepted Handover during Voice call

A.17.8.1 Performance Counter Table

Measurement Name	OR.MOBILITY.AcceptHoVoice
Description	This counter provides the number of the accepted Intra RAT handover during voice call.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever source gNB decides to perform Intra RAT HO procedure by the received (transmitted) following messages during voice call: - transmitted RRC: RRC Reconfiguration (Intra-cell HO) - HO procedure other than those above: received RRC: Measurement report
Measurement Result	Integer number (U32)
Measurement Type	OR.MOBILITY.AcceptHoVoice
Measurement Object Class	NRCeLU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.17.9 Completed Handover during Voice call

A.17.9.1 Performance Counter Table

Measurement Name	OR.MOBILITY.CompHoVoice
Description	This counter provides the number of the completed Intra RAT handover during voice call.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever source gNB detects successful Intra RAT handover procedure triggered by source gNB by the received following messages during voice call: - received RRC: RRC Reconfiguration Complete (Intra gNB-CU Inter gNB-DU HO, Intra gNB-DU Inter-cell HO) - HO procedure other than those above: received Xn-AP: UE

	CONTEXT RELEASE (Inter gNB-CU HO w/ Xn), NG-AP: UE CONTEXT RELEASE COMMAND (Inter gNB-CU HO w/o Xn)
Measurement Result	Integer number (U32)
Measurement Type	OR.MOBILITY.CompHoVoice
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.17.10 Accepted Inter RAT Handover to LTE during Voice call

A.17.10.1 Performance Counter Table

Measurement Name	OR.MOBILITY.AcceptHotoLteVoice
Description	This counter provides the number of the accepted Inter RAT handover to LTE during voice call.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever source gNB decides to perform Inter RAT HO to LTE procedure by the received following messages during voice call: - received RRC: Measurement report that triggers Inter RAT HO to LTE
Measurement Result	Integer number (U32)
Measurement Type	OR.MOBILITY.AcceptHotoLteVoice
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.17.11 Completed Inter RAT Handover to LTE during Voice call

A.17.11.1 Performance Counter Table

Measurement Name	OR.MOBILITY.CompHotoLteVoice
Description	This counter provides the number of the completed Inter RAT handover during voice call.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever source gNB detects successful Inter RAT handover procedure triggered by source gNB by the received following messages during voice call: - received NG-AP: UE CONTEXT RELEASE COMMAND from the AMF at Inter RAT HO to LTE
Measurement Result	Integer number (U32)
Measurement Type	OR.MOBILITY.CompHotoLteVoice
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.18 Monitoring of RRC re-establishment for SA

A.18.1 Number of initial RRC re-establishment requests when UE context can be retrieved

A.18.1.1 Performance Counter Table

Measurement Name	OR.REEST.IniReqUecontext
Description	This counter provides the number of the received RRC: RRC Reestablishment Request messages via F1-C which is in the first time for each ReestabUE-Identity when UE context can be retrieved.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Reestablishment Request message is received via F1-C when UE context can be retrieved, if RRC: RRC Reestablishment Request of the same ReestabUE-Identity is not received for a certain period of time from the reception timing.
Measurement Result	Integer number (U32)
Measurement Type	OR.REEST.IniReqUecontext
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.18.2 Number of RRC re-establishment request completions for initial RRC re-establishment requests when UE context can be retrieved

A.18.2.1 Performance Counter Table

Measurement Name	OR.REEST.IniReqCompUecontext
Description	This counter provides the number of the received RRC: RRC Reconfiguration Complete messages via F1-C which corresponds to the initial RRC: RRC Reestablishment Request when UE context can be retrieved.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Reconfiguration Complete message which corresponds to the initial RRC: RRC Reestablishment Request is received via F1-C when UE context can be retrieved.
Measurement Result	Integer number (U32)
Measurement Type	OR.REEST.IniReqCompUecontext
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.18.3 Number of retransmission RRC re-establishment requests when UE context can be retrieved

A.18.3.1 Performance Counter Table

Measurement Name	OR.REEST.RetaransReqUecontext
Description	This counter provides the number of the received RRC: RRC Reestablishment Request messages via F1-C which is the retransmission for each ReestabUE-Identity when UE context can be retrieved.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Reestablishment Request message is received via F1-C when UE context can be retrieved, if RRC: RRC Reestablishment Request message of the same ReestabUE-Identity is received for a certain period of time from the reception timing.
Measurement Result	Integer number (U32)
Measurement Type	OR.REEST.RetaransReqUecontext
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.18.4 Number of RRC re-establishment request completions for retransmission RRC re-establishment request when UE context can be retrieved

A.18.4.1 Performance Counter Table

Measurement Name	OR.REEST.RetaransReqCompUecontext
Description	This counter provides the number of the received RRC: RRC Reconfiguration Complete messages via F1-C which corresponds to the retransmission RRC: RRC Reestablishment Request when UE context can be retrieved.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Reconfiguration Complete message which corresponds to the retransmission RRC: RRC Reestablishment Request is received via F1-C when UE context can be retrieved.
Measurement Result	Integer number (U32)
Measurement Type	OR.REEST.RetaransReqCompUecontext
Measurement Object Class	NRCelLCUs
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.18.5 Number of initial RRC re-establishment requests when UE context cannot be retrieved

A.18.5.1 Performance Counter Table

Measurement Name	OR.REEST.IniReqNotUecontext
------------------	-----------------------------

Description	This counter provides the number of the received RRC: RRC Reestablishment Request messages via F1-C which is in the first time for each ReestabUE-Identity when UE context cannot be retrieved.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Reestablishment Request message is received via F1-C when UE context cannot be retrieved, if RRC: RRC Reestablishment Request of the same ReestabUE-Identity is not received for a certain period of time from the reception timing.
Measurement Result	Integer number (U32)
Measurement Type	OR.REEST.IniReqNotUecontext
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

A.18.6 Number of RRC re-establishment request completions for initial RRC re-establishment requests when UE context cannot be retrieved

3

A.18.6.1 Performance Counter Table

4

Measurement Name	OR.REEST.IniReqCompNotUecontext
Description	This counter provides the number of the received RRC: RRC Reconfiguration Complete messages via F1-C which corresponds to the initial RRC: RRC Reestablishment Request when UE context cannot be retrieved.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Setup Complete message which corresponds to the initial RRC: RRC Reestablishment Request is received via F1-C when UE context cannot be retrieved.
Measurement Result	Integer number (U32)
Measurement Type	OR.REEST.IniReqCompNotUecontext
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

5

A.18.7 Number of retransmission RRC re-establishment requests when UE context cannot be retrieved

7

A.18.7.1 Performance Counter Table

8

Measurement Name	OR.REEST.RetReqNotUecontext
Description	This counter provides the number of the received RRC: RRC Reestablishment Request messages via F1-C which is the retransmission for each ReestabUE-Identity when UE context cannot be retrieved.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Reestablishment Request message is received via F1-C when UE context cannot be retrieved, if RRC: RRC Reestablishment Request message of the same ReestabUE-Identity is received for a certain period of time from the reception timing.

Measurement Result	Integer number (U32)
Measurement Type	OR.REEST.RetransReqNotUecontext
Measurement Object Class	NRCeU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.18.8 Number of RRC re-establishment request completions for retransmission RRC re-establishment requests when UE context cannot be retrieved

A.18.8.1 Performance Counter Table

Measurement Name	OR.REEST.RetReqCompNotUecontext
Description	This counter provides the number of the received RRC: RRC Reconfiguration Complete messages via F1-C which corresponds to the retransmission RRC: RRC Reestablishment Request when UE context cannot be retrieved.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Reconfiguration Complete message which corresponds to the retransmission RRC: RRC Reestablishment Request is received via F1-C when UE context cannot be retrieved.
Measurement Result	Integer number (U32)
Measurement Type	OR.REEST.RetransReqCompNotUecontext
Measurement Object Class	NRCeUs
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.18.9 Number of initial RRC re-establishment requests during Voice call when UE context can be retrieved

A.18.9.1 Performance Counter Table

Measurement Name	OR.REEST.IniReqUecontextVoice
Description	This counter provides the number of the received RRC: RRC Reestablishment Request messages via F1-C which is in the first time for each ReestabUE-Identity during voice call when UE context can be retrieved.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Reestablishment Request message is received via F1-C during voice call when UE context can be retrieved, if RRC: RRC Reestablishment Request of the same ReestabUE-Identity is not received for a certain period of time from the reception timing.
Measurement Result	Integer number (U32)
Measurement Type	OR.REEST.IniReqUecontextVoice
Measurement Object Class	NRCeU
Switching Technology	Packet Switched
Generation	5GS

Purpose	Network Operator's Traffic Engineering Community
---------	--

A.18.10 Number of RRC re-establishment request completions for initial RRC re-establishment requests during Voice call when UE context can be retrieved

A.18.10.1 Performance Counter Table

Measurement Name	OR.REEST.IniReqCompUecontextVoice
Description	This counter provides the number of the received RRC: RRC Reconfiguration Complete messages via F1-C which corresponds to the initial RRC: RRC Reestablishment Request during voice call when UE context can be retrieved.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Reconfiguration Complete message which corresponds to the initial RRC: RRC Reestablishment Request is received via F1-C during voice call when UE context can be retrieved.
Measurement Result	Integer number (U32)
Measurement Type	OR.REEST.IniReqCompUecontextVoice
Measurement Object Class	NRCeLLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.18.11 Number of retransmission RRC re-establishment requests during Voice call when UE context can be retrieved

A.18.11.1 Performance Counter Table

Measurement Name	OR.REEST.RetransReqUecontextVoice
Description	This counter provides the number of the received RRC: RRC Reestablishment Request messages via F1-C which is the retransmission for each ReestabUE-Identity during voice call when UE context can be retrieved.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Reestablishment Request message is received via F1-C during voice call when UE context can be retrieved, if RRC: RRC Reestablishment Request message of the same ReestabUE-Identity is received for a certain period of time from the reception timing.
Measurement Result	Integer number (U32)
Measurement Type	OR.REEST.RetransReqUecontextVoice
Measurement Object Class	NRCeLLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.18.12 Number of RRC re-establishment request completions for retransmission RRC re-establishment request during Voice call when UE context can be retrieved

A.18.12.1 Performance Counter Table

Measurement Name	OR.REEST.RetransReqCompUecontextVoice
Description	This counter provides the number of the received RRC: RRC Reconfiguration Complete messages via F1-C which corresponds to the retransmission RRC: RRC Reestablishment Request during voice call when UE context can be retrieved.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever RRC: RRC Reconfiguration Complete message which corresponds to the retransmission RRC: RRC Reestablishment Request is received via F1-C during voice call when UE context can be retrieved.
Measurement Result	Integer number (U32)
Measurement Type	OR.REEST.RetransReqCompUecontextVoice
Measurement Object Class	NRCeLLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.19 Monitoring of connection status for SA

A.19.1 Normally Released Calls

A.19.1.1 Performance Counter Table

Measurement Name	OR.CONNECTSTATUS.NormalRelCalls
Description	This counter provides the number of the normally released calls.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the number of the normally released calls. Normal released calls is as follows: - received NG-AP: UE Context Release Command (Cause: Normal release or Deregister) - the expiry of a UE inactivity timer, the gNB suspend the RRC connection
Measurement Result	Integer number (U32)
Measurement Type	OR.CONNECTSTATUS.NormalRelCalls
Measurement Object Class	NRCeLLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.19.2 Abnormally Released Calls

A.19.2.1 Performance Counter Table

Measurement Name	OR.CONNECTSTATUS.AbnormalRelCalls
Description	This counter provides the number of the abnormally released calls.
Collection Method	CC (Cumulative Counter)

Condition	Measurement subcounter is incremented by the number of the abnormally released calls. - Trigger timing: when gNB release the UE Context except for the Normally Released calls (A.19.1) and Another Cell HO UE Contexts (A.19.3).
Measurement Result	Integer number (U32)
Measurement Type	OR.CONNECTSTATUS.AbnormalRelCalls
Measurement Object Class	NRCelLCU NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.19.3 Another Cell HO UE Contexts

3

A.19.3.1 Performance Counter Table

Measurement Name	OR.CONNECTSTATUS.AnotherCellHoUes
Description	This counter provides the number of the UE Contexts which has transferred to another cell.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the number of the UE Contexts which has transferred to another cell. Another Cell released call is as follows: - received Xn-AP: UE Context Release (Inter gNB-CU HO w/ Xn, RRC Connection Re-establishment (Inter gNB-CU)) - received NG-AP: UE Context Release Command (Inter gNB-CU HO w/o Xn, Inter-RAT HO to LTE, EPS fallback) - received RRC: RRC Reconfiguration Complete (Intra gNB-DU Inter-cell HO, RRC Connection Re-establishment (Intra gNB-DU Inter-cell)) - received F1-AP: UE Context Release Command (Intra gNB-CU Inter gNB-DU HO, RRC Connection Re-establishment (Intra gNB-CU Inter gNB-DU))
Measurement Result	Integer number (U32)
Measurement Type	OR.CONNECTSTATUS.AnotherCellHoUes
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

4

5

A.19.4 Normally Released Calls during voice call

6

A.19.4.1 Performance Counter Table

Measurement Name	OR.CONNECTSTATUS.NormalRelCallsVoice
Description	This counter provides the number of the normally released calls during voice call.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the number of the normally released calls during voice call. Normal released calls are as follows: - received NG-AP: UE Context Release Command (Cause: Normal release or Deregister)

	- the expiry of a UE inactivity timer, the gNB suspend the RRC connection
Measurement Result	Integer number (U32)
Measurement Type	OR.CONNECTSTATUS.NormalRelCallsVoice
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.19.5 Abnormally Released Calls during voice call

3

A.19.5.1 Performance Counter Table

Measurement Name	OR.CONNECTSTATUS.AbnormalRelCallsVoice
Description	This counter provides the number of the abnormally released calls during voice call.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the number of the abnormally released calls during voice call. - Trigger timing: when gNB release the UE Context except for the Normally Released calls (A.19.4) and Another Cell HO UE contexts (A.19.6) during voice call.
Measurement Result	Integer number (U32)
Measurement Type	OR.CONNECTSTATUS.AbnormalRelCallsVoice
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

4

5

A.19.6 Another Cell HO UE Contexts during voice call

6

A.19.6.1 Performance Counter Table

Measurement Name	OR.CONNECTSTATUS.AnotherCellHoUesVoice
Description	This counter provides the number of the UE Contexts which has transferred to another cell during voice call.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by the number of the UE Contexts which has transferred to another cell during voice call. Another Cell released calls are as follows: - received Xn-AP: UE Context Release (Inter gNB-CU HO w/ Xn, RRC Connection Re-establishment (Inter gNB-CU)) - received NG-AP: UE Context Release Command (Inter gNB-CU HO w/o Xn, Inter-RAT HO to LTE, EPS fallback) - received RRC: RRC Reconfiguration Complete (Intra gNB-DU Inter-cell HO, RRC Connection Re-establishment (Intra gNB-DU Inter-cell)) - received F1-AP: UE Context Release Command (Intra gNB-CU Inter gNB-DU HO, RRC Connection Re-establishment (Intra gNB-CU Inter gNB-DU))
Measurement Result	Integer number (U32)
Measurement Type	OR.CONNECTSTATUS.AnotherCellHoUesVoice
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched

Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.20 Monitoring of procedure for NR-DC

A.20.1 Number of S-NG-RAN Node Addition procedure attempted for each Cell

A.20.1.1 Performance Counter Table

Measurement Name	OR.NRDCPROCEDURE.SnAddAttemptCell
Description	This counter provides the number of the transmitted Xn-AP: S-NODE ADDITION REQUEST ACKNOWLEDGE message.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever Xn-AP: S-NODE ADDITION REQUEST ACKNOWLEDGE message is transmitted as PSCell.
Measurement Result	Integer number (U32)
Measurement Type	OR.NRDCPROCEDURE.SnAddAttemptCell
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.20.2 Number of S-NG-RAN Node Addition procedure successfully for each Cell

A.20.2.1 Performance Counter Table

Measurement Name	OR.NRDCPROCEDURE.SnAddSuccessCell
Description	This counter provides the number of the completed S-NG-RAN Node addition Procedure.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever S-NG-RAN Node addition procedure is completed as PSCell (i.e. Xn-AP: S-NODE RECONFIGURATION COMPLETE message (SN addition complete) is received and RACH processing with UE Context is completed).
Measurement Result	Integer number (U32)
Measurement Type	OR.NRDCPROCEDURE.SnAddSuccessCell
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.20.3 Number of S-NG-RAN Node Addition procedure attempted for each neighbour gNB

A.20.3.1 Performance Counter Table

Measurement Name	OR.NRDCPROCEDURE.SnAddAttemptNeiGnb
------------------	-------------------------------------

Description	This counter provides the number of the transmitted Xn-AP: S-NODE ADDITION REQUEST ACKNOWLEDGE messages. This counter is split into subcounters per neighbour gNB.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever Xn-AP: S-NODE ADDITION REQUEST ACKNOWLEDGE message is transmitted when the gNB configured by the message is group of subcounter. <i>neighbourgNB</i> .
Measurement Result	Integer number (U32)
Measurement Type	OR.NRDCPROCEDURE.SnAddAttemptNeiGnb. <i>neighbourgNB</i> where <i>neighbourgNB</i> is neighbour gNB index: 0: #0 1: #1 ... 63: #63
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.20.4 Number of S-NG-RAN Node Addition procedure successfully for each neighbour gNB

A.20.4.1 Performance Counter Table

Measurement Name	OR.NRDCPROCEDURE.SnAddSuccessNeiGnb
Description	This counter provides the number of the completed S-NG-RAN Node addition Procedure. This counter is split into subcounters per neighbour gNB.
Collection Method	CC (Cumulative Counter)
Condition	Measurement counter is incremented by 1 whenever S-NG-RAN Node addition procedure is completed (i.e. Xn-AP: S-NODE RECONFIGURATION COMPLETE message (SN addition complete) is received and RACH processing with UE Context is completed) when the gNB configured by the message is group of subcounter. <i>neighbourgNB</i> .
Measurement Result	Integer number (U32)
Measurement Type	OR.NRDCPROCEDURE.SnAddSuccessNeiGnb. <i>neighbourgNB</i> where <i>neighbourgNB</i> is neighbour gNB index: 0: #0 1: #1 ... 63: #63
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.20.5 Number of S-NG-RAN Node modification procedure attempted for each Cell

A.20.5.1 Performance Counter Table

Measurement Name	OR.NRDCPROCEDURE.SnModAttemptCell
Description	This counter provides the number of the received Xn-AP: S-NODE MODIFICATION REQUEST messages or transmitted Xn-AP: S-NODE MODIFICATION REQUIRED messages.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever Xn-AP: S-NODE MODIFICATION REQUEST message is received or Xn-AP: S-NODE MODIFICATION REQUIRED message is transmitted as PSCell. Double counting is not performed if the procedure is changed to the MN initiated SN Modification procedure after transmitted Xn-AP: S-NODE MODIFICATION REQUIRED message (e.g. Measurement gap Coordination (SN initiated)). This subcounter will not be incremented if received Xn-AP: S-NODE MODIFICATION REQUEST message which True is set for SCG Configuration Query IE.
Measurement Result	Integer number (U32)
Measurement Type	OR.NRDCPROCEDURE.SnModAttemptCell
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.20.6 Number of S-NG-RAN Node modification procedure successfully for each Cell

A.20.6.1 Performance Counter Table

Measurement Name	OR.NRDCPROCEDURE.SnModSuccessCell
Description	This counter provides the number of the completed S-NG-RAN Node modification Procedure.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever SN Modification procedure is completed as PSCell (i.e. Xn-AP: S-NODE RECONFIGURATION COMPLETE message (SN modification complete) is received or Xn-AP: S-NODE MODIFICATION CONFIRM is received and RACH processing with UE Context is completed)
Measurement Result	Integer number (U32)
Measurement Type	OR.NRDCPROCEDURE.SnModSuccessCell
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.20.7 Number of S-NG-RAN Node modification procedure attempted for each neighbour gNB

A.20.7.1 Performance Counter Table

Measurement Name	OR.NRDCPROCEDURE.SnModAttemptNeiGnb
Description	This counter provides the number of the received Xn-AP: S-NODE MODIFICATION REQUEST messages or transmitted Xn-AP: S-NODE MODIFICATION REQUIRED messages. This counter is split into subcounters per neighbour gNB.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever Xn-AP: S-NODE MODIFICATION REQUEST message is received or Xn-AP: S-NODE MODIFICATION REQUIRED message is transmitted, when the gNB configured by the message is group of subcounter. <i>neighbourgNB</i> . Double counting is not performed if the procedure is changed to the MN initiated SN Modification procedure after transmitted Xn-AP: S-NODE MODIFICATION REQUIRED message (e.g. Measurement gap Coordination (SN initiated)). This subcounter will not be incremented if received Xn-AP: S-NODE MODIFICATION REQUEST message which True is set for SCG Configuration Query IE.
Measurement Result	Integer number (U32)
Measurement Type	OR.NRDCPROCEDURE.SnModAttemptNeiGnb. <i>neighbourgNB</i> where <i>neighbourgNB</i> is neighbour gNB index: 0: #0 1: #1 ... 63: #63
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.20.8 Number of S-NG-RAN Node modification procedure successfully for each neighbour gNB

A.20.8.1 Performance Counter Table

Measurement Name	OR.NRDCPROCEDURE.SnModSuccessNeiGnb
Description	This counter provides the number of the completed S-NG-RAN Node modification Procedure. This counter is split into subcounters per neighbour gNB.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever SN Modification procedure is completed (i.e. Xn-AP: S-NODE RECONFIGURATION COMPLETE message (SN modification complete) is received or Xn-AP: S-NODE MODIFICATION CONFIRM is received and RACH processing with UE Context is completed) when the gNB configured by the message is group of subcounter. <i>neighbourgNB</i> .
Measurement Result	Integer number (U32)

Measurement Type	OR.NRDCPROCEDURE.SnModSuccessNeiGnb. <i>neighbourNB</i> where <i>neighbourNB</i> is neighbour gNB index: 0: #0 1: #1 ... 63: #63
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.20.9 Number of S-NG-RAN Node Release procedure attempted for each Cell

A.20.9.1 Performance Counter Table

Measurement Name	OR.NRDCPROCEDURE.SnRelAttemptCell
Description	This counter provides the number of the received Xn-AP: S-NODE RELEASE REQUIRED messages or transmitted Xn-AP: S-NODE RELEASE REQUEST messages.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever Xn-AP: S-NODE RELEASE REQUIRED message is received or Xn-AP: S-NODE RELEASE REQUEST message is transmitted as PSCell.
Measurement Result	Integer number (U32)
Measurement Type	OR.NRDCPROCEDURE.SnRelAttemptCell
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.20.10 Number of S-NG-RAN Node Release procedure successfully for each Cell

A.20.10.1 Performance Counter Table

Measurement Name	OR.NRDCPROCEDURE.SnRelSuccessCell
Description	This counter provides the number of the completed S-NG-RAN Node Release Procedure.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever SN Release procedure is completed as PSCell (i.e. Xn-AP: UE CONTEXT RELEASE message is transmitted)
Measurement Result	Integer number (U32)
Measurement Type	OR.NRDCPROCEDURE.SnRelSuccessCell
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.20.11 Number of S-NG-RAN Node Release procedure attempted for each neighbour gNB

A.20.11.1 Performance Counter Table

Measurement Name	OR.NRDCPROCEDURE.SnRelAttemptNeiGnb
Description	This counter provides the number of the received Xn-AP: S-NODE RELEASE REQUIRED messages or transmitted Xn-AP: S-NODE RELEASE REQUEST messages. This counter is split into subcounters per neighbour gNB.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever Xn-AP: S-NODE RELEASE REQUIRED message is received or Xn-AP: S-NODE RELEASE REQUEST message is transmitted, when the gNB configured by the message is group of subcounter. <i>neighbourgNB</i> .
Measurement Result	Integer number (U32)
Measurement Type	OR.NRDCPROCEDURE.SnRelAttemptNeiGnb. <i>neighbourgNB</i> where <i>neighbourgNB</i> is neighbour gNB index: 0: #0 1: #1 ... 63: #63
Measurement Object Class	NRCelLCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.20.12 Number of S-NG-RAN Node Release procedure successfully for each neighbour gNB

A.20.12.1 Performance Counter Table

Measurement Name	OR.NRDCPROCEDURE.SnRelSuccessNeiGnb
Description	This counter provides the number of the completed S-NG-RAN Node Release Procedure. This counter is split into subcounters per neighbour gNB.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever SN Release procedure is completed (i.e. Xn-AP: UE CONTEXT RELEASE message is transmitted)
Measurement Result	Integer number (U32)
Measurement Type	OR.NRDCPROCEDURE.SnRelSuccessNeiGnb. <i>neighbourgNB</i> where <i>neighbourgNB</i> is neighbour gNB index: 0: #0 1: #1 ... 63: #63

Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.20.13 Number of SCGFailureInformation received

A.20.13.1 Performance Counter Table

Measurement Name	OR.NRDCPROCEDURE.NumSCGFailInfoRec
Description	This counter provides the number of SCGFailureInformation received.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever RRC: SCG Failure Information message is received.
Measurement Result	Integer number (U32)
Measurement Type	OR.NRDCPROCEDURE.NumFailInfoRec
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.21 Monitoring of CA for SA

A.21.1 Number of SCell Addition procedure attempted

A.21.1.1 Performance Counter Table

Measurement Name	OR.SACAPROCEDURE.ScclAddAttempt
Description	This counter provides the number of the attempted SCell addition procedure.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever Special Cell decides to perform SCell addition procedure by the following trigger. - received RRC: Measurement report for SCell addition. - SCell addition triggered internal the gNB other than above trigger.
Measurement Result	Integer number (U32)
Measurement Type	OR.SACAPROCEDURE.ScclAddAttempt
Measurement Object Class	NRCellCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.21.2 Number of SCell Addition procedure successfully

A.21.2.1 Performance Counter Table

Measurement Name	OR.SACAPROCEDURE.ScclAddSuccess
Description	This counter provides the number of the completed SCell addition

	procedure.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever Special Cell detects successful SCell addition procedure by the received following messages. – received RRC: RRC Reconfiguration Complete for SCell addition.
Measurement Result	Integer number (U32)
Measurement Type	OR.SACAPROCEDURE.ScclAddSuccess
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

1

2

A.21.3 Number of SCell Change procedure attempted

3

A.21.3.1 Performance Counter Table

Measurement Name	OR.SACAPROCEDURE.ScclChangeAttempt
Description	This counter provides the number of the attempted SCell change procedure.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever Special Cell decides to perform SCell change procedure by the following trigger. – received RRC: Measurement report for SCell change. – SCell change triggered internal the gNB other than above trigger.
Measurement Result	Integer number (U32)
Measurement Type	OR.SACAPROCEDURE.ScclChangeAttempt
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

4

5

A.21.4 Number of SCell Change procedure successfully

6

A.21.4.1 Performance Counter Table

Measurement Name	OR.SACAPROCEDURE.ScclChangeSuccess
Description	This counter provides the number of the completed SCell change procedure.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever Special Cell detects successful SCell change procedure by the received following messages. – received RRC: RRC Reconfiguration Complete for SCell change.
Measurement Result	Integer number (U32)
Measurement Type	OR.SACAPROCEDURE.ScclChangeSuccess
Measurement Object Class	NRCelCU
Switching Technology	Packet Switched

Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.21.5 Number of SCell Deletion procedure attempted

A.21.5.1 Performance Counter Table

Measurement Name	OR.SACAPROCEDURE.ScclDeletionAttempt
Description	This counter provides the number of the attempted SCell deletion procedure.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever Special Cell decides to perform SCell deletion procedure by the following trigger. - received RRC: Measurement report for SCell deletion. - SCell deletion triggered internal the gNB other than above trigger.
Measurement Result	Integer number (U32)
Measurement Type	OR.SACAPROCEDURE.ScclDeletionAttempt
Measurement Object Class	NRCelICU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

A.21.6 Number of SCell Deletion procedure successfully

A.21.6.1 Performance Counter Table

Measurement Name	OR.SACAPROCEDURE.ScclDeletionSuccess
Description	This counter provides the number of the completed SCell deletion procedure.
Collection Method	CC (Cumulative Counter)
Condition	Measurement subcounter is incremented by 1 whenever Special Cell detects successful SCell deletion procedure by the received following messages. received RRC: RRC Reconfiguration Complete for SCell deletion.
Measurement Result	Integer number (U32)
Measurement Type	OR.SACAPROCEDURE.ScclDeletionSuccess
Measurement Object Class	NRCelICU
Switching Technology	Packet Switched
Generation	5GS
Purpose	Network Operator's Traffic Engineering Community

Revision History

Date	Revision	Author	Description
2022.03.25	03.00	O-RAN-WG5	Draft for voting
2022.07.21	04.00	O-RAN-WG5	Updates to v03.00: - Additional clarification on alignment with O-RAN O1 specification document - Editorial fixes
2023.03.08	05.00	O-RAN-WG5	Updates to v04.00: - Updates to Performance counters in Annex A - Addition of information model of CuCountGroup - Editorial fixes
2023.07.06	06.00	O-RAN-WG5	Updates to v05.00: - Editorial modification to align with ODR - Clarifications and corrections for PM counters in Annex A
2023.11.01	07.00	O-RAN-WG5	Updated to v06.00: - Removed description of rpc: list-available-files - Added O-CU-CP QoS IM - Clarification on counters defined in A.10.1 to A.10.12, A.11.1 to A.11.8, A14.1 to A14.12, A.15.1 to A.15.20, A.16.3 to A.16.5, A.19.2 to A.19.3 - Addition of new counters in A.1.11, A.1.12, A.3.3, A.3.4, A.11.9 to A.11.15, A.11.21 to A.11.31, A.16.6 to A.16.10, A.17.3 to A.17.11, A.18.9 to A.18.12, A.19.4 to A.19.6, A.20.1 to A.20.13, A.21.1 to A.21.6 - Minor editorial changes